
AP

Calculus CD

*This is a crazy continuation
of the calculus sequence*

FUNG SAN GAAN

AP Calculus CD

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Week 1	4	Week 9	26
smooth manifold, charts, atlas, transition functions, smooth structure, smooth maps, coordinates, diffeomorphisms, gluing lemma, partition of unity.		vector bundles, tangent and cotangent bundles, gradient, vector bundle homomorphism, and local trivializations, line integrals	
Week 2	7	Week 10	30
directional derivative, derivation, differentials, tangent space, tangent bundles, manifolds with boundary		covectors and tensors on manifolds, conservative and exact covectors, closed covector fields, Riemannian metrics, distances and angles on manifolds	
Week 3	9	Week 11	33
submersions, immersions, embeddings, rank theory, submanifolds, slice charts, level sets, sections, local sections, covering maps		differential forms, exterior derivatives, form products, form pullbacks, Lie derivative, orientation	
Week 4	13	Week 12	37
Sard's theorem, transversality, two versions of the Whitney embedding theorem		integration on manifolds, Stokes theorem, calc III revisit, de Rham cohomology.	
Week 5	16	Week 13	41
Lie group, Lie group homomorphisms, Lie group actions, equivariant maps, representations of Lie groups, semi-direct products of Lie groups.		more de Rham cohomology and its applications, degree theory, singular (co)homology, symplectic manifolds, Hamiltonian vector fields	
Week 6	18	Week 14	46
frames, F -related vector fields, Lie algebras, properties of Lie brackets, left-invariant vector fields, $\text{Lie}(G) \simeq T_e G$		fall break	
Week 7	22	Week 15	46
midterm		delayed indefinitely	
Week 8	22	Appendix	46
integral curves, flows, relationships between vector fields and flows, connected sums, doubles, collar neighborhoods, complete vector fields		Index	

General Notice: First 13 weeks' content covers all the math518 topics (and possibly more). For this reason, week 15 is delayed indefinitely and won't be available even if it's finished in the future.

These notes are adapted from Dan's lecture, FA24.

If you spot any error in these notes, please reach out to me via email b192.168.1.91.03@gmail.com, or you can direct message me on Discord at 3fsg.

Life's not perfect.

—Dan

Week 1

Definition of topological manifold, charts, atlas, transition functions, definition of manifold, smooth structure, smooth manifold, smooth maps, functions, coordinates, local diffeomorphism, explicit way of building a manifold, gluing lemma, partition of unity

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Vocabulary 1.1 (topological manifold). An n -dimensional topological manifold is a topological space M :

1. M is Hausdorff.
2. M is second countable.
3. M is locally Euclidean, i.e., locally homeomorphic to $\mathbb{R}^n$. For all point $p \in M$, there exists an open $U \subseteq M$ with $U \xrightarrow{\varphi} \varphi(U) \subseteq \mathbb{R}^n$. φ here is a homeomorphism.

Vocabulary 1.2 (chart). A chart on a topological n -manifold M is an open subset $U \subseteq M$ and a homeomorphism

$$U \xrightarrow{\varphi} \varphi(U) \subseteq \mathbb{R}^n$$

Vocabulary 1.3 (atlas). A collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ that covers M .

We get two things from charts

1. **Local coordinates/local functions.**

$$M \supseteq U \xrightarrow{\varphi} \varphi(U) \subseteq \mathbb{R}^n \xrightarrow{p_i} \mathbb{R}$$

gives a function $x_i : U \rightarrow \mathbb{R}$ for $i = 1, \dots, n$. So $\varphi(q) = (x_1(q), \dots, x_n(q))$ for all $q \in U$.

2. **Transition maps.** For charts (U, φ) and (V, ψ) we have $U \cap V \subseteq M$. Moreover,

$$\begin{array}{ccc} & U \cap V & \\ \varphi \swarrow & & \searrow \psi \\ \mathbb{R}^n \supseteq \varphi(U \cap V) & \xrightarrow{\psi \circ \varphi^{-1}} & \psi(U \cap V) \subseteq \mathbb{R}^n \end{array}$$

is a homeomorphism by construction, where $\psi \circ \varphi^{-1}$ is called a transition function.

Vocabulary 1.4 (compatible). Two charts (U, φ) and (V, ψ) are smoothly compatible if the transition map $\psi \circ \varphi^{-1}$ is ∞ -differentiable with an ∞ -differentiable inverse. Such a map is called a diffeomorphism.

Vocabulary 1.5 (smooth atlas). A smooth atlas is an atlas where any pair of charts is smoothly compatible. A smooth atlas is maximal if it is not a proper subset of any smooth atlas.

Vocabulary 1.6 (smooth stuff). A smooth structure on a topological manifold is a maximal smooth atlas. A smooth manifold is a topological manifold with smooth structures.

Lemma (happy). Let M be a topological manifold,

1. every smooth atlas is contained in a *unique maximal* smooth atlas,
2. any pair of smooth atlases are contained in the same maximal atlas if and only if the union is a smooth atlas.

Proof sketch. ① Given a smooth atlas $\mathcal{A}$, let $\overline{\mathcal{A}}$ denote the atlas of all charts that are smoothly compatible with the charts in $\mathcal{A}$. We *claim* that $\mathcal{A}$ is maximal. ② If $\mathcal{A}$ is compatible with $\mathcal{A}'$, then $\mathcal{A}' \subseteq \overline{\mathcal{A}}$. By symmetry, $\mathcal{A} \subseteq \overline{\mathcal{A}'}$. By uniqueness of maximal atlases, $\overline{\mathcal{A}} = \overline{\mathcal{A}'}$. $\square$

Example. Here are some common smooth stuff

1. $\mathbb{R}^n$ with smooth atlas given by the identity map $\varphi = \text{id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$.
2. Sphere S^n gets a smooth structure through the stereographic projection.
3. Products of smooth manifolds are smooth.
4. $\mathbb{R}$ with an atlas given by $\psi : \mathbb{R} \rightarrow \mathbb{R}, \psi(x) = x^3$. Since $\psi^{-1}(x) = x^{1/3}$ is not differentiable at $x = 0$. This atlas is not smoothly compatible with the atlas in (1).

Vocabulary 1.7 (smooth maps between M, N). Let M and N be smooth manifolds. A function $F : M \rightarrow N$ is smooth if for every $p \in M$, there exists a smooth chart (U, φ) for M whose domain contains p and (V, ψ) of N containing $F(p)$ such that $F(U) \subseteq V$ and the composite function $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Remark. If you like pictures, here is how it looks

$$\begin{array}{ccc} U \subseteq M & \xrightarrow{F} & N \supseteq V \\ \varphi \downarrow & & \psi \downarrow \\ \varphi(U) & \xrightarrow{\psi \circ F \circ \varphi^{-1}} & \psi(V) \end{array}$$

Whenever we need to show smooth maps between manifolds, we should check if the dashed map is smooth for any pair of charts U of M and V of N .

Example. Here are some smooth maps between manifolds:

1. Any smooth map $\mathbb{R}^m \rightarrow \mathbb{R}^n$ is a map of smooth manifolds for the standard smooth structure (identity charts).
2. $S^n := \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}$ have coordinate projections $S^n \rightarrow \mathbb{R}$. Those are also smooth.

Lemma (general stuff). ① Smooth maps between smooth manifolds are continuous. ② The composition of smooth maps is smooth.

Vocabulary 1.8 (smooth functions on M). The collection of smooth maps $M \rightarrow \mathbb{R}$ for a smooth manifold M is called the smooth function on M . Denoted as $C^\infty(M)$. Observe that $C^\infty(M)$ is an algebra over $\mathbb{R}$.

Vocabulary 1.9 (diffeomorphism). A diffeomorphism between smooth manifolds M and N is a smooth map $F : M \rightarrow N$ with a smooth inverse (i.e. F is a bijection and the inverse function is smooth).

Vocabulary 1.10 (local diffeomorphism). A local diffeomorphism $F : M \rightarrow N$ is a smooth map such that for all $p \in M$, there exist open sets $U \subseteq M$ and $V \subseteq N$, and a diffeomorphism $F|_U : U \rightarrow V$.

Example. Here are some diffeomorphisms:

1. $f : \mathbb{R} \rightarrow S^1 \subseteq \mathbb{R}^2 \simeq \mathbb{C}$, $f(x) = e^{ix} \in \mathbb{C}$. This has a local inverse (log) and hence is a local diffeomorphism.
2. Equip $\mathbb{R}$ with the standard smooth structure and let $\psi(x) = x^3$.

$$\begin{array}{ccc} M = \mathbb{R} & \xrightarrow{F} & N = \mathbb{R} \\ \varphi = \text{id} \downarrow & & \psi \downarrow \\ \varphi(U) & \xrightarrow[\text{id}]{} & \psi(V) \end{array}$$

Then $F : M \rightarrow N$ defined by $F(x) = x^{1/3}$ is a diffeomorphism. This shows that the new smooth structure on $\mathbb{R}$ is different than the id atlas.

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Example. More examples of smooth manifolds:

1. Given a smooth manifold M , an open subset $N \subseteq M$ is again a smooth manifold with charts obtained by intersecting the charts of M with the subset N .
2. $GL_n(\mathbb{R}) \subseteq M_{n \times n}(\mathbb{R}) \simeq \mathbb{R}^{n^2}$. It follows that $GL_n(\mathbb{R})$ is a smooth manifold.
3. Given $U \subseteq \mathbb{R}^n$ open and $\varphi : U \rightarrow \mathbb{R}$ smooth map with regular value $c \in \mathbb{R}$ (so $d\varphi \neq 0$ at c). Then $\varphi^{-1}(c)$ is a smooth manifold (by the implicit function theorem). Maybe a sub-example: $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$, $\varphi(x) = |x|^2$, and $\varphi^{-1}(1) = S^{n-1}$.

Lemma (smooth manifold chart lemma). Given sets M and $\{\varphi_\alpha : M \supseteq U_\alpha \rightarrow \mathbb{R}^n\}_{\alpha \in A}$ such that

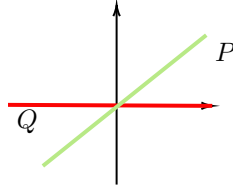
1. $\varphi_\alpha : U_\alpha \rightarrow \varphi(U_\alpha) \subseteq \mathbb{R}^n$ is a bijection.
2. $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$ is open.
3. $\varphi_\alpha(U_\alpha \cap U_\beta)$ open.
4. $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a smooth map.
5. Countably many of the U_α cover M .

6. For $p, q \in M$, $p \neq q$, either $p, q \in U_\alpha$ or $p \in U_\alpha$ and $q \in U_\beta$ with $U_\alpha \cap U_\beta = \emptyset$.

Then M has the structure of a smooth manifold with smooth atlas given by $\varphi_\alpha : U_\alpha \rightarrow M$.

Proof. We claim $\{\varphi_\alpha^{-1}(V) | \alpha \in A, V \subseteq \mathbb{R}^n \text{ open}\}$ is a basis for a topology on M . Then M is locally Euclidean. Further, M is second countable by (5) and Hausdorff by (6). Finally, the topological manifold M has a smooth structure given by the maximal smooth atlas determined by the smooth atlas from (4). $\square$

Example (smooth manifold chart lemma in action). Let $M = G_k(\mathbb{R}^n) = \{k\text{-dimensional subspaces in } \mathbb{R}^n\}$. Pick $Q \subseteq \mathbb{R}^n$ of $\dim(Q) = n - k$. Set $U_Q = \{P \in G_k(\mathbb{R}^n) | P \cap Q = \{0\}\}$. For example, when $k = 1, n = 2$



We claim that $U_Q \simeq \text{hom}(Q, Q^\perp) \simeq (n - k) \times k$ matrices $\simeq \mathbb{R}^{k(n-k)}$. This map takes a linear map $T : Q \rightarrow Q^\perp$ to its graph $\text{Graph}(T) \subseteq Q \oplus Q^\perp = \mathbb{R}^n$. Obtain a cover $\{U_Q\}$ of $G_k(\mathbb{R}^n)$ and bijections $U_Q \simeq \mathbb{R}^{k(n-k)}$. The tricky part to check in the lemma is (4). But it boils down to Cramer's rule. Conclude that $G_k(\mathbb{R}^n)$ has the structure of a smooth manifold. We have seen special cases of G_k when $k = 1$: $G_1(\mathbb{R}^n) = \mathbb{RP}^{n-1}$ and $G_1(\mathbb{C}^n) = \mathbb{CP}^{n-1}$.

Lemma (gluing lemma). For smooth manifolds M, N and an open (smooth) cover $\{U_\alpha\}_{\alpha \in A}$ of M , given $\varphi_\alpha : U_\alpha \rightarrow N$ such that for $\alpha, \beta \in A$,

$$\varphi_\alpha|_{U_\alpha \cap U_\beta} = \varphi_\beta|_{U_\beta \cap U_\alpha} : U_\alpha \cap U_\beta \rightarrow N$$

then there is a unique smooth map $F : M \rightarrow N$ such that $F|_{U_\alpha} = \varphi_\alpha$.

Before we talk about the partition of unity, first recall that for a topological space X , an open cover is a collection of open subsets $\{U_\alpha\}_{\alpha \in A}$ such that $\bigcup_{\alpha \in A} U_\alpha = X$.

Vocabulary 1.11 (partition of unity). Given a topological space X and some open cover $\{U_\alpha\}_{\alpha \in A}$, a partition of unity assigned to the cover is a collection of continuous functions $\psi_\alpha : X \rightarrow \mathbb{R}$ so that

1. $0 \leq \psi_\alpha(x) \leq 1, \forall x \in X$.
2. $\text{supp}(\psi_\alpha) := \{x \in X | f(x) \neq 0\} \subseteq U_\alpha$.
3. the set of supports $\{\text{supp}(\psi_\alpha)\}$ is locally finite i.e. $\psi_\alpha(p) \neq 0$ for only finitely many α .
4. $\sum_{\alpha \in A} \psi_\alpha(x) = 1, \forall x \in X$. By (3), only finitely many nonzero summands.

Theorem 1. Topological manifolds admit a partition of unity for any open cover. Moreover, smooth manifolds have a partition of unity for any open cover with smooth functions $\psi_\alpha : M \rightarrow \mathbb{R}$.

Proof idea. Consider manifold without boundaries, and similar idea can be used for the general case. Find a countable, locally finite refinement $\{B_i\}$ of X using basis elements of U_α . Define $M \rightarrow \mathbb{R}$ by

$$f_i = \begin{cases} H_i \circ \varphi_i & \text{on } B'_i \\ 0 & \text{on } M \setminus \bar{B}_i \end{cases}$$

where $B'_i \supseteq \bar{B}_1$ in X_α , $\varphi : B'_i \rightarrow \mathbb{R}^n$, $\varphi_i : \bar{B}_i \mapsto \bar{B}_{r_i}(0)$, $\varphi_i : B'_i \mapsto B_{r'_i}(0)$ for some $r_i < r'_i$, $H_i : \mathbb{R}^n \rightarrow \mathbb{R}$, H_i positive in $B_{r_i}(0)$ and zero otherwise. Define $g_i = \sum f_i / f_i$. Reindexing g_i and sum over some g_i gives the desired ψ_α . $\square$

Vocabulary 1.12 (exhaustion function). For a topological space X , an exhaustion function is a continuous function $f : X \rightarrow \mathbb{R}$ such that $f^{-1}((-\infty, c)) = X_c \subseteq X$ is a compact subspace for all finite c .

Proposition 1. Any smooth manifold M admits a smooth exhaustion function $f : M \rightarrow \mathbb{R}$.

Proof idea. Consider some $j \in \mathbb{N}$ and $\sum_{j=1}^\infty j\psi_j = f : M \rightarrow \mathbb{R}$ for $\psi_j : M \rightarrow \mathbb{R}$. They form a partition of unity subordinate to a precompact countable cover $\{U_j\}$. $\square$

Week 2

What is a directional derivative? What is a derivation? Directional derivative is a derivation. Properties of (total)differentials. Definition of tangent space at a point p , differential, tangent bundle, manifolds with boundary.

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We can view tangent vectors in two different ways: ① Geometric tangent vectors come from the velocity of a curve, while ② algebraic tangent vectors allow you to take directional derivatives of functions.

Vocabulary 2.1 (curve). A curve γ is a smooth map $\gamma : \mathbb{R} \rightarrow M$. Given a smooth function $f : M \rightarrow \mathbb{R}$, the derivative at $\gamma(0) = p \in M$ is given by

$$\left. \frac{d}{dt} \right|_{p=0} (f \circ \gamma) = D_{\gamma'}|_p f.$$

More generally, given a chart $\mathbb{R}^n \supseteq \varphi(U) \xrightarrow{\varphi^{-1}} U \subseteq M \rightarrow \mathbb{R}$. One can use composition to calculate directional derivatives.

Vocabulary 2.2 (derivation). For a smooth manifold, a derivation at $p \in M$ is a linear map $X : C^\infty(M) \rightarrow \mathbb{R}$ such that $X(fg) = X(f)g(p) + f(p)X(g)$.

Example. Here are some examples of derivations:

1. Derivatives along curves give derivations:

$$\left. \frac{d}{dt} \right|_p ((fg) \circ \gamma) = \left. \frac{d}{dt} \right|_p ((f \circ \gamma)(g \circ \gamma)) = \left(\left. \frac{d}{dt} \right|_p (f \circ \gamma) \right) (g \circ \gamma)(p) + (f \circ \gamma)(p) \left. \frac{d}{dt} \right|_p (g \circ \gamma).$$

2. Directional derivatives in $\mathbb{R}^n$ are given by geometric tangent vectors: Given vector v based at p , v_p yields a map $D_v|_p : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$, which takes the directional derivative in the direction v at p :

$$D_v|_p f = D_v f(p) = \lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t} = \left. \frac{d}{dt} \right|_{t=0} f(p + tv).$$

We can easily check that this is indeed a derivation.

Vocabulary 2.3 (tangent space). For a smooth manifold M , the tangent space at $p \in M$, denoted $T_p M$, is the vector space of derivations at p .

Now we should check if this is the “right thing” to define. We have seen that geometric tangent vectors can give us derivations in the previous example. It is reasonable to ask: Is there a correspondence between the set of all geometric tangent vectors $\{v_p\}$ and the set of directional derivatives $\{D_v|_p\}$? Is the whole set of derivations $T_p M$ produced by these directional derivatives?

Lemma. Given derivation $X : C^\infty(M) \rightarrow \mathbb{R}$, we have

1. $X(c) = 0$ for any constant c ¹.
2. If $f(p) = g(p) = 0$, then $X(fg) = 0$ ².

Proof. ① Assume $f \equiv 1$, $X(1) = 2X(1) \Rightarrow X(1) = 0$. ② Follows from the definition. □

Proposition 2. The map $v_p \mapsto D_v|_p$ is an isomorphism from $\mathbb{R}^n_p = \{v_p \mid v \in \mathbb{R}^n\}$ onto $T_p \mathbb{R}^n$.

Proof. The assignment $v \mapsto D_v$ is linear (property of directional derivative). For injectivity, we check the kernel. Suppose D_v is the 0 derivative. Write $v = \sum v_i e_i$ in standard basis. $D_v = \sum v_i \frac{\partial}{\partial x_i}$. Then

$$D_v(x_j) = \sum v_i \frac{\partial x_j}{\partial x_i} = v_j$$

¹In calculus language: derivative vanishes on constants.

²In calculus language: ...as well as 2nd order zeros.

Hence $D_v = 0$ if and only if $v_i = 0$ for all i . Then $v = 0$. Now we check surjectivity. Given X a derivation, i.e. $X \in T_p \mathbb{R}^n$. Use the above calculation to define $v = \sum v_i e_i$, where $X(x_j) = v_j$. We claim that X is a directional derivative D_v . Let $f \in C^\infty(\mathbb{R}^n)$, use Taylor

$$f(x) = f(p) + \sum \frac{\partial f}{\partial x_i}(p)(x_i - p) + o((x_i - p)^2)$$

X and D_v both vanish on $f(p)$ and $o((x_i - p)^2)$. So $Xf(x) = 0 + \sum \frac{\partial f}{\partial x_i}(p)(X(x_i) - X(p_i)) = \sum \frac{\partial f}{\partial x_i}(p)v_i = D_v f$. $\square$

Corollary. $T_p \mathbb{R}^n = \text{span}(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n})$. Further, $\dim(T_p \mathbb{R}^n) = n$.

The tangent space $T_p M$ is the vector space of derivations at $p \in M$. From now on, we will use the Einstein notation. This seriously sucks, but we need to get used to it.

Vocabulary 2.4 (differential). Given a smooth map $F : M \rightarrow N$. Define $dF_p : T_p M \rightarrow T_{F(p)} N$ by $(dF_p X)(f) = X(f \circ F)$, $X \in T_p M$ and $f \in C^\infty(N)$. Note that $f \circ F \in C^\infty(M)$.

Lemma. Here are some properties of differentials.

1. dF_p is well-defined.
2. $d(G \circ F)_p = dG_{F(p)} \circ dF_p$. Remember that $G \circ F$ is a map $M \rightarrow k$.
3. dF_p is linear.
4. $d(\text{id}_M)_p = \text{id}_{T_p M}$.
5. For $F : M \rightarrow N$, a local diffeomorphism, dF_p is an isomorphism.

Proposition 3. If M is an n -dimensional manifold, then $T_p M$ is an n -dimensional vector space.

Proof. Given a chart (U, φ) containing p , we have a local diffeomorphism $\mathbb{R}^n \supseteq \varphi(U) \xrightarrow{\varphi^{-1}} U \subseteq M$. So $\text{span}(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}) = T_{\varphi(p)} \mathbb{R}^n \simeq T_{\varphi(p)} M$. Conclude that $T_p M$ is n -dimensional. $\square$

Remark. The dimension of $T_p M$ is not the only thing we get from this. We also get a basis for $T_p M$ given by the image of $\frac{\partial}{\partial x^i}$. These are called *coordinate vectors*. They depend on the choice of chart and we write $\frac{\partial}{\partial x^i} \in T_p M$ when the chart is clear. Then for any $v \in T_p M$, $v = v^i \frac{\partial}{\partial x^i}$, where $v^i \in \mathbb{R}$ are components of v (relative to coordinate vectors). In coordinates, dF_p recovers the multivariable chain rule: Pick (U, φ) for M and (V, ψ) for N ,

$$\begin{array}{ccc} M & \xrightarrow{F} & N \\ \downarrow & & \downarrow \\ U \subseteq \mathbb{R}^n & \xrightarrow{F|_U} & V \subseteq \mathbb{R}^m \end{array}$$

corresponds to a map $dF_p : T_{\varphi(p)} \mathbb{R}^n \simeq T_p M \rightarrow T_{\psi(F(p))} \mathbb{R}^m \simeq T_{F(p)} N$. Thus

$$dF_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial F^j}{\partial x^i}(p) \frac{\partial}{\partial y^j} \Big|_{F(p)} \quad \text{and} \quad dF_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial(\psi \circ F \circ \varphi^{-1})^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial y^j} \Big|_{F(p)} \quad (\star)$$

for any $F : M \rightarrow N$. Of course, we can put dF_p into a matrix form with first row being $\frac{\partial F^1}{\partial x^1}(p), \dots, \frac{\partial F^1}{\partial x^n}(p)$. Now we consider a special case. Suppose $(U, \varphi), (V, \psi)$ are charts on M whose domains contain p . Then we get two bases of $T_p M$. But

$$T_p M \simeq T_{\varphi(p)} \mathbb{R}^n = \text{span} \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right) \simeq \text{span} \left(\frac{\partial}{\partial y^1}, \dots, \frac{\partial}{\partial y^n} \right) \simeq T_{\psi(p)} \mathbb{R}^n \simeq T_p M.$$

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Vocabulary 2.5 (tangent bundle). As a set, the tangent bundle of a manifold M is

$$TM = \bigsqcup_{p \in M} T_p M.$$

Example. The tangent bundle $T\mathbb{R}^n = \bigsqcup_{p \in \mathbb{R}^n} T_p \mathbb{R}^n \simeq \mathbb{R}^{2n} \ni (p, v)$.

Proposition 4. For any smooth manifold M , TM has a canonical structure of smooth manifold of dimension $2 \cdot \dim(M)$.

Proof sketch. Given a chart (U, φ) on M , consider $\pi^{-1}(U) \subseteq TM$ where $\pi : TM \rightarrow M$ is defined by $\pi(p, v) = p$. Then define $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ by

$$\tilde{\varphi} \left(v^i \frac{\partial}{\partial x^i} \Big|_p \right) = \underbrace{(x^1(p), \dots, x^n(p))}_{\text{coordinates for } p \text{ in } (U, \varphi)}, \underbrace{(v^1(p), \dots, v^n(p))}_{\text{coordinates of } V}.$$

To show transition maps on TM are smooth, we use the fact that they are matrices of partial derivatives of smooth transition maps for charts on M . Hence they are matrices of smooth functions. $\square$

Example. Some examples of tangent bundles. Maybe it would be helpful if we associate them with pictures.

1. $T\mathbb{R} \simeq \mathbb{R}^2$.
2. $TS^1 \simeq S^1 \times \mathbb{R}$.
3. $TS^2 \not\simeq S^2 \times \mathbb{R}^2$.

Vocabulary 2.6 (total differential). Given a smooth map $F : M \rightarrow N$, the total differential of F is $dF = \bigsqcup_{p \in M} dF_p : TM \rightarrow TN$ given by a dF_p map $(p, v) \mapsto (F(p), dF_p(v))$.

Proposition 5. The map $dF : TM \rightarrow TN$ is a map of smooth manifolds.

Proof. In coordinates,

$$dF(x^1, \dots, x^n, v^1, \dots, v^n) = \left(F^1(x), \dots, F^m(x), \frac{\partial F^1}{\partial x^i} v^i, \dots, \frac{\partial F^m}{\partial x^i} v^i \right)$$

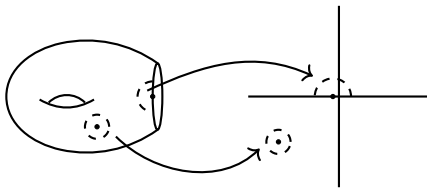
This defines a smooth map between open subsets of $\mathbb{R}^{2n}$ by the fact that F is smooth in local coordinates. $\square$

Lemma. Here are some properties of total differentials

1. $d(G \circ F) = dG \circ dF$.
2. $d(\text{id}_M) = \text{id}_{TM}$.
3. For $F : M \rightarrow N$, a diffeomorphism if and only if dF is a diffeomorphism.

Vocabulary 2.7 (manifold with boundary). $\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n | x^n \geq 0\}$.

Remark. Naturally we have $\partial\mathbb{H}^n \simeq \mathbb{R}^{n-1} \hookrightarrow \mathbb{H}^n \subseteq \mathbb{R}^n$ and $\text{int}(\mathbb{H}^n) \simeq \mathbb{R}^n$. A topological manifold with boundary is a Hausdorff 2nd countable space that is locally homeomorphic to $\mathbb{H}^n$, i.e. for some chart (U, φ) of M , $\varphi(U) \subseteq \mathbb{H}^n$.



Vocabulary 2.8 (boundary/interior points and transition maps). For $p \in M$ such that there exists a chart (U, φ) with $\varphi(p) \in \partial\mathbb{H}^n$, we call p a boundary point. Otherwise p is an interior point. Define smoothly compatible charts as before. Call them (U, φ) and (V, ψ) . Then $\varphi(U), \psi(V) \subseteq \mathbb{H}^n$ and the map $\varphi(U) \rightarrow \psi(V)$ must be smooth.

Remark. Here I'd like to skip all the category nonsense. Check the appendix.

Week 3

Went over submersions, immersions, embeddings (all the rank stuff). Spent a lot of time on submanifolds. Also did slice charts, level sets, sections, local sections, and covering maps.

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Vocabulary 3.1 (constant rank). A smooth map $F : M \rightarrow N$ has constant rank r if for all $p \in M$, $dF_p : T_p M \rightarrow T_{F(p)} N$ has rank r .

Vocabulary 3.2 (submersion and immersion). A submersion is $F : M \rightarrow N$ such that dF_p is surjective for all $p \in M$. An immersion is $F : M \rightarrow N$ such that dF_p is injective for all $p \in M$. In other words, if $\dim(N)$ is fixed, then submersion would imply $\text{rk}(dF_p) = \dim(N)$ and immersion would imply $\text{rk}(dF_p) = \dim(M)$.

Example. Here are some examples of immersions/submersions:

1. The map $S^n \hookrightarrow \mathbb{R}^{n+1}$ gives an injection on the tangent space hence has constant rank n . It is an immersion.
2. Projection map $\mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ is a submersion.
3. Projection map $S^1 \times \mathbb{R} \rightarrow \mathbb{R}$ is a submersion.
4. Inclusion map $S^1 \simeq S^1 \times \{0\} \hookrightarrow S^1 \times \mathbb{R}$ is an immersion.

Proposition 6. ① If dF_p is a surjection, then there exists a neighborhood U of p such that $F|_U$ is a submersion. ② If dF_p is injective, then there exists a neighborhood U of p such that $F|_U$ is an immersion.

Proof sketch. Having full rank is an open condition. This means the set of full rank matrices is open. So given a point $p \in M$ for some full rank dF_p , there exists some neighborhood U of p where dF_p has full rank. $\square$

Theorem 2. Let $F : M \rightarrow N$ be smooth and $p \in M$ such that dF_p is an isomorphism. Then there exists a neighborhood U of p such that $F|_U$ is a diffeomorphism.

Proof. Since dF_p is invertible, $\dim(M) = \dim(N)$. Now choose charts (U, φ) on M and (V, ψ) on N . By the inverse function theorem for $\mathbb{R}^n$, $\psi \circ F \circ \varphi^{-1}$ is a diffeomorphism in a neighborhood of $\varphi(p)$. This shows F is a diffeomorphism in a neighborhood of p . $\square$

Corollary. The map $F : M \rightarrow N$ is a local diffeomorphism if and only if dF_p is invertible for all $p \in M$.

Proposition 7. Here are some facts about local diffeomorphism

1. Closed under compositions.
2. Closed under taking products: diffeomorphisms $M_1 \rightarrow N_1$ and $M_2 \rightarrow N_2$ would imply a diffeomorphism $M_1 \times M_2 \rightarrow N_1 \times N_2$.
3. Local diffeomorphism imply local homeomorphism. And the map is open.
4. Closed under restriction to open submanifold $F|_U$.
5. Bijective diffeomorphism is a local diffeomorphism. Counter example, consider the map $\mathbb{R} \rightarrow S^1 \subseteq \mathbb{C}$ given by $\vartheta \mapsto e^{i\vartheta}$. This is a local diffeomorphism but not a (global) diffeomorphism.

Theorem 3 (rank theorem). Suppose M and N are smooth manifolds of dimensions m and n , respectively, and $F : M \rightarrow N$ is a smooth map with constant rank r . For each $p \in M$, there exist smooth charts (U, φ) for M centered at p and (V, ψ) for N centered at $F(p)$ such that $F(U) \subseteq V$, in which F has a coordinate representation of the form

$$\psi \circ F \circ \varphi^{-1}(x^1, \dots, x^r, x^{r+1}, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0). \quad (1)$$

In particular, if F is a smooth submersion, this becomes $\psi \circ F \circ \varphi^{-1}(x^1, \dots, x^n, x^{n+1}, \dots, x^m) = (x^1, \dots, x^n)$, and if F is a smooth immersion, it is $\psi \circ F \circ \varphi^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^m, 0, \dots, 0)$.

Proof idea. Since the theorem is local, after choosing some coordinate charts we can assume

$$\frac{\partial F^j}{\partial x^i} = \left[\begin{array}{c|c} r \times r \text{ invertible} & \text{stuff} \\ \hline \text{stuff} & (m-r) \times (n-r) \text{ singular} \end{array} \right] = \left[\begin{array}{c|c} \partial Q / \partial x & \partial Q / \partial y \\ \hline \partial R / \partial x & \partial R / \partial y \end{array} \right]$$

Write $F(x, y) = (Q(x, y), R(x, y))$. By the inverse function theorem, $\frac{\partial Q}{\partial x} = \text{id}$, and $\frac{\partial Q}{\partial y} = \frac{\partial R}{\partial x} = \frac{\partial R}{\partial y} = 0$. $\square$

Theorem 4 (global rank). Let M and N be smooth manifolds, and suppose $F : M \rightarrow N$ is a smooth map of constant rank.

1. If F is surjective, then it is a smooth submersion.
2. If F is injective, then it is a smooth immersion.
3. If F is bijective, then it is a diffeomorphism.

Vocabulary 3.3 (embedding). A smooth map is an embedding if it is a topological embedding (homeomorphic onto its image) and an immersion.

Example. Some examples about embedding/nonembedding

1. $S^n \subseteq \mathbb{R}^{n+1}$.
2. $S^1 \times S^1 \subseteq \mathbb{R}^4$.
3. Knots: $S^1 \hookrightarrow \mathbb{R}^3$.
4. A line intersecting itself gives an immersion $(0, 1) \hookrightarrow \mathbb{R}^2$ but it is **not** an embedding.

Vocabulary 3.4 (section). A smooth map $\pi : M \rightarrow N$ admits a section if there is a smooth map $\sigma : N \rightarrow M$ such that $\pi \circ \sigma = \text{id}_N$. Pictorially,

$$M \xleftarrow{\sigma} \searrow \pi \rightarrow N$$

Example (section). For a projection map $\pi : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$, the inclusion $\sigma : \mathbb{R}^n \hookrightarrow \mathbb{R}^{n+k}$ is a section.

Vocabulary 3.5 (local sections). A map $\pi : M \rightarrow N$ admits local sections if for all $p \in N$, there exists some neighborhood U of p such that $\pi|_U : \pi^{-1}(U) \rightarrow U$ admits sections. In other words, there exists a right inverse $\sigma : U \rightarrow \pi^{-1}(U)$ such that $\pi|_U \circ \sigma = \text{id}_U$.

Example (local but not global section). The Möbius band M has a map $f : \partial M \rightarrow S^1$ given by $z \mapsto z^2$. We identify $z \in S^1$ as $z \in U(1) \subseteq \mathbb{C}$. This is a local section because we can invert $f(z) = z^2$. It has no global section since $f^{-1}(z) = z^{1/2}$ is not continuous at 0.

Theorem 5 (local section theorem). Suppose M and N are smooth manifolds and $\pi : M \rightarrow N$ is a smooth map. Then π is a smooth submersion if and only if every point of M is in the image of a smooth local section of π .

Vocabulary 3.6 (smooth covering map). A smooth covering map $F : M \rightarrow N$ is a smooth map that ① is surjective ② for all $p \in N$ there exists a neighborhood $U \subseteq N$ of p such that every component of $F^{-1}(U)$ is mapped diffeomorphically to U .

Example. Examples of smooth covering maps

1. $f_n : S^1 \rightarrow S^1$ by $z \mapsto z^n$. Similar to previous example, we identify elements of S^1 with complex numbers in $U(1)$. For each $p \in S^1$, take a small interval $I \ni p$. Then $f_n^{-1}(I)$ is a disjoint union of intervals.
2. $\exp : \mathbb{R} \rightarrow S^1$ given by $t \mapsto \exp(it)$ is an ∞ -fold covering.

Proposition 8 (local section theorem for smooth covering maps). Smooth covering maps admit local sections, hence are submersions by the local section theorem. More precisely, suppose E and M are smooth manifolds with or without boundary, and $\pi : E \rightarrow M$ is a smooth covering map. Given any evenly covered open subset $U \subseteq M$, any $q \in U$, and any p in the fiber of π over q , there exists a unique smooth local section $\sigma : U \rightarrow E$ such that $\sigma(q) = p$.

Proposition 9. Covering maps are local diffeomorphisms.

Vocabulary 3.7 (smooth submanifold). Smooth submanifolds are smooth manifolds that are subsets of other smooth manifolds.

Vocabulary 3.8 (codimension). Let $S \subseteq M$ be a submanifold. We define the codimension of S , denoted $\text{codim}(S)$, to be $\text{codim}(S) = \dim(M) - \dim(S)$.

Example. Some examples of submanifolds

1. $S^n \subseteq \mathbb{R}^{n+1}$ has codimension 1.
2. A *slice* of $M \times N$ is $M \times \{p\}$ for some $p \in N$. Then $M \times \{p\} \subseteq M \times N$ is a $\text{codim}(M \times \{p\}) = \dim(N)$ -submanifold. More generally, if U is an open subset of $\mathbb{R}^n$ and $k \leq n$, a k -dimensional slice of U (or a k -slice) is any subset of the form

$$S = \{(x^1, \dots, x^n) \in U \mid x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

3. $S^1 \times \mathbb{R}$ is simply a cylinder. Each circle drawn on the cylinder represents a slice $S^1 \times \{t_i\} \subseteq S^1 \times \mathbb{R}$.

$$\cdots \quad \overline{\quad \bigcirc \quad \bigcirc \quad \bigcirc \quad} \quad \cdots$$

$t_{i-1} \quad t_i \quad t_{i+1}$

4. "Standard" slices in $\mathbb{R}^{n+k}$ are given by setting some coordinates to zero: $\mathbb{R}^n \hookrightarrow \mathbb{R}^{n+k}$. More generally, any vector $v \in \mathbb{R}^k$ gives a slice $\mathbb{R}^n \times \{v\} \subseteq \mathbb{R}^{n+k}$.
5. The graph of a function $F : M \rightarrow N$ denoted $\text{Graph}(F) = \{x, F(x)\} \subseteq M \times N$ is a submanifold.
6. For a manifold M with boundary, $\partial M \subseteq M$ is a codimension 1 manifold.

Vocabulary 3.9 (all the k -slice business). Let M be a smooth n -manifold, and let (U, φ) be a smooth chart on M . If S is a subset of U such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then we say that S is a k -slice of U . Given a subset $S \subseteq M$ and a nonnegative integer k , we say that S satisfies the *local k -slice condition* if each point of S is contained in the domain of a smooth chart (U, φ) for M such that $S \cap U$ is a single k -slice in U . Any such chart is called a *slice chart* for S in M , and the corresponding coordinates $(x^1, \dots, x^n)$ are called *slice coordinates*.

Corollary (of the rank theorem). Locally any embedded submanifold is a slice:

$$\begin{array}{ccc}
 S & \xhookrightarrow{\text{codim } n \text{ submfd}} & M \\
 \downarrow \pi & & \downarrow \pi \\
 U & & V \\
 \downarrow \varphi & & \downarrow \psi \\
 \varphi(U) \subset \mathbb{R}^n & \xrightarrow{x \mapsto (x, v)} & \psi(V) \subset \mathbb{R}^{n+k}
 \end{array}$$

Any fixed $v \in \mathbb{R}^n$ determines a slice.

Now we should think about the converse...

Theorem 6. Given a subset $S \subseteq M$ for which there exist charts on M : $\{U_\alpha, \varphi_\alpha\}$ such that ① $S \subseteq \bigcup_\alpha U_\alpha \subseteq M$ ② in each chart $(U_\alpha, \varphi_\alpha)$, $S \cap U_\alpha \hookrightarrow U_\alpha$ is a slice, then $S \subseteq M$ is a submanifold.

Remark. Combining the above results, we have that S satisfies the k -slice condition $\iff S$ is embedded.

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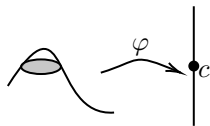
Vocabulary 3.10 (level set). A level set of a smooth map $\varphi : M \rightarrow N$ is $\varphi^{-1}(v) \subseteq M$ for some $c \in N$.

Example. Some examples of level sets

1. Consider $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\varphi(x, y) = xy$. So

$$\varphi^{-1}(c) \begin{cases} \text{smooth manifold } y = \frac{c}{x} & \text{if } c \neq 0 \\ \text{singular } x + y \text{ axis} & \text{if } c = 0 \end{cases}$$

2. Height function



Proposition 10. If $\varphi : M \rightarrow N$ has constant rank r , then for all $c \in N$ the level set $\varphi^{-1}(c)$ is an embedded submanifold of codimension r .

Proof. Write $m = \dim(M)$, $n = \dim(N)$, and $k = m - r$. Let $S := \varphi^{-1}(c) \subseteq M$. From the rank theorem, for each $p \in S$ there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \varphi(p)$ in which φ has a coordinate representation of the form (1), and therefore $S \cap U$ is the slice

$$\{(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \in U : x^1 = \dots = x^r = 0\}.$$

Thus S satisfies the local k -slice condition, so it is an embedded submanifold of dimension k . □

Vocabulary 3.11 (regular value). A regular value of a smooth map $\varphi : M \rightarrow N$ is a point $c \in N$ such that for all $x \in M$ with $\varphi(x) = c$, the differential $d\varphi_x : T_x M \rightarrow T_c N$ is surjective.

Vocabulary 3.12 (critical value). All $c \in N$ that are not regular.

Theorem 7 (regular level set theorem). For $c \in N$ a regular value of a smooth map $\varphi : M \rightarrow N$, the level set $\varphi^{-1}(c) \subseteq M$ is an embedded submanifold.

Proof. The set of points $p \in M$ with $\text{rk}(d\varphi_p) = \dim(N)$ (i.e. full rank) is open by Proposition 6. Let $U \subseteq M$ be that open set. Since c is a regular value, $\varphi^{-1}(c) \in U$. It follows that $\varphi|_U : U \rightarrow N$ is a smooth submersion. By Proposition 10, $\varphi^{-1}(c)$ is an embedded submanifold of U . Since the composition of smooth embeddings $\varphi^{-1}(c) \hookrightarrow U \hookrightarrow M$ is a smooth embedding, $\varphi^{-1}(c)$ is an embedded submanifold of M . $\square$

Example. Before we discuss tangent spaces to submanifolds, note that for some embedding $i : S \hookrightarrow M$, $di : TS \rightarrow TM$ implies $T_p S \subseteq T_p M$ for all $p \in S$. A concrete example is $S^1 \hookrightarrow \mathbb{R}^2$ implies $T_a S^1 \subseteq T_a \mathbb{R}^2$. Now we should generalize this observation into a proposition.

Proposition 11. Let $S \subseteq M$ be an embedded submanifold and let $p \in S$. As a subspace of $T_p M$, the tangent space $T_p S$ is

$$T_p S = \{v \in T_p M : vf = 0 \text{ whenever } f \in C^\infty(M) \text{ and } f|_S = 0\}.$$

Proof. We show both inclusions:

- ($\subseteq$) Suppose $v \in T_p S \subseteq T_p M$. This means that $v = di_p(w)$ for some $w \in T_p S$, where $i : S \hookrightarrow M$. If f is any smooth real-valued function on M that vanishes on S , then $f \circ i \equiv 0$, so $vf = di_p(w)f = w(f \circ i) = 0$.
- ($\supseteq$) Use slice coordinates near p . If $w \in T_p S \subseteq T_p M$, then

$$w \in \text{span} \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^k} \right) \subseteq \text{span} \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right) = T_p M$$

Take $v \in T_p M$, we write $v = v^i \frac{\partial}{\partial x^i}$. Let $vf = 0$ for some f in the statement. To show last $n - k$ coordinates vanish, we need to choose a function that can do that. Let U be some neighborhood of p . Consider $f(x) = \psi_U(x)x^j$ for $x \in U$ and $f(x) = 0$ elsewhere. It is clear that ψ_U is a bump function whose support is in U . Note that $f(x) = \psi_U(x)x^j$ restricts to 0 on $S \cap U$ for $j > k$ because that is how slice charts work. So

$$0 = vf = \sum_i v^i \frac{\partial}{\partial x^i} (x^i \psi_U) = \sum_i v^i \frac{\partial x^i}{\partial x^i} \psi_U = v^j \psi_U$$

This shows $v^j = 0$ for $j > k$ and $v \in \text{span}(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^k}) = T_p S$. $\square$

Vocabulary 3.13 (measure zero). A subset $A \subseteq M$ is measure zero if for any chart (U, φ) , $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has measure zero.

Theorem 8 (Sard's). For $F : M \rightarrow N$ smooth, the set of critical values has measure zero.

Corollary. If $\dim(M) < \dim(N)$, then $F(M) \subseteq N$ is measure zero for any smooth map $F : M \rightarrow N$.

Proof. Every value of $F : M \rightarrow N$ is a critical value (for dimensional reasons). Hence by Sard's $F(M) \subseteq N$ is measure zero. $\square$

Remark. Very false without the smoothness hypothesis. For example, the space filling curves $\mathbb{R} \rightarrow \mathbb{R}^2$ (continuous but not smooth).

Week 4

Sard's Theorem, transversality, and two versions of the Whitney embedding theorem.

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Theorem 9 (Whitney embedding theorem, weakest version). For a compact manifold M , there exists an embedding $M \hookrightarrow \mathbb{R}^N$ for sufficiently large $N \in \mathbb{N}$.

Proof. Use compactness to choose a finite atlas $\{U_\alpha, \varphi_\alpha\}_{\alpha=1}^k$ of M . Choose a partition of unity $\{\psi_\alpha\}_{\alpha=1}^k$ subordinate to this cover, i.e. $\psi_\alpha : M \rightarrow \mathbb{R}$ such that $\text{supp}(\psi_\alpha) \subseteq U_\alpha$ and $\sum_\alpha \psi_\alpha(x) \equiv 1$ for all x . Then define $\varphi : M \rightarrow \mathbb{R}^{k(m+1)}$, where $m = \dim(M)$, by

$$\varphi(x) = \underbrace{(\psi_1(x)\varphi_1(x), \dots, \psi_k(x)\varphi_k(x))}_{\in \mathbb{R}^m}, \underbrace{(\psi_1(x), \dots, \psi_k(x))}_{\in \mathbb{R}^k}$$

In particular, $\psi_j(x)\varphi_j(x) = 0$ if $x \notin U_j$. Now we need to check ① φ is injective and ② φ is an immersion, i.e. $d\varphi$ is injective.

- ① Assume $\varphi(x_1) = \varphi(x_2)$. Last k entries being equal means $\varphi_i(x_1) = \varphi_i(x_2)$ for any i . There exists some j such that $\varphi_j(x_1) = \varphi_j(x_2) \neq 0$. Then $x_1, x_2 \in \text{supp}(\psi_j) \subseteq U_j$. But φ_j is a diffeomorphism on U_j , $\varphi_j(x_1) = \varphi_j(x_2)$ implies $x_1 = x_2$. This shows φ is injective.
- ② Choose $p \in M$. We want to show $d\varphi_p$ is injective (i.e. it has trivial kernel). Let $X \in T_p M$, consider

$$d\varphi_p(X) = (X(\psi_1)\varphi_1(p) + \psi_1(p)d\varphi_1(X), \dots, X(\psi_k)\varphi_k(p) + \psi_k(p)d\varphi_k(X), X(\psi_1), \dots, X(\psi_k))$$

Here we wrote $d\varphi_i$ instead of $X(\varphi_i)$ because φ_i are single variable functions. If $d\varphi_p(X) = 0$, then $X(\psi_j) = 0$ for all j . It follows that $\psi_j(p)d\varphi_j(X) = 0$. Find a j such that $\psi_j(p) \neq 0$ implies $d\varphi_j(X) = 0$. But $d\varphi_j$ is an isomorphism, we must have $X = 0$. This shows φ is an immersion.

The result follows from the above claims. $\square$

Theorem 10 (Whitney embedding theorem, better version). Any compact n -manifold can be embedded into $\mathbb{R}^{2n+1}$ and immersed in $\mathbb{R}^{2n}$.

Proof. Consider the composition $M \xrightarrow{\varphi} \mathbb{R}^N \xrightarrow{P_v} \mathbb{R}^{N-1}$, where $v \in \mathbb{R}^N - \{0\}$. Here φ is an embedding and P_v is the projection to the subspace orthogonal to v . In other words, $\ker(P_v) = \text{span}(v)$. Note that P_v only depends on $[v] \in \mathbb{RP}^{N-1}$. Consider

- ① $\chi : (M \times M) - \Delta_M \rightarrow \mathbb{RP}^{N-1}$ given by $(p_1, p_2) \mapsto [\varphi(p_1) - \varphi(p_2)]$.
- ② $\tau : TM - M \rightarrow \mathbb{RP}^{N-1}$ given by $(p, X) \mapsto [d\varphi_p(X)]$ for some nonzero $X \in T_p M$.

The diagonal $\Delta_M \subseteq M \times M$ is given by (p, p) and TM means the zero section: $(p, X) \in TM$ such that $X \neq 0$. Observe that

- ① $\varphi \circ P_v$ fails to be injective when $[\varphi(p_1) - \varphi(p_2)] = [v]$ since P_v cannot tell differences within $[v]$. If $2n = \dim(M \times M) < N - 1$, then the differential of χ cannot be surjective. It follows that all values are critical. By Sard's theorem, the set of such $[v]$ has measure zero.
- ② $\varphi \circ P_v$ fails to be an immersion if and only if $[d\varphi_p(X)] = [v]$. This is a measure zero condition for $\dim(TM) = 2n < N - 1$. This happens when we choose $[v]$ in the image of τ .

So if we choose $[v]$ outside of the images of χ, τ , the composition $P_v \circ \varphi$ is an embedding. The set of "bad points" is a union of two measure zero sets; hence, it is measure zero as well. Now we can say *almost* every projection P_v results in an embedding for $2n < N - 1$. Finally, we can reduce the dimension of an immersion by restricting τ to vectors of unit length in TM , then

$$SM \subseteq (TM - M) \xrightarrow{\tau} \mathbb{RP}^{N-1}$$

Therefore, $2n - 1 < N - 1$ and $2n < N$. $\square$

Observation. Given an embedding $\varphi : M \hookrightarrow \mathbb{R}^N$, we have $d\varphi : TM \hookrightarrow T\mathbb{R}^N = \mathbb{R}^N \times \mathbb{R}^N = \mathbb{R}^{2N}$.

Vocabulary 4.1 (normal bundle). $NM = \{(x, v) \in \mathbb{R}^N \times \mathbb{R}^N | x \in M \text{ and } v \cdot w = 0, \forall w \in T_x M \subseteq T_x \mathbb{R}^N\}$.

Proposition 12. The normal bundle $NM \hookrightarrow \mathbb{R}^N \times \mathbb{R}^N$ is an embedding.

Proof. Use slice charts. Locally, $\varphi : M \hookrightarrow \mathbb{R}^N$ looks like $\mathbb{R}^k \hookrightarrow \mathbb{R}^k \times \mathbb{R}^{N-k} = \mathbb{N}$. Then

$$N\mathbb{R}^k = \mathbb{R}^k \times \mathbb{R}^{N-k} \hookrightarrow \mathbb{R}^N \times \mathbb{R}^N$$

This idea gives slice charts for $NM \subseteq \mathbb{R}^{2N}$. It is an embedded submanifold. $\square$

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Vocabulary 4.2 (tubular neighborhood). Let $E : NM \rightarrow \mathbb{R}^n$ given by $E(x, v) = x + v$. For $\delta : M \rightarrow \mathbb{R}_{>0}$, take $V_\delta = \{(x, v) \in NM \mid \delta(x) > |v|\}$. A tubular neighborhood of M is of the form V_δ , on which E is diffeomorphic to its image.

Theorem 11 (tubular neighborhood theorem). Let M be an embedded submanifold of $\mathbb{R}^n$. Then there exists a tubular neighborhood of M .

Proof. Observe that $dE_{(x,0)}$ is invertible. By the inverse function theorem, there exists a neighborhood of $M \subseteq NM$, on which E is a local diffeomorphism. For arbitrary V_δ , the map might fail to be injective. But note that we can shrink V_δ if needed. $\square$

Vocabulary 4.3 (intersect transversally). Submanifolds $S, S' \subseteq M$ intersect transversally if for all $p \in S \cap S'$, $\text{span}\{T_p S, T_p S'\} = T_p M$, denoted as $S \pitchfork S'$.

Vocabulary 4.4 (more general transverse). Given $S \subseteq M$ a submanifold and a smooth map $F : N \rightarrow M$. The map F is transverse to S if for all $x \in F^{-1}(S)$, $\text{span}\{T_{F(x)} S, dF_x(T_x N) = \text{im}(dF_x)\} = T_{F(x)} M$. Denote this as $F \pitchfork S$.

Remark. ① For $N = S'$ and $F : S' \rightarrow M$ is an embedding, then $F \pitchfork S$ if and only if $S' \pitchfork S$ (also $F(S') \pitchfork S$). ② If F is a submersion, then $dF_x(T_x N) = T_{F(x)} M$. Thus F is transverse to *any* submanifold $S \subseteq M$.

Theorem 12. Given $F : N \rightarrow M$, $S \subseteq M$ submanifold. If $F \pitchfork S$, then $F^{-1}(S) \subseteq N$ is a submanifold.

Proof sketch. Using slice coordinates, we may work locally on S :

$$\begin{array}{ccccc}
 & T_p N & & N & \\
 & \downarrow & & \downarrow F & \\
 S = \mathbb{R}^k & \xleftarrow{\quad} & \mathbb{R}^k \times \mathbb{R}^{n-k} & \xrightarrow{\sim} & \mathbb{R}^n = M \\
 & \downarrow & & \downarrow \pi & \\
 T_x \mathbb{R}^k & \longrightarrow & T_x \mathbb{R}^k \times T_x \mathbb{R}^{n-k} & & \\
 & & \downarrow d\pi & & \\
 & & T_y \mathbb{R}^{n-k} & & \mathbb{R}^{n-k}
 \end{array}$$

Then $F^{-1}(S) = (\pi \circ F)^{-1}(0)$. This is because S gets mapped to zero coordinates, as we can see in this diagram. Transversality tells us 0 is a regular value. By Theorem 7, $F^{-1}(S)$ is an embedded submanifold. $\square$

Corollary. If $S, S' \subseteq M$ are submanifolds and $S \pitchfork S'$, then $S \cap S' \subseteq M$ is a submanifold.

Theorem 13. For $S \subseteq M$ submanifold and $F : N \times X \rightarrow M$ transverse to S , $F|_{N \times \{x\}} : N \rightarrow M$ is transverse to S for almost all $x \in X$. Almost all means the complement of a measure zero set.

Vocabulary 4.5 (smooth homotopy). Two maps $f, g : N \rightarrow M$ are smoothly homotopic if there is a smooth map $F : N \times [0, 1] \rightarrow M$ such that $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in N$.

Corollary. Given a map $f : N \rightarrow M$ and a submanifold $S \subseteq M$, f is homotopic to a map g with $g \pitchfork S$.

Vocabulary 4.6 (isotopic). Smooth embeddings $f, g : N \hookrightarrow M$ are isotopic if there exists a smooth map $F : N \times [0, 1] \rightarrow M$ such that

1. $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in N$.
2. $F(-, t)$ is an embedding for all $t \in [0, 1]$.

Theorem 14 (Whitney approximation theorem). Suppose N is a smooth manifold with or without boundary, M is a smooth manifold (without boundary), and $F : N \rightarrow M$ is a continuous map. Then F is homotopic to a smooth map. If F is already smooth on a closed subset $A \subseteq N$, then the homotopy can be taken to be relative to A .

Lemma. If N and M are smooth manifolds with or without boundary, smooth homotopy is an equivalence relation on the set of all smooth maps from N to M .

Week 5

Definition of Lie group, examples of Lie groups, Lie group homomorphisms, Lie group actions, equivariant maps, representations of Lie groups, and semi-direct products of Lie groups

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Vocabulary 5.1 (Lie group). A Lie group G is a manifold that is also a group such that multiplication and inversion are smooth. Multiplication is a map $G \times G \rightarrow G$ given by $(g, h) \mapsto g \cdot h$. A homomorphism of Lie groups is a group homomorphism $G \rightarrow H$ that is also a smooth map.

Example. Here are some examples of Lie groups

1. $GL_n(\mathbb{R}) \subseteq \mathbb{R}^{n^2}$, $GL_n(\mathbb{C}) \subseteq \mathbb{C}^{n^2}$. They are all open.
2. $S^1 : e^{i\alpha} \cdot e^{i\beta} = e^{i\alpha+\beta}$ has a homomorphism $\mathbb{R} \rightarrow S^1$ given by $\alpha \mapsto e^{i\alpha}$.
3. Any countable group equipped with discrete topology can be viewed as a 0-dimensional manifold.

Vocabulary 5.2 (Lie subgroup). A subgroup H of a Lie group G is an embedded submanifold $H \hookrightarrow G$.

Example. Some interesting Lie subgroups

1. $GL_n^+(\mathbb{R}) \leq GL_n(\mathbb{R})$ are matrices with positive determinants.
2. For any $n \in \mathbb{N}$, $\mathbb{Z}/n\mathbb{Z} \hookrightarrow S^1$ given by $[k] \mapsto e^{2\pi i \frac{k}{n}}$ makes it a subgroup of S^1 .
3. $T : GL_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$ given by $A \mapsto A^T A$ is smooth. We claim that $I_n \in M_n(\mathbb{R})$ is a regular value. It follows that $T^{-1}(I_n) \subseteq GL_n(\mathbb{R})$ is an embedded submanifold. But $T^{-1}(I_n)$ is the orthogonal group $O(n)$. So $O(n) \subseteq GL_n(\mathbb{R})$ is a Lie subgroup. Similarly, we can conclude $U(n) \leq GL_n(\mathbb{R})$ is a Lie subgroup.

Vocabulary 5.3 (translation). On a Lie group G , left and right translation maps $L_g : G \rightarrow G$ and $R_g : G \rightarrow G$ are given by $L_g(h) = gh$ and $R_g(h) = hg$. They are diffeomorphisms: $(L_g)^{-1} = L_{g^{-1}}$, $(R_g)^{-1} = R_{g^{-1}}$. They are not homomorphisms unless $g = e$.

Vocabulary 5.4 (conjugation). Let G be a Lie group. The conjugation is defined to be $C_g : G \rightarrow G$ with $C_g(n) = gng^{-1}$. This gives an isomorphism of Lie groups for each $g \in G$ with $(C_g)^{-1} = C_{g^{-1}}$. Recall that a subgroup is *normal* if $C_g(H) = H$ for all $g \in G$.

Theorem 15. Every Lie group homomorphism $F : G \rightarrow H$ has constant rank.

Proof. Our plan is to show the rank of dF_g is the same as dF_e . For all $g \in G$, $F(L_g h) = F(gh) = F(g)F(h) = L_{F(g)}F(h) = (L_{F(g)} \circ F)(h)$. Now we have shown $F \circ L_g = L_{F(g)} \circ F$. Consider

$$\begin{array}{ccc} T_e G & \xrightarrow{dF_e} & T_e H \\ \simeq \downarrow (dL_g)_e & & \simeq \downarrow (dL_{F(g)})_e \\ T_g G & \xrightarrow{dF_g} & T_{F(g)} H \end{array}$$

The diagram commutes by taking the differential of $F \circ L_g = L_{F(g)} \circ F$. Since $(dL_g)_e$ and $(dL_{F(g)})_e$ are invertible, $\text{rk}(dF_e) = \text{rk}(dF_g)$ for all g . $\square$

Corollary. The kernel of a homomorphism of Lie groups $F : G \rightarrow H$ is an embedded Lie subgroup $\ker(F) \triangleleft G$ that is normal.

Proof. We only need to show this is a submanifold since other stuff follows from group theory. Note that $\ker(F) = F^{-1}(e)$. Since F is constant rank, $F^{-1}(e) \subseteq G$ is a submanifold. $\square$

Example (det homomorphism). Recall that $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^\times = \mathbb{R} - \{0\}$, where $\mathbb{R}^\times$ is a multiplicative Lie group. The kernel $\ker(\det) = SL_n(\mathbb{R}) \triangleleft GL_n(\mathbb{R})$. Conclude that the special linear group is normal in $GL_n(\mathbb{R})$. Similarly, $SO(n) \triangleleft O(n)$ and $SU(n) \triangleleft U(n)$.

Warning. The image of a Lie group homomorphism can be nasty. For example, $\mathbb{R} \hookrightarrow T^2 = \mathbb{R}^2/\mathbb{Z}^2$ given by

$$\mathbb{R} \rightarrow \mathbb{R}^2 \rightarrow T^2 \quad x \mapsto (x, \sqrt{2}x) \mapsto (e^{2\pi x}, e^{2\pi\sqrt{2}x})$$

These maps curve with irrational slopes. For a closed curve $(x, \lambda x)$, there exists some x such that $(x, \lambda x) = (m, n) \in \mathbb{Z}^2$. The image of this homomorphism is dense.

Vocabulary 5.5 (group action). A Lie group G acts on a manifold M on the left by a smooth map

$$\vartheta : G \times M \rightarrow M \quad (g, p) \mapsto g \cdot p = \vartheta_g(p) = \vartheta(g, p)$$

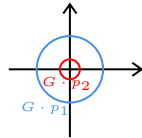
such that ① $g_1 \cdot (g_2 \cdot p) = (g_1 g_2) \cdot p$ and ② $e \cdot p = p$.

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Vocabulary 5.6 (orbit and stabilizer). The orbit of a point is $G \cdot p = \{x \in M | g \cdot p = x\} \subseteq M$. The stabilizer is $\text{stab}(p) = \{g \in G | g \cdot p = p\} \leq G$.

Example. Here are some examples of group actions:

1. $S^1 \xrightarrow{z^n} S^1 \curvearrowright \mathbb{R}^2$ by rotation. Every $e^{i\vartheta} \in S^1$ corresponds to an action $\begin{bmatrix} \cos(n\vartheta) & -\sin(n\vartheta) \\ \sin(n\vartheta) & \cos(n\vartheta) \end{bmatrix} \curvearrowright \mathbb{R}^2$. For $p_1, p_2 \in \mathbb{R}^2$, their orbits can be visualized as



Note that $\text{stab}(p) = \{e\}$ if $p \neq (0,0)$ and $\text{stab}((0,0)) = S^1$.

2. $S^1 \curvearrowright S^2$. This can be interpreted as rotations in the xy -plane. Concretely,

$$S^1 \simeq SO(2) = \begin{bmatrix} \cos(\vartheta) & -\sin(\vartheta) & 0 \\ \sin(\vartheta) & \cos(\vartheta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \leq O(3) \curvearrowright S^2$$

3. More general examples include $GL_n(\mathbb{R}) \curvearrowright \mathbb{R}^n$ and $O(n) \curvearrowright S^{n-1}$.

Vocabulary 5.7 (transitive and free action). The action is transitive if $G \cdot p = M$ and any $p, q \in M$, there exists some $g \in G$ such that $g \cdot p = q$. The action is free if $\text{stab}(p) = \{e\}$ for all p .

Vocabulary 5.8 (equivariant map). A smooth manifold with smooth action by a Lie group G is called a G -manifold. Given G -manifolds M and N , a smooth map $F : M \rightarrow N$ is equivariant if $F(\vartheta^M(g \cdot p)) = \vartheta^N(g, F(p))$. Officially, $F(g \cdot p) = g \cdot F(p)$.

$$\begin{array}{ccc} G \times M & \xrightarrow{\text{id}_G \times F} & G \times N \\ \downarrow \vartheta^M & & \downarrow \vartheta^N \\ M & \xrightarrow{F} & N \end{array}$$

Lemma. Suppose $F : M \rightarrow N$ is G -equivariant and M is a transitive G -manifold, then F has constant rank.

Proof. We use the same argument that is in the proof of the Lie group homomorphism theorem³. $\vartheta_g(p) = q$ has solution $g \in G$ for all $p, q \in M$. That is, for all $p, q \in M$ there exists g such that $g \cdot p = q$ due to transitivity. Then

$$\begin{array}{ccc} T_p M & \xrightarrow{dF_p} & T_{F(p)} N \\ \simeq \downarrow (d\vartheta_g^M)_p & & \simeq \downarrow (d\vartheta_g^N)_{F(p)} \\ T_q M & \xrightarrow{dF_q} & T_{F(q)} N \end{array}$$

$(d\vartheta_g^N)_{F(p)}$ exists because F is equivariant. Since $(d\vartheta_g^N)_{F(p)}$ and $(d\vartheta_g^M)_p$ are isomorphisms, $\text{rk}(dF_p) = \text{rk}(dF_q)$. $\square$

³Every Lie group homomorphism has constant rank.

Proposition 13. For a group action $\vartheta : G \times M \rightarrow M$ and any $p \in M$, $\text{stab}(p) \leq G$ is an embedded Lie subgroup. If $\text{stab}(p) = \{e\}$, then $G \cdot p \subseteq M$ is an immersed submanifold.

Proof. Consider $\vartheta(-, p) : G \rightarrow M$ for a fixed $p \in M$. We claim that $\vartheta(-, p)$ is G -equivariant for the left action on itself. Further, $G \curvearrowright G$ being transitive implies $\vartheta(-, p)$ has constant rank. Hence, $\text{stab}(p) \subseteq G$ is an embedded submanifold. If $\text{stab}(p) = \{e\}$, then $\vartheta(-, p)$ is an immersion by slice chart arguments. $\square$

Vocabulary 5.9 (semidirect product). Let H, N be Lie groups. Define an action $\vartheta : H \times N \rightarrow N$ through Lie group homomorphisms ($\vartheta_h : N \rightarrow N$). This makes $N \rtimes_{\vartheta} H$ a Lie group with underlying manifold $N \times H$ and multiplication $(n, h) \cdot (n', h') = (n, \vartheta_h(n'), hh')$.

Remark. Some facts about semidirect products: ① $N \triangleleft N \rtimes_{\vartheta} H$ via $N \hookrightarrow N \rtimes_{\vartheta} H$ given by $n \mapsto (n, e_H)$. ② $H \leq N \rtimes_{\vartheta} H$. Normal if and only if ϑ is a transitive action. ③ If $N \triangleleft G$ and $H \leq G$, then $NH = G$. ④ $N \cap H = \{e\}$ implies $G \simeq N \rtimes H$.

Example. Act $O(n)$ on $\mathbb{R}^n$, where $\mathbb{R}^n$ is a multiplicative group. The semidirect product $\mathbb{R}^n \rtimes O(n)$ is called the Euclidean group. Multiplication is defined as $(v, A) \cdot (w, B) = (v + Aw, AB)$. Then we get an action $\mathbb{R}^n \rtimes O(n) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $(v, A) \cdot x = v + Ax$. The Euclidean group preserves the Euclidean distance between any two points. In other words, for any $x, y \in \mathbb{R}^n$ and $g \in \mathbb{R}^n \rtimes O(n)$, $|x - y| = |g \cdot x - g \cdot y|$.

Vocabulary 5.10 (representation). A representation of a Lie group G is a smooth homomorphism $\rho : G \rightarrow GL(V)$, where V is a vector space.

Vocabulary 5.11 (faithful). A representation is faithful if it is injective.

Example. Some examples of Lie group representations:

1. Any subgroup of $GL(V)$ has a “standard” representation such as $O(n) \hookrightarrow GL_n(\mathbb{R})$, $U(n) \hookrightarrow GL_n(\mathbb{C})$, etc.
2. A finite group G has a faithful representation with $V = \mathbb{C}^{|G|} = \text{span}\{e_g | g \in G\}$, where $\rho(g) \cdot e_h = e_{gh}$.

Vocabulary 5.12 (vector field). A smooth vector field on a manifold M is a smooth map $X : M \rightarrow TM$ such that $\pi \circ X = \text{id}_M$:

$$TM \begin{array}{c} \nwarrow X \searrow \\ \xrightarrow{\pi} \end{array} M$$

Written in coordinate basis for any $p \in U$:

$$X_p = X^i(p) \frac{\partial}{\partial x^i} \Big|_p \in T_p M$$

Remark. Two facts about vector fields:

1. Smoothness of vector fields can be checked in charts of M , where

$$X|_U = X^i \frac{\partial}{\partial x^i}$$

Note that $X^i \in C^\infty(U)$ if and only if X is smooth on U .

2. Let $\mathfrak{X}(M)$ be the set of all vector fields on M . This is a module over the ring $C^\infty(M)$ of smooth functions: $X, Y \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$ implies $fX + gY \in \mathfrak{X}(M)$. More concretely, $(fX + gY)_p = f(p)X_p + g(p)Y_p$.

Week 6

Local and global frames, F -related vector fields, Lie algebras, properties of Lie brackets, left-invariant vector fields, $\text{Lie}(G) \simeq T_e G$, examples of Lie brackets.

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Vocabulary 6.1 (global derivation). For each $X \in \mathfrak{X}(M)$, we get $X(fg) = (Xf)g + fXg$ for $f, g \in C^\infty(M)$.

Proposition 14. The following conditions are equivalent:

1. A vector field X is smooth (i.e. a smooth function $X : M \rightarrow TM$ and $\pi \circ X = \text{id}_M$).

2. For all $f \in C^\infty(M)$, Xf is smooth. Recall that $(Xf)_p = (X_p f)$.
3. For all $f \in C^\infty(U)$, $X|_U f \in C^\infty(U)$. This is essentially (2) but more local.

Proof. It suffices to work locally. Then $X|_U = X^j \frac{\partial}{\partial x^j}$ is smooth if and only if $X^j \in C^\infty(M)$ is smooth. (1 $\Rightarrow$ 2.): applying this to $x \in U$,

$$Xf(x) = \left(X^i(x) \frac{\partial}{\partial x^i} \Big|_x \right) f = X^i(x) \frac{\partial f}{\partial x^i}(x)$$

X is smooth implies $X|_U$ is smooth. Therefore, each component function $X^i|_U$ is smooth. It follows that Xf is smooth. (2 $\Rightarrow$ 3): take coordinate function x^j on U . Then for a bump function on U , $\psi x^j \in C^\infty(M)$ and $X(\psi x^j)$ smooth. Note that near the center of the chart U , $X(\psi x^j)|_U = X|_U(x^j) = X^j \in C^\infty(U)$. (3 $\Rightarrow$ 1) suppose Xf is smooth whenever f is smooth on an open subset of M . If x^i are any smooth local coordinates on $U \subseteq M$, $x^i \in C^\infty(U)$. Applying X to one of these functions,

$$X(x^i) = X^j \frac{\partial}{\partial x^j}(x^i) = X^i$$

Since $X(x^i)$ is smooth by assumption, all component functions of X are smooth. $\square$

Proposition 15. The set of derivations $D : C^\infty(M) \rightarrow C^\infty(M)$ is isomorphic to $\mathfrak{X}(M)$ as vector spaces⁴.

Proof. We already have a map from vector fields to derivations given by $(Xf)(p) = X_p f \in \mathbb{R}$. Conversely, given a derivation $D : C^\infty(M) \rightarrow C^\infty(M)$, we need to produce a vector field X such that $Df = Xf$. Construct a derivation at any $p \in M$ by

$$\begin{array}{ccc} & \xrightarrow{X_p} & \\ C^\infty(M) & \xrightarrow{D} C^\infty(M) & \xrightarrow{\text{ev}_p} \mathbb{R} \end{array}$$

We want to check $X_p f = (Df)(p)$. This expression is linear on f because D is linear. It satisfies the product rule because D is a derivation. And lastly, $Xf = Df$ is smooth if f is smooth. Proposition 15 gives that this vector field is smooth. $\square$

Vocabulary 6.2 (local frame and global frame). Given vector fields $X_1, \dots, X_k$ on M , these are linearly independent if for all $p \in M$, $\{(X_i)_p\}_{i=1}^k \subseteq T_p M$ are linearly independent. In this case, $(X_1, \dots, X_k)$ form a local frame. Moreover, if they form a basis of $T_p M$, then $(X_1, \dots, X_k)$ forms a global frame ($k = \dim(M)$).

Vocabulary 6.3 (parallelizable). A manifold M is parallelizable if it admits a global frame.

Remark. Most manifolds are not parallelizable as they do not have global frames. But ① all Lie groups are parallelizable, and ② most spheres are not except for S^1, S^3, S^7 .

Observation. Local frames always exist. In other words, there exists $U \subseteq M$ such that $TM|_U = TU$ admits a global frame.

Question. Consider $F : M \rightarrow N$ with $dF_p : T_p M \rightarrow T_{F(p)} N$ for all $p \in M$. When does $dF_p(X_p) \in T_{F(p)} N$ define a vector field on N ?

Answer. Rarely. It defines a vector field in some special cases.

Vocabulary 6.4 (F -related). Let $F : M \rightarrow N$, $X \in \mathfrak{X}(M)$, and $Y \in \mathfrak{X}(N)$. X, Y are F -related if $dF_p(X_p) = Y_{F(p)}$ for all $p \in M$. This happens if ① F is a diffeomorphism or ② F is an inclusion from a submanifold.

Remark. Suppose we have maps $M \xrightarrow{F} N \xrightarrow{f} \mathbb{R}$. An equivalent definition of being F -related is that for all $f \in C^\infty(N)$, $X(f \circ F) = (Yf \circ F)$.

Proposition 16 (① F is a diffeomorphism). If $F : M \rightarrow N$ is a diffeomorphism, then for all $X \in \mathfrak{X}(M)$, there exists a unique $Y \in \mathfrak{X}(N)$ such that X is F -related to Y .

⁴Even better: they are isomorphic as $C^\infty(M)$ -modules.

Proof. For Y to be F -related, it has to satisfy $dF_p(X_p) = Y_{F(p)}$ for every $p \in M$. Given a diffeomorphism F , we define Y to be $Y_q = dF_{F^{-1}(q)}(X_{F^{-1}(q)})$. Clearly Y is the unique (rough) vector field that is F -related to X .

Observe that $Y : N \rightarrow TN$ is the composition of the following smooth maps $N \xrightarrow{F^{-1}} M \xrightarrow{X} TM \xrightarrow{dF} TN$ given by

$$q \xrightarrow{F^{-1}} F^{-1}(q) \xrightarrow{X} X^i(F^{-1}(q)) \frac{\partial}{\partial x^i} \Big|_{F^{-1}(q)} \xrightarrow{dF} dF_{F^{-1}(q)}(X_{F^{-1}(q)})$$

It follows that Y is smooth. $\square$

Vocabulary 6.5 (pushforward). We write $F_*X = Y$ for this vector field $Y \in \mathfrak{X}(N)$ appeared in Proposition 17. F_*X is the pushforward of X along the diffeomorphism F .

Proposition 17 (② F is an inclusion from a submanifold). For an embedding $i : S \hookrightarrow M$, a vector field $Y \in \mathfrak{X}(M)$ is i -related to a vector field $X \in \mathfrak{X}(S)$ if and only if for all $p \in S$, $Y_p \in T_p S \subseteq T_p M$.

Remark. Given $X, Y \in \mathfrak{X}(M)$, we get maps $X, Y : C^\infty(M) \rightarrow C^\infty(M)$. The composition $X \circ Y$ is rarely a derivation. Also note that we are able to treat elements of $\mathfrak{X}(M)$ as derivations by Proposition 16. For example, $\frac{\partial}{\partial x} \circ \frac{\partial}{\partial y}$ is not a derivation.

Vocabulary 6.6 (Lie bracket). The Lie bracket of $X, Y \in \mathfrak{X}(M)$ is $[X, Y] = XY - YX$. Both XY and YX are compositions of derivations. Its coordinate formula looks like

$$[X, Y] = (XY^j - YX^j) \frac{\partial}{\partial x^j}.$$

Proposition 18. Lie bracket is a well-defined operation in $\mathfrak{X}(M)$. Equivalently, the formula for $[X, Y] : C^\infty(M) \rightarrow C^\infty(M)$ is a derivation.

Proof. This will require some calculation and cancellations. Here we go:

$$\begin{aligned} ([X, Y])(fg) &= (XY - YX)(fg) = X(Y(fg)) - Y(X(fg)) = X((Yf)g + f(Yg)) - Y((Xf)g + f(Xg)) \\ &= (X(Yf))g + (Yf)(Xg) + (Xg)(Yg) + f(XYg) - ((Y(Xf))g + (Xf)Yg + (Yf)(Xg) + fY(Xg)) \\ &= (XYf)g + f(XYg) - (YXf)g - f(YXg) = ([X, Y]f)g + f([X, Y]g). \end{aligned} \quad \square$$

Vocabulary 6.7 (Lie algebra). A Lie algebra is a vector space V with an assignment $[-, -] : V \times V \rightarrow V$ with the properties that

1. It is bilinear: $a, b \in \mathbb{R}$ and $X, Y, Z \in V$, $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ and $[Z, aX + bY] = a[Z, X] + b[Z, Y]$.
2. It is skew: $[X, Y] = -[Y, X]$ for all $X, Y \in V$.
3. It satisfies the Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Theorem 16. $\mathfrak{X}(M)$ with a Lie bracket is a Lie algebra.

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Our main example for Lie algebra comes from any smooth manifold M . Consider $V = \mathfrak{X}(M)$ with Lie bracket $[X, Y] = XY - YX$. Next, we will think about the Lie algebra of a Lie group.

Proposition 19. The Lie bracket has the following additional properties:

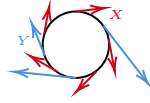
1. $[X, fY] = f[X, Y] + (Xf)Y$.
2. For any smooth map $F : M \rightarrow N$ and F -related vector fields $X_1, X_2 \in \mathfrak{X}(M)$, $Y_1, Y_2 \in \mathfrak{X}(N)$, we have $[X_1, X_2]$ is F -related to $[Y_1, Y_2]$ if X_1 is F -related to Y_1 and X_2 is F -related to Y_2 .

Proof. (1) is given by a direct computation from the definition of the Lie bracket. For (2), X, Y are F -related if and only if $X(f \circ F) = Yf \circ F$ for all $f \in C^\infty(N)$ by the alternative definition of being F -related. In this case, $X_1(X_2(f \circ F)) = X_1(Y_2f \circ F) = (Y_1Y_2f) \circ F \Rightarrow [X_1, X_2](f \circ F) = ([X_1, X_2]f) \circ F$ if and only if $[X_1, X_2]$ is F -related to $[Y_1, Y_2]$. $\square$

Vocabulary 6.8 (left-invariant and left translation). Let G be a Lie group. $X \in \mathfrak{X}(G)$ is left-invariant if for all $g \in G$, $X = (L_g)_*X \in \mathfrak{X}(G)$. Left translation $L_g : G \rightarrow G$ is given by $L_g(h) = gh$. In more detail,

$$(dL_g)_h : T_h G \rightarrow T_{gh} G \quad (dL_g)_h(X_h) = X_{gh} \quad (\odot)$$

Sometimes we write $()^L$ for left-invariant vector fields.

Figure 1: Picture of two vector fields X and Y on S^1 .

Example. Let $G = S^1 = U(1) = SO(2)$ be a Lie group. Suppose $X, Y \in \mathfrak{X}(G)$, it can be easily observed that X is left-invariant while Y is not left-invariant. The collection of left-invariant vector fields on G is denoted $\text{Lie}(G)$ or $\mathfrak{g}$. But we will stick to the former.

Theorem 17. The evaluation map $\Phi : \text{Lie}(G) \rightarrow T_e G$ given by $X \mapsto X_e$ is an isomorphism of vector spaces. Thus, $\text{Lie}(G)$ is finite-dimensional, with dimension equal to $\dim(G)$.

Proof. The map is linear because $\Phi(X + Y) = (X + Y)_e = X_e + Y_e$ and it also works with scalars. Then we show Φ is injective. Let $\Phi(X) = 0$ and $X \in \text{Lie}(G)$. For all $g \in G$, $X_g = d(L_g)_e(X_e) = 0$ because X is left-invariant. $X = 0$ because it evaluates to 0 at every g . Next, we need to show Φ is surjective. Given any $v \in T_e G$, define a rough vector field

$$v^L|_g := d(L_g)_e(v)$$

If we can find some left-invariant $X \in \text{Lie}(G)$ such that $X_e = v$, then X is given by the formula above by iterating over all $g \in G$. Next, we want to show v^L is a smooth vector field. By Proposition 15, we only need to show $v^L f$ is smooth whenever $f \in C^\infty(G)$. Choose a smooth curve $\gamma : (-\delta, \delta) \rightarrow G$ such that $\gamma(0) = e$ and $\gamma'(0) = v$. Then for all g ,

$$(v^L f)(g) = v^L|_g f = d(L_g)_e(v)f = v(f \circ L_g) = \gamma'(0)(f \circ L_g) = \frac{d}{dt} \Big|_{t=0} (f \circ L_g \circ \gamma)(t).$$

We see that v^L is the derivative of a map consists of three smooth maps. As a result, v^L is smooth. Now we verify v^L is left-invariant. In other words, we want to show $d(L_h)_g(v^L|_g) = v^L|_{hg}$ for all h, g . We have

$$d(L_h)_g(v^L|_g) = d(L_h)_g \circ d(L_g)_e(v) = d(L_h \circ L_g)_e(v) = v^L|_{hg}.$$

These imply $\Phi(v^L) = v$ hence Φ is indeed surjective. In particular, this shows that $\text{Lie}(G)$ is finite dimensional with $\dim(\text{Lie}(G)) = \dim(G)$. $\square$

Theorem 18. Every Lie group is parallelizable, meaning that there exists a global frame $X^1, \dots, X^{\dim(G)} \in \mathfrak{X}(G)$. Note that this collection spans $T_g G$ for all $g \in G$.

Proof. By our previous observation, local frames always exist. In particular, we can choose a local frame $\{X^i\}$ such that $\{X_e^1, \dots, X_e^{\dim(G)}\}$ forms a basis of $T_e G$. But Theorem 16 tells us $T_e G \simeq \text{Lie}(G)$. As a result, we can left translate this basis by applying $(dL_g)_e$ for every g . This gives $X_g^i = (dL_g)_e X_e^i$ which is a local frame for G at g . Since g is arbitrary, $\{X^i\}$ becomes a global frame. $\square$

Theorem 19. A Lie group homomorphism $F : G \rightarrow H$ determines a map $F_* : \text{Lie}(G) \rightarrow \text{Lie}(H)$. In other words, left-invariant vector fields on G are F -related to left-invariant vector fields on H .

Proof. If $Y \in \text{Lie}(H)$ is F -related to X , $Y_e = dF_e(X_e)$. Since it is left-invariant, it comes with the form $Y = (dF_e(X_e))^L$. Since F is a homomorphism, $F(gh) = F(g)F(h)$. Recall the definition for left/right translation. It follows that $F(L_g h) = L_{F(g)} F(h)$ and $F \circ L_g = L_{F(g)} \circ F$. Differentiating both sides, we get $dF \circ d(L_g) = d(L_{F(g)}) \circ dF$. At $e \in G$, the square on the right commutes. Then $dF_e(X_g) = dF_e(d(L_{F(g)})_e(X_e)) = d(L_{F(g)})_e(dF_e(X_e)) = d(L_{F(g)})_e(Y_e) = Y_{F(g)}$. This proves X and Y are F -related. We will show properties of the Lie bracket later. $\square$

Proposition 20. The bracket of left-invariant vector field is left-invariant.

Proof. Recall that $(L_g)_* X \stackrel{\circ}{=} X$ for all $g \in G$. So $(L_g)_*[X_1, X_2] = [(L_g)_*X_1, (L_g)_*X_2] \stackrel{\circ}{=} [X_1, X_2]$. $\square$

Theorem 20. There is a functor Lie from Lie groups and Lie group homomorphisms to Lie algebras and Lie algebra homomorphisms:

$$G \mapsto \text{Lie}(G) \quad (F : G \rightarrow H) \mapsto (F_* : \text{Lie}(G) \rightarrow \text{Lie}(H)) \Rightarrow F_*[X, Y] = [F_*X, F_*Y]$$

where $\text{Lie}(G)$ is the collection of left-invariant vector fields under Lie bracket. Let id_* be the identity in $\text{Lie}(G)$. As a consequence of being a functor, $(F_1 \circ F_2)_* = (F_1)_* \circ (F_2)_*$ for $F_1 : G_1 \rightarrow G_2$ and $F_2 : G_2 \rightarrow G_3$.

Remark. A Lie group is *abelian* if $[-, -] = 0$. Intuitively, treat $[-, -]$ as some form of “commutator”. For any abelian Lie group G , $[-, -] : \text{Lie}(G) \times \text{Lie}(G) \rightarrow \text{Lie}(G)$ is the zero map.

Theorem 21. A Lie subgroup $i : H \hookrightarrow G$ determines a Lie subalgebra $A \subseteq \text{Lie}(G)$ with $A = i_* \text{Lie}(H) = \{X \in \text{Lie}(G) \mid X_e \in T_e H \subseteq T_e G\}$.

Proof. Clearly $T_e H \subseteq T_e G$. The square

$$\begin{array}{ccc} T_e H & \hookrightarrow & T_e G \\ \downarrow (dL_h)_e & & \downarrow (dL_h)_e \\ T_h H & \hookrightarrow & T_h G \end{array}$$

commutes. So left-invariant vector fields are compatible under inclusion. Therefore, $\text{Lie}(H) \subseteq \text{Lie}(G)$ compatible with the Lie bracket. $\square$

Example. Here are some examples of Lie brackets.

1. For S^1 , $\text{Lie}(S^1) \simeq T_e S^1 \simeq \mathbb{R}$. The isomorphism is given by $a \frac{\partial}{\partial \vartheta} \mapsto a$. Consult the vector field X in Figure 1 for a visualization.

$$\left[a \frac{\partial}{\partial \vartheta}, b \frac{\partial}{\partial \vartheta} \right] = \left(a \frac{\partial}{\partial \vartheta} \right) \left(b \frac{\partial}{\partial \vartheta} \right) - \left(b \frac{\partial}{\partial \vartheta} \right) \left(a \frac{\partial}{\partial \vartheta} \right) = ab \frac{\partial}{\partial \vartheta} \frac{\partial}{\partial \vartheta} - ba \frac{\partial}{\partial \vartheta} \frac{\partial}{\partial \vartheta} = 0$$

So $[-, -] : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is the zero map. By the remark above, S^1 is abelian.

2. Let $G = GL_n(\mathbb{R})$ as an open set of $M_{n \times n}(\mathbb{R}) = \mathbb{R}^{n^2}$. We then have $\text{Lie}(G) = T_e G = M_{n \times n}(\mathbb{R})$ with Lie bracket given by $[A, B] = AB - BA$ for some $A, B \in T_e GL_n(\mathbb{R}) \simeq M_{n \times n}(\mathbb{R})$.
3. Let $G = O(n)$. Recall that we showed $O(n)$ was a smooth manifold via regular value: $GL_n(\mathbb{R}) \xrightarrow{\Phi} M_{n \times n}(\mathbb{R}) \subseteq \mathbb{R}^{n^2}$, $A \mapsto A^T A$. So $O(n) = \Phi^{-1}(\text{id}_n) \subseteq GL_n(\mathbb{R})$. This also gives $T_{I_n} O(n) = \ker(d\Phi_{I_n})$. Compute $d\Phi_{\text{id}}(B) = B^T + B$. Hence $\text{Lie}(O(n)) = \{B \in M_{n \times n}(\mathbb{R}) \mid B^T = -B\}$ is the set of skew symmetric $n \times n$ matrices. The bracket on $\text{Lie}(O(n))$ is the commutator $AB - BA$ for some $A, B \in \text{Lie}(O(n))$ by Theorem 20.

Week 7

Midterm.

Week 8

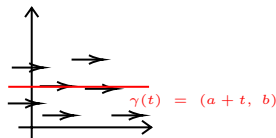
Definitions and examples of integral curves and flows. Conditions for existence and properties of flows. Relationships between vector fields and flows. Techniques such as attaching manifolds, connected sums, and doubles. Applications of the collar neighborhood theorem. Flow naturality and completeness of vector fields.

10/15

Vocabulary 8.1 (integral curve). For $V \in \mathfrak{X}(M)$, a curve $\gamma : I \rightarrow M$ is an integral curve for V if $\frac{d}{dt} \big|_t \gamma(t) = \gamma'(t) = V_{\gamma(t)}$. $\gamma'(t) \in T_{\gamma(t)} M$ as taking the derivative on a point of a curve gives us a tangent vector.

Example. Some integral curves

1. We start with an easy one. If $V_p = 0$ at $p \in M$, then $\gamma : \mathbb{R} \rightarrow M$ such that $\gamma(t) = p$ for all $t \in \mathbb{R}$ is an integral curve. Note that $V_p = 0$ is a singular point of V .
2. Consider $1 \cdot \frac{\partial}{\partial x} + 0 \cdot \frac{\partial}{\partial y}$ on $\mathbb{R}^2$,



3. Consider the vector field $-y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$ on $\mathbb{R}^2$. It is outlined in figure 1. The integral curve is the circle.

Remark. In local coordinates, the existence and uniqueness of an integral curve is an initial value problem. We want $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^n$ such that $\gamma(0) = p \in \mathbb{R}^n$ and $\gamma'(0) = V_{\gamma(0)}$. More generally, $\gamma'(t) = V_{\gamma(t)}$. In coordinates, $(\gamma')^i(t) = V^i(\gamma^1(t), \dots, \gamma^n(t))$. Such initial value problems of n equations have unique solutions for some $\epsilon > 0$. This gives the existence and uniqueness of integral curves for $V \in \mathfrak{X}(M)$, but note that the domain of a curve $\gamma : I \rightarrow M$ might be small. For example, $I \neq \mathbb{R}$ but $I = (-\epsilon, \epsilon) \subsetneq \mathbb{R}$. Intuitively, we are tracing arrows.

Vocabulary 8.2 (flow). A global flow on M is an action $\vartheta : \mathbb{R} \times M \rightarrow M$. Here we consider $\mathbb{R}$ as a Lie group under addition. So for each t , $\vartheta_t : M \rightarrow M$. $\vartheta_t = \vartheta(t, -)$ is a diffeomorphism and $\vartheta_t \circ \vartheta_s = \vartheta_{t+s}$. As expected, $\vartheta_0 = \text{id}_M$.

Remark. Given a flow $\vartheta : \mathbb{R} \times M \rightarrow M$, we obtain curves $\gamma_p(t) = \vartheta(t, p)$, $\gamma_p : \mathbb{R} \rightarrow M$, and $\gamma_p(0) = p$. For a fixed p , $\gamma_p(t)$ represents the orbit of p under the $\mathbb{R}$ -action. We get a vector field $V \in \mathfrak{X}(M)$ from a flow via

$$V_p = \left. \frac{d}{dt} \right|_{t=0} (\vartheta_t(p)) = \left. \frac{d}{dt} \right|_{t=0} (\vartheta(t, p))$$

V defined as such is smooth and curves γ_p are integral curves of V . We have a way of passing from (flows) $\Rightarrow$ (vector fields + integral curves). It is reasonable to ask: When does a vector field come from a flow?

Vocabulary 8.3 (flow domain). A flow domain is an open submanifold $\mathcal{D} \subseteq \mathbb{R} \times M$ such that for all $p \in M$,

$$\mathcal{D}^{(p)} := \{t \in \mathbb{R} \mid (t, p) \in \mathcal{D}\} \subseteq \mathbb{R}$$

is an open interval containing zero.

Example. Not all vector fields determine flows:

1. Let $M = \mathbb{R}^2$ and $V = x^2 \frac{\partial}{\partial x}$. Taking the integral curve of V starting at $(1, 0)$, we get $\gamma(t) = (\frac{1}{1-t}, 0)$. This integral curve is only defined on $(-\infty, 1) \subsetneq \mathbb{R}$.
2. Consider $1 \cdot \frac{\partial}{\partial x} + 0 \cdot \frac{\partial}{\partial y}$ on $\mathbb{R}^2 \setminus \{0\}$. $\gamma(t)$ cannot be extended to $(0, 0)$.

Vocabulary 8.4 (local flow). A local flow is a continuous map $\vartheta : \mathcal{D} \rightarrow M$ such that ① $\vartheta(0, p) = p$ and ② $\vartheta(t, \vartheta(s, p)) = \vartheta(t+s, p)$.

Vocabulary 8.5 (infinitesimal generator). The infinitesimal generator of a flow ϑ is the vector field $V_p = \left. \frac{d}{dt} \right|_{t=0} (\vartheta_t(p))$.

Theorem 22. Given some $V \in \mathfrak{X}(M)$, there exists a unique maximal local flow $\vartheta : \mathcal{D} \rightarrow M$ with infinitesimal generator V . It is maximal in the sense that for all local $\vartheta' : \mathcal{D}' \rightarrow M$, $\mathcal{D}' \subseteq \mathcal{D}$.

proof idea. Apply the existence and uniqueness for ODE a few times. □

Vocabulary 8.6 (complete vector field). A vector field $V \in \mathfrak{X}(M)$ is complete if it is the infinitesimal generator of a global flow. Or equivalently, integral curves of V are defined on all of $\mathbb{R}$. Or $\mathcal{D} = \mathbb{R} \times M$ is a flow domain of V .

Lemma (uniform time lemma). Let $V \in \mathfrak{X}(M)$ with local flow $\vartheta : \mathcal{D} \rightarrow M$. If there exists an $\epsilon > 0$ for every $p \in M$ such that $(-\epsilon, \epsilon) \times M \subseteq \mathcal{D}$, then $\mathbb{R} \times M = \mathcal{D}$. In other words, if the domain of $\vartheta(t, p)$ contains $(-\epsilon, \epsilon) \times \{p\}$ for every p , then the flow is global ($\iff V$ is complete).

Proof. We argue by contradiction. Let $\gamma_p : I \rightarrow M$ be an integral curve with maximal connected domain $I \subsetneq \mathbb{R}$. Say $I = (a, b)$ and $\gamma_p(0) = p$, then consider the curve $\tilde{\gamma}_p = \gamma_p(t + t_0)$ so that $t_0 \in (b - \epsilon, b + \epsilon)$. By assumption, the domain of $\tilde{\gamma}_p$ is at least $(-\epsilon, \epsilon)$. But this allows us to extend the domain of γ_p past b . This is a contradiction. □

Proposition 21 (naturality of integral curves). Let M, N be smooth manifolds, $F : M \rightarrow N$ a smooth map. Vector fields $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F -related if and only if F takes integral curves of X to integral curves of Y . Moreover, $F \circ \gamma$ is an integral curve of Y .

Proof. Let $\gamma : I \rightarrow M$ be an integral curve of X . Then $F \circ \gamma : I \rightarrow N$, and

$$(F \circ \gamma)'(t) = dF_{\gamma(t)}(\gamma'(t)) = dF_{\gamma(t)}(X_{\gamma(t)}) = Y_{(F \circ \gamma)(t)}$$

confirming that $F \circ \gamma$ is an integral curve of Y at $F(\gamma(t))$. Conversely, suppose F takes integral curves to integral curves. Assume γ is an integral curve at p for X , then

$$Y_{F(p)} = (F \circ \gamma)'(0) = dF_p(\gamma'(0)) = dF_p(X_p),$$

verifying the relationship between X and Y under the map F . $\square$

Vocabulary 8.7 (compactly supported vector field). A compactly supported vector field $V \in \mathfrak{X}(M)$ is zero outside of the supported compact subset $K \subseteq M$. In other words, $\text{supp}(V) = \{x \in M | V_x \neq 0\} \subseteq K \subseteq M$.

Theorem 23. Compactly supported vector fields have global flows if and only if they are complete if and only if they have flow domain $\mathbb{R} \times M$.

Proof. Suppose the support of vector field V is K . Outside of K , we take the trivial flow, i.e. $\vartheta(t, p) = p$ for all $t \in \mathbb{R}$ and $p \in M \setminus K$. Now define ϑ on K . For this, take open neighborhood U_p for each $p \in K$. Some ODE stuff tells us the flow $(-\epsilon_p, \epsilon_p) \times U_p \rightarrow M$ of V exists for some ϵ_p . By compactness of K , we get a finite cover $\{U_p\}$. Take ϵ to be the smallest ϵ_p . By finiteness, $\epsilon > 0$. By the uniform time lemma, this gives a global flow on $\bigcup U_p \supseteq K$. Finally, we patch the flow on K with the flow on $M \setminus K$ using the existence and uniqueness to get a flow $\vartheta : \mathbb{R} \times M \rightarrow M$. $\square$

Corollary. Vector fields on compact manifolds are complete.

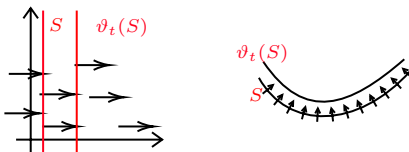
Theorem 24. Every left-invariant vector field on a Lie group G has a global flow.

Proof. Let G be a Lie group, let $X \in \text{Lie}(G)$, and let $\vartheta : \mathcal{D} \rightarrow G$ denote the flow of X . There is some $\epsilon > 0$ such that $\vartheta(t, p)$ is defined on $(-\epsilon, \epsilon)$. Let $g \in G$ be arbitrary. Because X is L_g -related to itself, it follows from proposition 22 that the curve $L_g \circ \vartheta : (-\epsilon, \epsilon) \rightarrow G$ is an integral curve. $\square$

10/17

Example. To visualize flows, we can consider images of submanifolds $S \subseteq M$ under a flow.

1. Some pictures



2. Let $S^n \subseteq \mathbb{R}^{n+1}$ be a submanifold. The flow under the radial vector field $-\nabla|x|^2$ flows spheres of radius 1 to spheres of different radii. The picture here should be a sphere with arrows pointing outwards.

Theorem 25 (flowout). Let $S \subseteq M$ be a submanifold and $V \in \mathfrak{X}(M)$. The flow of V is $\vartheta : \mathcal{D} \rightarrow M$. Set $\mathcal{O} = \mathcal{D} \cap (\mathbb{R} \times S)$ and $\Phi = \vartheta|_{\mathcal{O}}$. If V is nowhere tangent to S , meaning that $V_p \notin T_p S \subseteq T_p M$ for all $p \in S$, then

1. $\Phi : \mathcal{O} \rightarrow M$ is an immersion.
2. $1 \cdot \frac{\partial}{\partial t} + 0 \cdot \frac{\partial}{\partial s} \in \mathfrak{X}(\mathcal{O})$ is Φ -related to V .
3. There exists $\delta : S \rightarrow \mathbb{R}_{>0}$ such that $\Phi : \mathcal{O}_\delta \rightarrow M$ is an embedding for

$$\mathcal{O}_\delta = \{(t, s) | s \in S, t < \delta(s)\} \subseteq \mathcal{O} \subseteq \mathbb{R} \times S$$

Vocabulary 8.8 (flowout). $\Phi(\mathcal{O}_\delta) \subseteq M$ is the flowout of S along V .

Remark. A special case would be $p \in M$ as a 0-dimensional submanifold. Then the flowout of p along a vector field V is an integral curve.

Vocabulary 8.9 (inward/outward-pointing vector). Suppose M is a manifold with $\partial M \neq \emptyset$. For any $p \in \partial M$, a vector $v_p \in T_p M$ with $v_p \notin T_p \partial M \subseteq T_p M$ is inward-pointing if there exists some $\gamma : [0, 1) \rightarrow M$ such that $\gamma(0) = p$ and $\gamma'(0) = v_p$. v_p is outward-pointing if there exists some $\gamma : (-1, 0] \rightarrow M$ such that $\gamma(0) = p$ and $\gamma'(0) = v_p$.

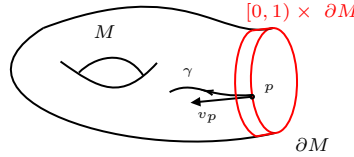


Figure 2: Inward-pointing vector v_p and collar neighborhood $[0, 1) \times \partial M$.

Remark. Given $V \in \mathfrak{X}(M)$ that is inward-pointing (i.e. for all $p \in \partial M$, $v_p \in T_p M$ is inward-pointing). We can consider the flow along V and the flowout of $\partial M \subseteq M$ along V .

Vocabulary 8.10 (collar neighborhood). A collar neighborhood of $\partial M \subseteq M$ is an open submanifold $[0, 1) \times \partial M$ with a map $F : [0, 1) \times \partial M \hookrightarrow M$ such that $F(0, -) : \partial M \hookrightarrow M$ is the inclusion of the boundary.

Theorem 26. Collar neighborhoods always exist.

proof idea. We can always construct an inward-pointing vector field on M . Then apply the flowout theorem to get an embedded submanifold $\mathcal{O} \simeq [0, 1) \times \partial M$. This gives the collar neighborhood. $\square$

Vocabulary 8.11 (attaching manifold). Let M, N be manifolds with nonempty boundaries. If there is a diffeomorphism $h : \partial M \rightarrow \partial N$, we can attach M and N together using h : $M \hookrightarrow M \cup_h N \hookleftarrow N$. We start with the topological space

$$M \cup_h N = (M \sqcup N) / \sim$$

where $\sim$ is the quotient for the equivalence relation generated by $x \sim h(x)$ whenever $h(x) \in \partial N$ for all $x \in \partial M$. To make this space into a smooth manifold, we need to specify a smooth atlas. For $p \in \text{int}(M) \sqcup \text{int}(N) \subseteq (M \sqcup N) / \sim$, use charts for M and N : $U \in \text{int}(M)$ and $V \in \text{int}(N)$. Now we consider points on the boundary. Using collar neighborhoods of ∂M and ∂N , we get

$$\begin{cases} [0, 1) \times \partial M \rightarrow M \rightarrow (M \sqcup N) / \sim \\ (-1, 0] \times \partial M \simeq [0, 1) \times \partial N \rightarrow N \rightarrow (M \sqcup N) / \sim \end{cases} \Rightarrow ((-1, 0] \times \partial M) \sqcup ([0, 1) \times \partial M) / \sim \rightarrow (M \sqcup N) / \sim$$

The last map is just $(-1, 1) \times \partial M \rightarrow (M \sqcup N) / \sim$. Then it follows that for any $p \in \partial M \hookrightarrow (M \sqcup N) / \sim$, define charts containing p from charts on $(-1, 1) \times \partial M$.

Vocabulary 8.12 (connected sum). Given connected manifolds M_1 and M_2 without boundary of the same dimension n . Consider open coordinate balls $D_1 \subseteq M_1$ and $D_2 \subseteq M_2$. Then $M_1 \setminus D_1$ and $M_2 \setminus D_2$ are manifolds with boundary $S^{n-1} = \partial \overline{D_i}$. Let $h : \partial \overline{D_1} \rightarrow \partial \overline{D_2}$ be a diffeomorphism between boundaries. The connected sum of M_1 and M_2 is

$$M_1 \# M_2 := (M_1 \setminus D_1) \cup_h (M_2 \setminus D_2)$$

Vocabulary 8.13 (double). Let M be a smooth manifold with boundary. The double of M is the manifold $D(M) = M \cup_{\text{id}} M$, where $\text{id} : \partial M \rightarrow \partial M$ is the identity map of ∂M . It is obtained from $M \sqcup M$ by identifying each boundary point in one copy of M with the same boundary point in the other. It is a smooth manifold without boundary. It is easy to check that $D(M)$ is compact if and only if M is compact, and connected if and only if M is connected. It is useful to extend the definition to manifolds without boundary by defining $D(M) = M \sqcup M$ when $\partial M = \emptyset$. In particular, M can be embedded into its double.

Example. Useful applications of doubles:

1. Suppose we want to embed a manifold with boundary into $\mathbb{R}^N$. Take the embedding of the double $M \cup_{\text{id}} M \hookrightarrow \mathbb{R}^N$. Note that $D(M)$ is a manifold without boundary. Use embeddings $M \hookrightarrow D(M) \hookrightarrow \mathbb{R}^N$.
2. To consider flows on M with $\partial M \neq \emptyset$, we can ask when does a flow on $D(M)$ restricts to $M \hookrightarrow D(M)$. The answer is when $V_{p \in \partial M} \in T_p \partial M \subseteq T_p M$ for some $V \in \mathfrak{X}(M)$.

Week 9

Definitions and properties of vector bundles, including tangent and cotangent bundles. Went over terms like gradient, vector bundle homomorphism, and local trivializations. Proved propositions related to integrals.

10/22

Example. Some motivating examples for vector bundles:

1. Consider the map $TM \rightarrow M$. In this case, TM is a manifold with special property that for each $p \in M$, $T_p M$ is a vector space.

2. $TM|_S := \bigsqcup_{p \in S} T_p M$. Also $T_p S \subseteq T_p M$ implies $TS \hookrightarrow TM|_S$. Or in picture:

$$\begin{array}{ccc} TM|_S & \hookrightarrow & TM \\ \downarrow & & \downarrow \\ S & \xrightarrow{\text{submfld}} & M \end{array}$$

Vocabulary 9.1 (vector bundle). The main idea is that suppose we have vector spaces E_p for each $p \in M$ and they vary smoothly with p . More precisely, a real vector bundle $E \rightarrow X$ for a topological space X is a topological space E with

1. a continuous surjection $\pi : E \rightarrow X$.
2. the structure of a vector space (check the axioms) on each fiber $\pi^{-1}(p) = E_p$, $p \in X$, such that for each p (and fixed k) there exists some open $U \subseteq X$ with $p \in U$ and a local homeomorphism φ satisfies the diagram on the right. And $\varphi(p', -) : \mathbb{R}^k \rightarrow E_{p'}$ is a linear isomorphism for all $p' \in U$.

$$\begin{array}{ccc} U \times \mathbb{R}^k & \xrightarrow{\varphi} & \pi^{-1}(U) \\ & \searrow \text{proj} & \downarrow \pi \\ & & U \end{array}$$

Remark. This comes with two variations: ① real $\leadsto$ complex, quaternionic. ② continuous $\leadsto$ smooth so that $\pi : E \rightarrow M$ is a smooth submersion and local trivializations are smooth. From here on, we assume all vector bundles are smooth so that

1. E, M are smooth manifolds.
2. $\pi : E \rightarrow M$ is a smooth map.
3. local trivializations are diffeomorphisms.

Vocabulary 9.2 (vector bundle homomorphism).

$E \xrightarrow{F} E'$ A vector bundle map or a vector bundle homomorphism is displayed on the left. Moreover, $\downarrow \pi$ $\downarrow \pi$ $\pi \circ F = f \circ \pi$ and the restriction $F_p : E_p \rightarrow E'_{f(p)}$ is linear for all $p \in M$. For example, $M \xrightarrow{f} M'$ $f : M \rightarrow M'$, $F := df : TM \rightarrow TM'$ is a map of vector bundles.

Example. More examples about vector bundles,

1. Trivial rank- k bundle $E := M \times \mathbb{R}^k \rightarrow M$. Locally, every vector bundle is *trivializable* meaning $E|_U \xrightarrow{\varphi} U \times \mathbb{R}^k$. More generally, $\pi : E \rightarrow M$ is trivializable if there exists an isomorphism of vector bundles ψ such that

$$\begin{array}{ccc} E & \xrightarrow{\psi} & M \times \mathbb{R}^k \\ \downarrow \pi & & \downarrow \text{proj} \\ M & \xrightarrow{\text{id}} & M \end{array}$$

2. Tangent bundles. We can restrict tangent bundles to submanifolds.
3. Mobius bundle. $E = \mathbb{R}^2 / \sim$ and a map $\mathbb{Z} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $(n, (x, y)) = n \cdot (x, y) = (x + n, (-1)^n y)$. $E \rightarrow \mathbb{R}/\mathbb{Z} \simeq S^1$ and the fibers are vector spaces. Lastly, the map $(-1)^n : \mathbb{R} \rightarrow \mathbb{R}$ is linear.
4. Rank-1 vector bundles are called *line bundles*.

Vocabulary 9.3 (bundle chart and transition function/map). E naturally carries a manifold structure: if (U, φ) is a chart for M and (U, ψ) is a bundle chart for E , we define a manifold chart for E by the composition

$$\pi^{-1}(U) \xrightarrow{\psi} U \times \mathbb{R}^k \xrightarrow{\varphi \times \text{Id}_{\mathbb{R}^k}} \varphi(U) \times \mathbb{R}^k \subseteq \mathbb{R}^{n+k}$$

It is natural to ask about the transition function between two trivializations. Let $U, V \subseteq M$ with $U \cap V \neq \emptyset$. Given a vector bundle $\pi : E \rightarrow M$, consider local trivializations φ_U and φ_V for $\pi^{-1}(U)$ and $\pi^{-1}(V) \subseteq E$ respectively. We have a commuting diagram

$$\begin{array}{ccccc} E|_{U \cap V} & & & & \\ \downarrow \simeq & \searrow & & & \\ U \cap V \times \mathbb{R}^k & \xleftarrow{\varphi_U} & \pi^{-1}(U \cap V) & \xrightarrow{\varphi_V} & U \cap V \times \mathbb{R}^k \\ & \searrow p & \downarrow \pi & \swarrow p & \\ & & U \cap V & & \end{array}$$

Note that $p \circ (\varphi_V \circ \varphi_U^{-1}) = p$ does not mean $p \circ (\varphi_V \circ \varphi_U^{-1})(x, v) = p(x, v)$. It only tells us that they are the same map. So $\varphi_V \circ \varphi_U^{-1}(x, v) = (x, \sigma(x, v))$ for some smooth $\sigma : (U \cap V) \times \mathbb{R}^k \rightarrow \mathbb{R}^k$. If $x \in U \cap V$ fixed, the map $\tau(x) : \mathbb{R}^k \rightarrow \mathbb{R}^k$ given by $v \mapsto \sigma(x, v)$ is a linear map. In particular, τ is invertible. Otherwise $\varphi_V \circ \varphi_U^{-1}$ will not be invertible. It turned out $\tau : U \cap V \rightarrow GL_k(\mathbb{R})$ is also smooth. Here, the transition function is denoted τ_{UV} .

Proposition 22 (vector bundle construction). Suppose we have M , this will allow us to build bundle from M . Given an open cover $\{U_i\}$ of M and functions $\tau_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL_k(\mathbb{R})$ satisfying

1. $\tau_{\alpha\alpha}$ is a constant map to $\text{id} = GL_k(\mathbb{R})$.
2. $\tau_{\alpha\beta} \circ \tau_{\beta\gamma} = \tau_{\alpha\gamma} : U_\alpha \cap U_\beta \cap U_\gamma \rightarrow GL_k(\mathbb{R})$.

We get a rank- k vector bundle from this.

proof sketch. Define $(\bigsqcup_i (U_i \times \mathbb{R}^k)) / \sim$ where $U_\alpha \times \mathbb{R}^k \ni (x, v) \sim (x', v') \in U_\beta \times \mathbb{R}^k$ when $x = x' \in M$ and $v = \tau_{\alpha\beta} v'$. We get $\pi : E \rightarrow M$ via $\pi[x, v] = x \in M$. Can check this is well-defined. $\square$

Vocabulary 9.4 (sum, tensor product, and dual). Given vector bundles E and E' over M , we can construct

1. $E \oplus E'$ with $(E \oplus E')_p := E_p \oplus E'_p$ for all $p \in M$. Transition functions are $\tau_i \oplus \tau'_i$.
2. $E \otimes E'$ with $(E \otimes E')_p := E_p \otimes E'_p$ for all $p \in M$. Transition functions are $\tau_i \otimes \tau'_i$.
3. E^* with $(E^*)_p = (E_p)^* = \text{hom}(E_p, \mathbb{R})$. Transition functions are $(\tau_i)^* = (\tau_i^{-1})^T$.

In general, these transition functions come from $\otimes$, $\oplus$, and $()^*$ of linear functions.

Vocabulary 9.5 (section). Given a vector bundle $\pi : E \rightarrow M$, its (global) sections are elements of the set

$$\Gamma(E) = \{\sigma : M \rightarrow E \text{ smooth} | \pi \circ \sigma = \text{id}_M\}$$

Remark. $\pi \circ \sigma = \text{id}_M$ if and only if σ is a function that has value in E_p at each $p \in M$, i.e. $\sigma(p) \in E_p$. On the other hand, $\Gamma(E)$ is a module over $C^\infty(M)$, meaning that for $f_1, f_2 \in C^\infty(M)$ and $\sigma_1, \sigma_2 \in \Gamma(E)$, $f_1 \sigma_1 + f_2 \sigma_2 \in \Gamma(E)$. To verify this, we need to work locally using trivializations.

Vocabulary 9.6 (local section of vector bundles). Let $\pi : E \rightarrow M$, a local section of E is a section of $E|_U$ for some open set $U \subseteq M$, i.e. $\sigma : U \rightarrow E$ such that $\pi \circ \sigma = \text{id}_U$.

Vocabulary 9.7 (frame of vector bundles). A collection $C = \{\sigma_1, \dots, \sigma_k\}$ of sections of a vector bundle $E \rightarrow M$ are *linearly independent* if for all $p \in M$, $\{\sigma_1(p), \dots, \sigma_k(p)\}$ are linearly independent in E_p . C is a (global) frame.

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Vocabulary 9.8 (cotangent bundle). Recall that the dual vector space of a finite dimensional vector space V is $V^* := \text{hom}(V, \mathbb{R})$. For any basis $v^1, \dots, v^n$ of V , we can find a basis $dv^1, \dots, dv^n$ of V^* characterized by

$$dv^j(v^i) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{everything else} \end{cases} = \delta^{ij}$$

For $E = TM$, the dual bundle $(TM)^* = T^*M$ is the cotangent bundle of M . We can apply this to $T_p M$ and $T_p^* M$. Given local coordinates $x^1, \dots, x^n$ on $U \subseteq M$, for each $p \in U$ the coordinate basis $\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p$ provides a basis for $T_p M$, which gives rise to a dual basis for $T_p^* M$: $dx^1|_p, \dots, dx^n|_p$. Similarly, it is characterized by

$$dx^j|_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \delta^{ij} \in C^\infty(U)$$

If $p \in T^*M$, $p = (x, t)$ where $x \in M$ and $t \in T_x^*M$.

Remark. More generally, for any $\omega \in \Gamma(T^*M) = \mathfrak{X}^*(M)$, we can pair any $X \in \Gamma(TM) = \mathfrak{X}(M)$ with $\omega(X) \in C^\infty(M)$. Recall that $\omega : M \rightarrow T^*M$ with coordinate functions $\omega_1, \dots, \omega_n$. It follows from the coordinate formula of (co)vector fields that

$$\omega|_U = \omega_i dx^i \quad \text{and} \quad \omega|_U \left(\frac{\partial}{\partial x^j} \right) = \omega_i dx^i \left(\frac{\partial}{\partial x^j} \right) = \omega_i \quad (\blacksquare)$$

for some $U \subseteq M$.

Vocabulary 9.9 (covector field of $f\omega$). If $f \in C^\infty(M)$ and $\omega \in \mathfrak{X}^*(M)$, the covector field $f\omega$ is defined by

$$(f\omega)_p = f(p)\omega_p$$

Vocabulary 9.10 (change of coordinate). Suppose $\tilde{x}^1, \dots, \tilde{x}^n$ is from another coordinate system with basis $\tilde{v}^1, \dots, \tilde{v}^n$ correspond to charts (U, φ) and (V, ψ) . The transition map is $\psi \circ \varphi^{-1}(x) = (\tilde{x}^1(x), \dots, \tilde{x}^n(x))$ for some $x \in \varphi(U \cap V)$. By $(\star)$, $d(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j} \Big|_{\psi(p)}$. Using the characterization of coordinate vectors:

$$\begin{aligned} \frac{\partial}{\partial x^i} \Big|_p &= (d\varphi_p)^{-1} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = d(\varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = d\psi_{\psi(p)}^{-1} \circ d(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \\ &= d\psi_{\psi(p)}^{-1} \left(\frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j} \Big|_{\psi(p)} \right) = \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j} \Big|_p \end{aligned}$$

Now applying this to ω , we get

$$\omega_i = \omega \left(\frac{\partial}{\partial x^i} \right) = \omega \left(\frac{\partial \tilde{x}^j}{\partial x^i} \frac{\partial}{\partial \tilde{x}^j} \right) = \frac{\partial \tilde{x}^j}{\partial x^i} \omega \left(\frac{\partial}{\partial \tilde{x}^j} \right) = \frac{\partial \tilde{x}^j}{\partial x^i} \tilde{\omega}_j$$

Vocabulary 9.11 (gradient). Just like what we have seen in calculus: $\text{grad}(f) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}$ where $f \in C^\infty(\mathbb{R}^n)$.

Remark. Does $\text{grad}(f)$ make sense for $f \in C^\infty(M)$? What happens under coordinate changes? Here is how we are going to generalize $\text{grad}(f)$ to manifolds: Given a function $f \in C^\infty(M)$. Define $df \in \Gamma(T^*M) = \mathfrak{X}^*(M)$ by

$$df(v) = v f \quad (\blacklozenge)$$

for some $v \in \Gamma(TM) = \mathfrak{X}(M)$. This determines a section of T^*M . Locally in some open set U :

$$df|_U = (df)_i dx^i \stackrel{\blacksquare}{=} df \left(\frac{\partial}{\partial x^i} \right) dx^i \stackrel{\blacklozenge}{=} \frac{\partial f}{\partial x^i} dx^i$$

To summarize, we have a map $d : C^\infty(M) \rightarrow \Gamma(T^*M)$ given by $f \mapsto df$. This map has properties that

1. $d(fg) = (df)g + f(dg)$.
2. f is constant implies $df = 0 \in \Gamma(T^*M)$.
3. If p is a critical point of $f : M \rightarrow \mathbb{R}$, then $df_p = 0$ ($\iff \frac{\partial f}{\partial x^i} = 0$ for all i).

Vocabulary 9.12 (1-form). Sections of T^*M are called 1-forms.

Vocabulary 9.13 (pullback by F). Given $F : M \rightarrow N$, we define $F^*\omega \in \Gamma(T^*M)$ for $\omega \in \Gamma(T^*N)$ (also known as the pullback of ω by F) by

$$(F^*\omega)_p(V_p) = \omega_{F(p)}(dF_p(V_p)) \quad \text{or} \quad (F^*\omega)_p = dF_p^*(\omega_{F(p)})$$

where $V \in \mathfrak{X}(M)$. Define $dF_p^* : T_{F(p)}^*N \rightarrow T_p^*M$ as the opposite of $dF_p : T_pM \rightarrow T_{F(p)}^*N$. This is called the point-wise pullback by F at p . Also as a reminder,

$$dF_p^*(\omega)(v) = \omega(dF_p(v))$$

for $\omega \in T_{F(p)}^*N$ and $v \in T_pM$. This is due to p273 of Lee.

Proposition 23. Let $F : M \rightarrow N$, $u \in C^\infty(N)$, and $\omega \in \Gamma(T^*N)$. Then

- ① $F^*(u\omega) = F^*uF^*\omega = (u \circ F)F^*\omega$.
- ② $F^*(du) = d(F^*u) = d(u \circ F) \in \Gamma(T^*M)$ when u is smooth. This says " d commutes with pullback".

Proof. Mostly definition/equation-chasing but we are doing it anyway

- ① $(F^*(\omega))_p \stackrel{1}{=} dF_p^*((u\omega)_{F(p)}) \stackrel{2}{=} dF_p^*(u(F(p))\omega_{F(p)}) \stackrel{3}{=} u(F(p))dF_p^*(\omega_{F(p)}) \stackrel{4}{=} u(F(p))(F^*\omega)_p \stackrel{5}{=} ((u \circ F)F^*\omega)_p$.
- ② $(F^*du)_p(v) \stackrel{6}{=} (dF_p^*(du_{F(p)}))(v) \stackrel{7}{=} du_{F(p)}(dF_p(v)) \stackrel{8}{=} dF_p(v)u \stackrel{9}{=} v(u \circ F) \stackrel{0}{=} d(u \circ F)_p(v)$

1, 4, 6, and 7 are due to Vocabulary 9.13. 2 and 5 are due to Vocabulary 9.9. 3 is true by the linearity of dF_p^* . 8 is due to the definition of du . 9 is due to Vocabulary 2.4. Lastly, 0 is due to the definition of $d(u \circ F)$. $\square$

Vocabulary 9.14 (line integral). Let $f : [a, b] \rightarrow \mathbb{R}$. According to the previous remark, $df = \frac{\partial f}{\partial t} dt = \frac{df}{dt} dt$. Now apply the fundamental theorem of calculus

$$\int_a^b df = \int_a^b \frac{df}{dt} dt = f(b) - f(a)$$

Proposition 24 (diffeomorphism invariance of the integral). Let ω be a smooth covector field on the compact interval $[a, b] \subseteq \mathbb{R}$. If $\varphi : [c, d] \rightarrow [a, b]$ is an increasing diffeomorphism (meaning that $t_1 < t_2$ implies $\varphi(t_1) < \varphi(t_2)$),

$$\int_{[c,d]} \varphi^*\omega = \int_{[a,b]} \omega.$$

Proof. The standard coordinate on $[c, d]$ is s and the coordinate on $[a, b]$ is t . Recall that we can write $\omega = \omega_j dy^j$ in coordinates. Now we apply proposition 24 twice: $\varphi^*\omega = \varphi^*(\omega_j dy^j) = \varphi^*(f(t)dt) = (f \circ \varphi)\varphi^*dt = (f \circ \varphi)d(t \circ \varphi)$. Since φ takes s to t , $d(t \circ \varphi)$ is just $d\varphi$. Evaluating it at s , we get $(f \circ \varphi)(s)d(\varphi(s)) = f(\varphi(s))dt = f(t)dt$. So

$$\int_{[c,d]} \varphi^*\omega = \int_a^b f(t)dt = \int_{[a,b]} \omega$$

$\square$

Vocabulary 9.15 (curve in manifold). A smooth curve in a manifold is a smooth $\gamma : [a, b] \rightarrow M$ with a compact domain. Meanwhile, a piecewise smooth curve in a manifold is a collection of smooth curves $\gamma_i : [a_i, b_i] \rightarrow M$ with $\gamma_i(b_i) = \gamma_{i+1}(a_{i+1})$.

Vocabulary 9.16 (line integral). If $\gamma : [a, b] \rightarrow M$ is a smooth curve and $\omega \in \Gamma(T^*M)$ on M , the line integral of ω over γ is

$$\int_\gamma \omega = \int_{[a,b]} \gamma^*\omega = \int_I f(t)dt$$

Recall that for a smooth map $F : M \rightarrow N$ and ω a covector field on N , $F^*\omega$ is defined to be a covector field on M by Vocabulary 9.13. So $\gamma^*\omega$ is a smooth covector field on $[a, b]$.

Example. Consider $M = S^1$, $d\vartheta \in \Gamma(T^*S^1)$ where $d\vartheta(\frac{\partial}{\partial\vartheta}) = 1$. Let $\gamma : [a, b] \rightarrow S^1$ be a closed curve. So $\gamma(a) = \gamma(b) = 1 \in S^1 \subseteq \mathbb{C}$. Then

$$\int_{\gamma} d\vartheta = 2\pi \cdot (\text{winding number})$$

Alternatively, we can consider $M = \mathbb{R}^2 \setminus \{0\} = \mathbb{C} \setminus \{0\}$. In this case, $\omega = (xdy - ydx)/r^2 = (xdy - ydx)/(x^2 + y^2)$. If $\gamma : [a, b] \rightarrow \mathbb{R}^2 \setminus \{0\}$, then

$$\int_{\gamma} \omega = 2\pi \cdot (\text{winding number})$$

Proposition 25. Let M be a smooth manifold with or without boundary. Suppose $\gamma : [a, b] \rightarrow M$ is a piecewise smooth curve, and $\omega, \omega_1, \omega_2 \in \mathfrak{X}^*(M)$.

1. For any $c_1, c_2 \in \mathbb{R}$, $\int_{\gamma}(c_1\omega_1 + c_2\omega_2) = c_1 \int_{\gamma} \omega_1 + c_2 \int_{\gamma} \omega_2$.
2. If γ is a constant map, then $\int_{\gamma} \omega = 0$.
3. If $\gamma_1 = \gamma|_{[a, c]}$ and $\gamma_2 = \gamma|_{[c, b]}$ with $a < c < b$, then $\int_{\gamma} \omega = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega$.
4. If $F : M \rightarrow N$ is any smooth map and $\eta \in \mathfrak{X}^*(N)$, then $\int_{\gamma} F^*\eta = \int_{F \circ \gamma} \eta$.

Week 10

Covered properties of covectors and tensors on manifolds, focusing on conservative and exact covectors, closed covector fields, and Riemannian metrics. Included how these concepts relate to differential forms, measuring distances and angles on manifolds.

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Vocabulary 10.1 (conservative). A covector field ω is conservative if for all closed curves γ , $\int_{\gamma} \omega = 0$.

Lemma. For a conservative covector field ω , $\int_{\gamma} \omega$ is independent of the choice of paths from $\gamma(a)$ to $\gamma(b)$.

proof sketch. Suppose we have two distinct paths γ, α that start and end at the same points. Let $\bar{\alpha}$ denote the reverse direction of α . Observe that γ and $\bar{\alpha}$ form a closed curve. Since $\int_{\gamma + \bar{\alpha}} \omega = 0$,

$$\int_{\gamma + \bar{\alpha}} \omega = \int_a^b \gamma^* \omega + \int_b^a \bar{\alpha}^* \omega = \int_a^b \gamma^* \omega - \int_a^b \bar{\alpha}^* \omega = 0$$

□

Theorem 27. $\omega \in \Gamma(T^*M)$ is conservative if and only if $\omega = df$ for some $f \in C^\infty(M)$. Note that f is not unique as $\omega = df = d(f + c)$ for any constant c . Recall that locally $df = \frac{\partial f}{\partial x^i} dx^i$.

proof sketch. ($\Leftarrow$) Suppose $\omega = df$, then we apply proposition 24 and the fundamental theorem of calculus:

$$\int_{\gamma} \omega = \int_{\gamma} dt = \int_a^b \gamma^* df = \int_a^b d\gamma^* f = \int_a^b \frac{d}{dt}(\gamma^* f) dt = f(\gamma(b)) - f(\gamma(a))$$

($\Rightarrow$) Suppose ω is conservative. Choose a point $p \in M$. Define $f(x) = \int_{\gamma_x} \omega$ where γ_x is a path from p to x . This is well-defined because the integral does not depend on the choice of path. We can verify that ① f as defined is smooth and ② $df = \omega$ using the fundamental theorem of calculus. □

Vocabulary 10.2 (exact). Let $\omega \in \Gamma(T^*M)$ be exact if $\omega = df$ for some $f \in C^\infty(M)$. By theorem 26, ω is exact if and only if it is conservative. Since f is smooth, it must satisfy $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ locally on some U . If $\omega = \omega_i dx^i = df$, then $\omega_i = \frac{\partial f}{\partial x^i}$. Plugging this into the equation above, we get

$$\frac{\partial \omega_j}{\partial x^i} = \frac{\partial \omega_i}{\partial x^j} \quad (\bullet)$$

Vocabulary 10.3 (closed covector field). A covector field is closed if its components in every smooth chart satisfy the equation above.

Lemma. Exact implies closed.

Proposition 26. Let ω be a smooth covector field on a smooth manifold M with or without boundary. The following are equivalent:

1. ω is closed.
2. ω satisfies $\bullet$ in some chart around every point.
3. For any open subset $U \subseteq M$ and smooth vector fields X, Y on U , $X(\omega(Y)) - Y(\omega(X)) = \omega([X, Y])$.

Proof. ((1.) $\iff$ (2.)) by definition. ((2.) $\implies$ (3.)) Now we write out ω , X , and Y in coordinates. Compute

$$X(\omega(Y)) = X \left(\omega_i dx^i \left(Y^k \frac{\partial}{\partial x^k} \right) \right) \stackrel{\blacksquare}{=} X(\omega_i Y^i) \stackrel{6.1}{=} Y^i X \omega_i + \omega_i X Y^i = Y^i X^j \frac{\partial}{\partial x^j} (\omega_i) + \omega_i X Y^i$$

Repeating the same calculation for $Y(\omega(X))$, we get

$$X(\omega(Y)) - Y(\omega(X)) = \omega_i (X Y^i - Y X^i) \stackrel{\blacksquare}{=} \omega \left(\frac{\partial}{\partial x^i} \right) (X Y^i - Y X^i) \stackrel{6.6}{=} \omega([X, Y])$$

((3.) $\implies$ (2.)) In any coordinate chart we can apply $X(\omega(Y)) - Y(\omega(X)) = \omega([X, Y])$ with $X = \frac{\partial}{\partial x^i}$ and $Y = \frac{\partial}{\partial x^j}$. But $[X, Y] = 0$ in this case. Remaining terms give (2.). $\square$

Proposition 27. For $F : M \rightarrow N$ a local diffeomorphism, $\omega \in \Gamma(T^*N)$ is closed implies $F^*\omega \in \Gamma(T^*M)$ is closed. And $\omega = df \in \Gamma(T^*N)$ is exact implies $F^*\omega \in \Gamma(T^*M)$ is exact. Remember $F^* : \Gamma(T^*N) \rightarrow \Gamma(T^*M)$.

Proof. For exactness, set $g = F^*f$. So $F^*\omega = F^*df = d(F^*f) = dg$ shows that $F^*\omega$ is exact. For closed covector fields, if (U, φ) is any smooth chart for N , then $(F^{-1}(U), \varphi \circ F)$ is a chart in M . The coordinate representation of F is the identity. So if ω satisfies $\bullet$, $F^*\omega$ has to satisfy it as well. Being closed is a local property. This can be checked in charts. Being exact is a global property: a function f with $df = \omega$ is defined on all of M . $\square$

Example. Let $M = S^1$. $\omega = d\vartheta$ is closed but not exact as $\int_{S^1} d\vartheta = 2\pi \neq 0$ implies $d\vartheta$ is not conservative.

Theorem 28. If ω is closed, it is locally exact. In other words, for any $p \in M$, there exists some open $U \subseteq M$ such that $\omega|_U = df_U$ where $f_U \in C^\infty(U)$.

proof sketch. Locally, define $f_U = \int_{\gamma_x} \omega|_U$ for γ_x path from p to $x \in U$. $\square$

Vocabulary 10.4 (tensor product). Given vector spaces V and W , the tensor product $V \otimes W$ is a vector space characterized by

$$\begin{array}{ccc} V \times W & \xrightarrow{\text{bilin}} & Z \\ \downarrow & \nearrow \text{!linear} & \\ V \otimes W & & \end{array}$$

where Z is some other vector space. Given a basis $v^1, \dots, v^m$ of V and a basis $w^1, \dots, w^n$ of W , we get a basis $v^i \otimes w^j$ of $V \otimes W$. Further, $\dim(V \otimes W) = \dim(V) \dim(W)$. If $x \in V \otimes W$, $x = \sum_{i,j} a_{ij} v^i \otimes w^j = (\sum_{i,j} a_{ij} v^i) \otimes w^j = v^i \otimes (\sum_{i,j} a_{ij} w^j)$. We also get $v \otimes \lambda w = \lambda v \otimes w$ for any $v \in V$ and $w \in W$. More generally, if $x \in V_1 \otimes \dots \otimes V_n$,

$$x = \sum_{i_1, \dots, i_n} a_{i_1, \dots, i_n} v_1^{i_1} \otimes \dots \otimes v_n^{i_n}$$

Proposition 28. We have the following isomorphisms

1. $V \otimes W \simeq W \otimes V$.
2. $V \otimes (W \otimes U) \simeq (V \otimes W) \otimes U$.
3. $V \otimes \mathbb{R} \simeq V$.

Vocabulary 10.5 (tensor). Covariant tensors are sections of $(T^*M)^{\otimes k} \rightarrow M$. Contravariant tensors are sections of $(TM)^{\otimes l} \rightarrow M$. A (k, l) -tensor is a section of $(T^*M)^{\otimes k} \otimes (TM)^{\otimes l} \rightarrow M$. In local coordinates,

$$A = \begin{cases} A_{i_1 \dots i_k} dx^{i_1} \otimes \dots \otimes dx^{i_k}, & A \in \Gamma((T^*M)^{\otimes k}); \\ A^{i_1 \dots i_k} \frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_k}}, & A \in \Gamma((TM)^{\otimes k}); \\ A_{j_1 \dots j_l}^{i_1 \dots i_k} \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_l}} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_l}, & A \in \Gamma((T^*M)^{\otimes k} \otimes (TM)^{\otimes l}). \end{cases}$$

Remark. A covariant tensor field pulls back along a smooth map $F : M \rightarrow N$

$$F^*A \stackrel{\text{loc}}{=} (F^*A_{i_1, \dots, i_k})(y^{i_1} \circ F) \otimes \dots \otimes d(y^{i_k} \circ F)$$

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Another perspective on tensors is in terms of *multilinear maps*. Recall that $\omega \in \Gamma(T^*M)$ and $X \in \mathfrak{X}(M) = \Gamma(TM)$ give $\omega(X) \in C^\infty(M)$. For any $f \in C^\infty(M)$, $\omega(fX) = f\omega(X)$. In coordinates, if $X = fX^i \frac{\partial}{\partial x^i}$ and $\omega = \omega_i dx^i$, then $\omega(fX) = \omega_i fX^i = f\omega_i X^i = f\omega(X)$. The very first equality comes from the calculation in Proposition 27. For $\omega \in \Gamma((T^*M)^{\otimes 2})$,

$$\omega(fX, Y) = \omega(X, fY) = f\omega(X, Y)$$

where $\omega(X, Y) \in C^\infty(M)$. This property is called $C^\infty(M)$ -multilinear. Similarly, contravariant tensors are $C^\infty(M)$ -multilinear functions on covector fields with an isomorphism $(T^*M)^* \simeq TM$.

Vocabulary 10.6 (pullback of covariant k -tensors). The pullback looks like $(F^*\omega)_p(X, Y) = \omega_{F(p)}(dF_p X, dF_p Y)$ according to Vocabulary 9.13.

Vocabulary 10.7 (symmetric and alternating covariant k -tensors). A k -tensor $\omega \in \Gamma((T^*M)^{\otimes k})$ is symmetric if

$$\omega(X_1, \dots, X_k) = c \sum \omega(X_{\sigma(1)}, \dots, X_{\sigma(k)})$$

It is alternating if

$$\omega(X_1, \dots, X_k) = c \sum (-1)^{|\sigma|} \omega(X_{\sigma(1)}, \dots, X_{\sigma(k)})$$

where $\sigma \in \mathfrak{S}_k$ and some constant coefficient c .

Vocabulary 10.8 (Riemannian metric). A Riemannian metric on a manifold M is a symmetric covariant 2-tensor $g \in \Gamma((T^*M)^{\otimes 2})$. At each $p \in M$, g gives an inner product:

$$g : M \rightarrow (T^*M)^{\otimes 2} \Rightarrow g_p : p \mapsto ((T_p M)^{\otimes 2} \rightarrow \mathbb{R}) = g_p(v, w) = \langle v, w \rangle_g \in \mathbb{R}$$

for $v, w \in T_p M$. Locally, g can be written as $g = g_{ij} dx^i \otimes dx^j$ such that $g_{ij} = g_{ji}$. So locally g is the data of a symmetric positive definite matrix. A manifold with g is a *Riemannian manifold*.

Vocabulary 10.9 (distance on Riemannian manifolds). For $v, w \in T_p M$, lengths and angles are defined as

$$|v| = (g_p(v, v))^{1/2} \quad \cos(\vartheta) = \frac{g_p(v, w)}{|v| \cdot |w|}$$

respectively. The length of a path $\gamma : [a, b] \rightarrow M$ is defined as

$$|\gamma| = \int_\gamma |\gamma'(t)| dt = \int_a^b |\gamma'(t)| dt$$

We claim that this does not depend on the parametrization of γ . The distance between two points $p, q \in M$ in a connected Riemannian manifold is

$$d(p, q) := \inf_{\gamma \text{ path } p \rightarrow q} |\gamma|$$

Remark. M as a topological space with the distance function defined above is a metric space.

Example. Take $M = \mathbb{R}^n$ with the usual dot product on $T_p M \simeq \mathbb{R}^n$. This is a Riemannian manifold where $g = \delta_{ij} dx^i \otimes dx^j = \delta_{ij} dx^i dx^j$. So $g = (dx^1)^2 + \dots + (dx^n)^2$ gives the usual norm. The distance function d agrees with the usual notion of distance in $\mathbb{R}^n$ as well.

Theorem 29. Every smooth manifold admits a Riemannian metric.

Proof. Let $\{(U_\alpha, \varphi_\alpha)\}$ be some charts. Suppose $\bar{g} = \delta_{ij} dx^i \otimes dx^j$ is the usual metric on $\mathbb{R}^n$. Since we know how a covariant tensor field pulls back along a smooth map,

$$g_\alpha := \varphi_\alpha^* \bar{g} = (\delta_{ij} \circ \varphi_\alpha) d(x^i \circ \varphi_\alpha) \otimes d(x^j \circ \varphi_\alpha) = \delta_{ij} dx^i \otimes dx^j$$

defines a metric on each U_α . Let $\{\psi_\alpha\}$ be the partition of unity subordinate to the cover $\{U_\alpha\}$. Define $g = \sum_\alpha \psi_\alpha g_\alpha$. Given $x, y \in T_p M$,

$$g_p(x, y) = \sum_\alpha \psi_\alpha(p) g_\alpha|_p(x, y)$$

Note that this is a finite sum of positive definite pairings with nonnegative coefficients that sum up to 1. It must be positive definite itself. $\square$

⁵For covectors α and β , $\alpha \otimes \beta = \frac{1}{2}(\alpha \otimes \beta + \beta \otimes \alpha) = \alpha \beta$.

Remark. Each smooth manifold of dimension greater than 0 admits infinitely many Riemannian structures. They form a contractible space of metrics on M .

Example. Any embedded manifold $M \hookrightarrow \mathbb{R}^N$ has a Riemannian structure by restricting the dot product on $\mathbb{R}^N$ to $T_p M \subseteq \mathbb{R}^N$ for each $p \in M$. More generally, the restriction of a Riemannian structure along an immersion $N \hookrightarrow (M, g)$ gives a Riemannian structure on N .

Vocabulary 10.10 (orthonormal frame). Given a local frame $X_1, \dots, X_n$ of $TM|_U$, we can apply Gram-Schmidt to these point-wise to get an orthonormal basis of $(T_p M, g_p)$ for $p \in U$. This actually gives an orthonormal frame, i.e. the orthonormal basis varies smoothly with p .

Proposition 29. A Riemannian structure provides an isomorphism of bundles: $TM \simeq T^*M$.

Proof. We first write down the map $TM \rightarrow T^*M$ by $(p, v) \mapsto (p, g_p(v, -))$. Note that $g_p(v, -) : T_p M \rightarrow \mathbb{R}$ is given by $w \mapsto g_p(v, w)$. To show this map is smooth, we need to observe how it acts on smooth vector fields. For any fixed $X \in \mathfrak{X}(M)$, $g(X, -)$ can be viewed as a linear function over $C^\infty(M)$ with respect to the second variable. Tensor characterization tells us $g(X, -)$ is in fact a smooth covector field. From there, we can verify $g(X, -) \in \Gamma(T^*M)$ showing it is smooth. Here we identify X with $(p, X_p) \in TM$. If $g_p(v, -) = 0$ for some $v \in T_p M$, then

$$g_p(v, v) = \langle v, v \rangle_g = 0$$

By the positive definiteness, $v = 0$. Since $\dim(TM) = \dim(T^*M)$, injectivity implies bijection. $\square$

Vocabulary 10.11 (musical isomorphism). Now we have these maps $TM \leftrightarrow T^*M$ given by $X \mapsto X^\flat$ and $\omega^\sharp \leftarrow \omega$. $X^\flat$ and $\omega^\sharp$ denote the image and the inverse map of $g(-, -)$ respectively. This uses the identification in proposition 30. In local coordinates, $g = g_{ij} dx^i dx^j$. So for some $X, Y \in \mathfrak{X}(M)$,

$$g_{ij} dx^i dx^j(X)(Y) = g_{ij} dx^i \left(X^i \frac{\partial}{\partial x^i} \right) dx^j \left(Y^j \frac{\partial}{\partial x^j} \right) = g_{ij} (X^i Y^j)$$

This implies $X^\flat = g(X, -) = g_{ij} X^i dx^j$. Suppose $\omega \in \mathfrak{X}^*(M)$ identified with $(p, \omega_p) \in T^*M$. $\omega^\sharp = g^{-1}(-, -)(\omega) = g^{ij} \omega_j \frac{\partial}{\partial x^i}$ where g^{ij} is the inverse of the matrix $g_{ij}(p)$.

Remark. We can define the gradient as

$$\text{grad}(f) = g^{-1}(-, -)(df) = (df)^\sharp = \left(\frac{\partial f}{\partial x^i} dx^i \right)^\sharp = \frac{\partial f}{\partial x^i} (dx^i)^\sharp = \frac{\partial f}{\partial x^i} g^{ij} \frac{\partial}{\partial x^j}$$

If on $\mathbb{R}$ with the usual metric, we have $g^{ij} = \delta^{ij}$. This is the grad we all know and love.

Week 11

Differential forms: Defined on manifolds for integration purposes, and evaluation. Exterior Derivative: Extends differentiation to forms. Wedge and Symmetric Products. Pullback and Lie Derivative: Transform forms between manifolds or along vector fields. Orientation and Oriented Manifolds.

11/5

Vocabulary 11.1 (differential k -form). A differential k -form is an alternating covariant k -tensor

$$\omega \in \Omega^k(M) := \Gamma(\Lambda^k(T^*M)) \subseteq \Gamma((T^*M)^{\otimes k})$$

$\omega(X_1, \dots, X_k) = (-1)^{|\sigma|} \omega(X_{\sigma(1)}, \dots, X_{\sigma(k)}) \in C^\infty(M)$ where $\sigma \in \mathfrak{S}_k$ is a permutation. The *vector space of k -form* is denoted $\Omega^k(M)$. There are 2 important structures on differential forms: ① **Wedge product** $\Omega^k(M) \otimes \Omega^l(M) \rightarrow \Omega^{k+l}(M)$, $(\omega, \eta) \mapsto \omega \wedge \eta$ and ② **exterior derivative** $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$. $|\omega| = k$. These satisfy:

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^{|\omega|} \omega \wedge d\eta$$

Example. Some k -forms

1. $\Omega^1(M) = \Gamma(T^*M)$, i.e. differential 1-form are covectors.
2. $\Omega^0(M) = C^\infty(M)$. This becomes $\Omega^1(M)$ under d . So we have a map $d : \Omega^0(M) \rightarrow \Omega^1(M)$. $df(V) = V(f)$ for any $V \in \mathfrak{X}(M)$. Locally,

$$df = \frac{\partial f}{\partial x^i} dx^i$$

Recall symmetric and alternating products. Suppose $\omega \in (V^*)^{\otimes k}$ and $\eta \in (V^*)^{\otimes l}$. More precisely, we get symmetric and alternating multilinear functions via

$$\begin{aligned} \text{sym}(\omega)(X_1, \dots, X_k) &= \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \omega(X_{\sigma(1)}, \dots, X_{\sigma(k)}) \\ \text{alt}(\omega)(X_1, \dots, X_k) &= \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} (-1)^{|\sigma|} \omega(X_{\sigma(1)}, \dots, X_{\sigma(k)}) \end{aligned}$$

If $\omega \otimes \eta \in (V^*)^{\otimes(k+l)}$, $\omega \otimes \eta(X_1, \dots, X_k, X_{k+1}, \dots, X_{k+l}) = \omega(X_1, \dots, X_k) \eta(X_{k+1}, \dots, X_{k+l})$.

Vocabulary 11.2 (alternating and symmetric product). We define the alternating and symmetric product of ω and η to be $\text{alt}(\omega \otimes \eta)$ and $\text{sym}(\omega \otimes \eta)$. Then $\omega \wedge \eta := \frac{(k+l)!}{k!l!} \text{alt}(\omega \otimes \eta)$.

Proposition 30. The wedge product of differential forms is defined point-wise by $\omega \wedge \eta$. It satisfies

1. Bilinearity: $(f\omega) \wedge \eta = \omega \wedge (f\eta)$ for any $f \in C^\infty(M)$.
2. Associativity: $\omega_1 \wedge (\omega_2 \wedge \omega_3) = (\omega_1 \wedge \omega_2) \wedge \omega_3$.
3. Anticommutativity: $\omega \wedge \eta = (-1)^{|\omega| \cdot |\eta|} \eta \wedge \omega$. This is negative if and only if at least one of ω, η has odd degree.
4. $F^*(\omega \wedge \eta) = F^*\omega \wedge F^*\eta$ for $F : N \rightarrow M$ smooth.
5. In any smooth chart, $F^* \left(\sum_{\mathcal{I}} \omega_{\mathcal{I}} dy^{i_1} \wedge \dots \wedge dy^{i_k} \right) = \sum_{\mathcal{I}} (\omega_{\mathcal{I}} \circ F) d(y^{i_1} \circ F) \wedge \dots \wedge d(y^{i_k} \circ F)$.

Example. Some examples of wedge products

1. Forms on $\mathbb{R}^3$.
 - (a) 1-forms: $C^\infty(M)$ -linear span of dx^1, dx^2, dx^3 .
 - (b) 2-forms: $dx^1 \wedge dx^2 = -dx^2 \wedge dx^1$, $dx^1 \wedge dx^3 = -dx^3 \wedge dx^1$, and $dx^2 \wedge dx^3 = -dx^3 \wedge dx^2$.
(**Warning:** $dx^1 \wedge dx^1 = 0$. In fact, any repeat will result in 0.)
 - (c) 3-forms: $dx^1 \wedge dx^2 \wedge dx^3 = (-1)^{|\sigma|} dx^{\sigma(1)} \wedge dx^{\sigma(2)} \wedge dx^{\sigma(3)}$.
2. Forms on $\mathbb{R}^n$. Define "multi-index" $\mathcal{I} = (i_1, \dots, i_k)$ with order $i_1 < \dots < i_k$.

$$\omega = \sum_{i_1 < \dots < i_k} \omega_{i_1, \dots, i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_{\mathcal{I}} \omega_{\mathcal{I}} dx^{\mathcal{I}}$$

At any point, differential k -forms are spanned by $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ of $dx^{\mathcal{I}}$'s.

Vocabulary 11.3 (exterior derivative). For any $\omega \in \Omega^1(M)$, $d\omega \in \Omega^2(M)$ is defined as

$$d\omega \stackrel{\text{loc}}{=} d(\omega_j dx^j) := \sum_{i,j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j = \sum_{i < j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j + \sum_{i > j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j = \sum_{i < j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} \right) dx^i \wedge dx^j$$

The negative sign is introduced by the i, j swapping. More generally, for $\omega \in \Omega^k(M)$,

$$d\omega \stackrel{\text{loc}}{=} d \left(\sum_{\mathcal{I}} \omega_{\mathcal{I}} dx^{\mathcal{I}} \right) = d \left(\sum_{i_1 < \dots < i_k} \omega_{i_1, \dots, i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \right) := \sum d\omega_{\mathcal{I}} \wedge dx^{\mathcal{I}} = \sum_{\mathcal{I}} \sum_{i=1}^k \frac{\partial \omega_{\mathcal{I}}}{\partial x^i} dx^i \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}$$

We then need to reorder the multi-indices and remove terms in the sum where dx^i appeared twice. Alternatively,

$$\begin{aligned} d\omega(X_1, \dots, X_{k+1}) &= \sum_{1 \leq i \leq k+1} (-1)^{i+1} X_i \omega(X_1, \dots, \hat{X}_i, \dots, X_{k+1}) \\ &\quad + \sum_{1 \leq i < j \leq k+1} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_{k+1}) \end{aligned}$$

Vocabulary 11.4 (closed k-form). An 1-form $\omega = \omega_i dx^i$ is closed if and only if $\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} = 0$. For any $\omega \in \Omega^1(M)$, recall that $d\omega \in \Omega^2(M)$ is defined as $d\omega = \sum_{i < j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} \right) dx^i \wedge dx^j$. Therefore, $d\omega = 0$ if and only if ω is closed if and only if $d\omega(X, Y) = X\omega(Y) - Y\omega(X) - \omega([X, Y])$. $\omega \in \Omega^k(M)$ is closed if $d\omega = 0$.

Proposition 31. There exists a unique $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ such that

1. d is $\mathbb{R}$ -linear. For any $c \in \mathbb{R}$, $d(c\omega) = cd\omega$.
2. $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^{|\omega|} \omega \wedge d\eta$. If $f \in \Omega^0(M)$, $d(f\eta) = df \wedge \eta + f d\eta$.
3. $d^2 = d \circ d = 0$.
4. $f \in \Omega^0(M) = C^\infty(M)$, $df = \frac{\partial f}{\partial x^i} dx^i$.
5. Recall from Proposition 24, if $F : M \rightarrow N$, $F^*(d\omega) = d(F^*\omega)$.

$$\begin{array}{ccc} \Omega^k(N) & \xrightarrow{F^*} & \Omega^k(M) \\ \downarrow d & & \downarrow d \\ \Omega^{k+1}(N) & \xrightarrow{F^*} & \Omega^{k+1}(M) \end{array}$$

proof sketch. (1.) It is clear that d is linear. (2.) follows from our coordinate-wise construction. (3.) starts with d^2 on 0-forms:

$$d(df) = d\left(\frac{\partial f}{\partial x^i} dx^i\right) \stackrel{11.3}{=} \frac{\partial^2 f}{\partial x^i \partial x^j} dx^i \wedge dx^j \stackrel{11.3}{=} \sum_{i < j} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \frac{\partial^2 f}{\partial x^j \partial x^i} \right) dx^i \wedge dx^j = 0.$$

(4.) is merely a reduction of the general case of Vocabulary 11.3. (5.) can be derived as such:

$$F^*(d(\omega_{\mathcal{I}} dx^{\mathcal{I}})) \stackrel{11.3}{=} F^*(d\omega_{\mathcal{I}} \wedge dx^{\mathcal{I}}) \stackrel{\text{prop 31}}{=} F^*(d\omega_{\mathcal{I}}) \wedge F^*(dx^{\mathcal{I}}) \stackrel{\text{prop 24}}{=} d(\omega^{\mathcal{I}} \circ F) \wedge d(x^{\mathcal{I}} \circ F) \stackrel{11.3}{=} dF^*(\omega_{\mathcal{I}} dx^{\mathcal{I}})$$

□

11/7

Vocabulary 11.5 (exact k -form). $\omega \in \Omega^k(M)$ is exact if there exists some $\eta \in \Omega^{k-1}(M)$ such that $d\eta = \omega$.

Vocabulary 11.6 (contraction operator). For $X \in \mathfrak{X}(M)$,

$$\iota_X : \Omega^k(M) \rightarrow \Omega^{k-1}(M) \quad \omega \mapsto \omega(X, -, \dots, -)$$

Example. Recall that the determinant map can be thought of as an alternating multilinear map

$$\det : \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_{n \text{ copies}} \rightarrow \mathbb{R}^n$$

where we put the n -column vectors together in a matrix and then take the matrix's determinant. Then if $v \in \mathbb{R}^n$ is any column vector, $\iota_v \det$ is going to be an alternating multilinear map that now takes in $(n-1)$ -column vectors $w_1, \dots, w_{n-1}$, and spits out the determinant of the matrix with columns $v, w_1, \dots, w_{n-1}$. In other words

$$\iota_v \det(w_1, \dots, w_{n-1}) = \det(v, w_1, \dots, w_{n-1}) = \det \begin{bmatrix} | & | & \dots & | \\ v & w_1 & \dots & w_{n-1} \\ | & | & \dots & | \end{bmatrix}$$

Even more concretely, if $\det : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $v = [1 \ 0]^T$, then $\iota_v \det([a \ b]^T) = \det \begin{bmatrix} 1 & a \\ 0 & b \end{bmatrix} = b$.

Lemma. $\iota_X \circ \iota_X = 0$ for any $X \in \mathfrak{X}(M)$.

Proof. $((\iota_X \circ \iota_X)\omega)(X_1, \dots, X_{k-2}) = \omega(X, X, X_1, \dots, X_{k-2}) = -\omega(X, X, X_1, \dots, X_{k-2})$ because we can switch the first two slots with a sign change. Since the first two slots are the same, $(\iota_X \circ \iota_X)\omega$ must be the zero map. □

Vocabulary 11.7 (Lie derivative of forms). The Lie derivative of k -forms is given by a map

$$\mathcal{L}_X : \Omega^k(M) \rightarrow \Omega^k(M) \quad \omega \mapsto \mathcal{L}_X \omega$$

Proposition 32 (Cartan's magic formula). For any $\omega \in \Omega^k(M)$ and $X \in \mathfrak{X}(TM)$,

$$\mathcal{L}_X \omega = \iota_X(d\omega) + d(\iota_X \omega)$$

In other words, $\mathcal{L}_X \omega = \iota_X \circ d + d \circ \iota_X$.

proof sketch. Note that $\mathcal{L}_X(\omega \wedge \eta) = (\mathcal{L}_X\omega) \wedge \eta + \omega \wedge (\mathcal{L}_X\eta)$. This allows us to verify the Cartan formula by checking in low degrees. For $f \in C^\infty(M) = \Omega^0(M)$, we get

$$\iota_X(df) + d(\iota_X f) = \iota_X(df) = df(X) = Xf = \mathcal{L}_X f$$

In higher degrees, $\omega = \omega_{\mathcal{I}} dx^{\mathcal{I}}$ locally. Some careful calculations yield $d \circ (\iota_X \omega_{\mathcal{I}} dx^{\mathcal{I}}) + \iota_X \circ (d\omega_{\mathcal{I}} dx^{\mathcal{I}}) = \mathcal{L}_X \omega$. $\square$

Corollary. $d(\mathcal{L}_X \omega) = \mathcal{L}_X(d\omega)$.

Proof. Indeed, we can expand both sides using proposition 33: $d((\iota_X \circ d + d \circ \iota_X)\omega) = ((\iota_X \circ d + d \circ \iota_X)d\omega)$. Proposition 32 says $d^2 = 0$. So $d(\iota_X \circ d)\omega = (d \circ \iota_X)d\omega$. $\square$

Remark. Set $M = \mathbb{R}^3$ and consider $\Omega^\bullet(M) := \bigoplus_{k \in \mathbb{N}} \Omega^k(M)$. Recall that $\iota_X(dx \wedge dy \wedge dz)$ is a 2-form obtained by inserting X into the first argument of $dx \wedge dy \wedge dz$. Suppose $X = X^1 \frac{\partial}{\partial x} + X^2 \frac{\partial}{\partial y} + X^3 \frac{\partial}{\partial z}$. Define $\beta : \mathfrak{X}(M) \rightarrow \Omega^2(\mathbb{R}^3)$ given by $\beta(X) = \iota_X(dx \wedge dy \wedge dz)$ as a $C^\infty(M)$ module map. So for some $Y, Z \in \mathfrak{X}(M)$,

$$\begin{aligned} \beta(X)(Y, Z) &= \iota_X(dx \wedge dy \wedge dz)(Y, Z) = dx \wedge dy \wedge dz(X, Y, Z) \\ &= X^1 dy(Y) dz(Z) - X^1 dy(Z) dz(Y) + dx(Z) X^2 dz(Y) - dx(Y) X^2 dz(Z) + dx(Y) dy(Z) X^3 - dx(Z) dy(Y) X^3 \\ &= (X^1 dy \wedge dz - X^2 dx \wedge dz + X^3 dx \wedge dy)(Y, Z) = \beta \left(X^1 \frac{\partial}{\partial x} + X^2 \frac{\partial}{\partial y} + X^3 \frac{\partial}{\partial z} \right)(Y, Z) \end{aligned}$$

It turned out $\beta : TM \rightarrow \Lambda^2(T^*\mathbb{R}^3)$ is a bundle isomorphism. Recall that $X^\flat = g(X, -) = g_{ij} X^i dx^j$ and $\omega^\# = g^{-1}(-, -)(\omega) = g^{ij} \omega_j \frac{\partial}{\partial x^i}$. In $\mathbb{R}^3$ this becomes $X^\flat = X^i dx^i$ and $\omega^\# = \omega_j \frac{\partial}{\partial x^j}$. Finally, we define another bundle isomorphism $\star : C^\infty(\mathbb{R}^3) \rightarrow \Omega^3(\mathbb{R}^3)$ by $\star(f) = f dx \wedge dy \wedge dz$. With all the setups:

$$\begin{array}{ccccccc} C^\infty(\mathbb{R}^3) & \xrightarrow{\text{grad}} & \mathfrak{X}(\mathbb{R}^3) & \xrightarrow{\text{curl}} & \mathfrak{X}(\mathbb{R}^3) & \xrightarrow{\text{div}} & C^\infty(\mathbb{R}^3) \\ \uparrow \text{id} & & \downarrow \flat & \uparrow \# & \downarrow \beta & & \downarrow \star \\ \Omega^0(\mathbb{R}^3) & \xrightarrow{d} & \Omega^1(\mathbb{R}^3) & \xrightarrow{d} & \Omega^2(\mathbb{R}^3) & \xrightarrow{d} & \Omega^3(\mathbb{R}^3) \end{array}$$

Vocabulary 11.8 (orientation). An orientation of a vector space is an equivalence class of ordered bases where $(e_1, \dots, e_n) \sim (e'_1, \dots, e'_n)$ if the matrix sending one basis to the other has a positive determinant.

Example. On $\mathbb{R}^n$, the standard basis gives the *standard orientation*.

Vocabulary 11.9 (smooth orientation). Given a smooth manifold M , one can consider an orientation on each tangent space $T_p M$ at point $p \in M$. Such a collection of point-wise orientations is smooth if for each $p \in M$ there exists U and a local frame $(E_1, \dots, E_n)$ of $TM|_U$ such that the orientations on $T_{p'} M$, $p' \in U$, come from the orientation determined by the ordered frame.

Vocabulary 11.10 (oriented manifold). An oriented manifold $(M, \mathcal{O})$ is a smooth manifold M together with a smoothly varying point-wise orientation of $T_p M$ for each $p \in M$. It is also given by charts whose transition functions are orientated, i.e. $\det(d(\varphi^{-1} \circ \psi)) > 0$.

Vocabulary 11.11 (evaluating k -form). Suppose we have 1-forms $\omega_1, \dots, \omega_k \in \Omega^1(M)$ and vector fields $V_1, \dots, V_k \in \mathfrak{X}(M)$. The evaluation of the k -form $\omega_1 \wedge \dots \wedge \omega_k$ on $V_1, \dots, V_k$ is given by

$$\omega_1 \wedge \dots \wedge \omega_k(V_1, \dots, V_k) = \det(\omega_i V_j)$$

Proposition 33. Two nonzero n -covectors determine the same orientation if they differ by a positive multiple.

Proof. Suppose we have two bases (e_i) and (e'_i) . Matrix A transforms e_i to e'_i for all i . They form n -covectors $\omega = e_1 \wedge \dots \wedge e_n$ and $\omega' = e'_1 \wedge \dots \wedge e'_n$. Let $V_i \in \mathfrak{X}(M)$. Now we can do the calculation:

$$\omega(V_1, \dots, V_n) \stackrel{11.11}{=} \det(e_i V_j) = \det(A e'_i V_j) = \det(A) \det(e'_i V_j) = \det(A) \omega'(V_1, \dots, V_n)$$

It turned out they differ by a multiple of $\det(A)$. This still holds for "if and only if". $\square$

Proposition 34. A nonvanishing n -form on a manifold determines an orientation. Conversely, an oriented manifold has a nonvanishing n -form that every point agrees with the orientation of $T_p M$. This n -form is $e_1 \wedge \dots \wedge e_n$.

Week 12

Integration on manifolds, Stokes theorem, Calc III revisit, and de Rham cohomology.

11/12

Vocabulary 12.1 (continuous orientation). A pointwise orientation (on $T_p M$) is said to be continuous if every point of M is in the domain of an oriented local frame.

Lemma. A parallelizable manifold is orientable.

Proof. Recall that a smooth manifold is parallelizable if it admits a smooth global frame. Suppose M is parallelizable, and let $(E_1, \dots, E_n)$ be a global smooth frame for M . Declare $[E_1|_p, \dots, E_n|_p]$ to be an ordered basis on $T_p M$ with positive orientation. Since we can restrict $(E_1, \dots, E_n)$ to any $p \in M$, M is in the domain of this global smooth frame. This shows the orientation is continuous. $\square$

Proposition 35. Given a codimension-1 submanifold $S \subseteq M$ (or a hypersurface) with inclusion map $i : S \hookrightarrow M$. A vector field X that is nowhere tangent to S induces an orientation on S .

proof sketch. Suppose $(E_1, \dots, E_n)$ is an oriented basis for $T_p S$. By assumption, $(X_p, E_1, \dots, E_n)$ is an oriented basis for $T_p M$. Let ω be a nonvanishing n -form on M . By Vocabulary 9.13, $i^* \eta(v) = \eta(di(v)) = \eta|_{T_p S}(v)$ for some $(n-1)$ -form η . Then $i^*(\iota_X \omega)$ is a nonvanishing $(n-1)$ -form on S . It gives an orientation on S . $\square$

Corollary. Given $\partial M \subseteq M$ and an orientation of M , we get an induced orientation on ∂M .

Proof. Take the outward pointing normal vector field for $\partial M \subseteq M$. $\square$

Vocabulary 12.2 (integrating k -forms). Let $D \subseteq \mathbb{R}^k$ be the domain of integration. The integral of a k -form $\omega \in \Omega^k(\mathbb{R}^k)$ over D is defined as

$$\int_D \omega = \int_D f dx^1 \wedge \dots \wedge dx^k := \int_D f dx^1 \dots dx^k = \int_D f dV$$

Simply speaking, to compute the integral of a form like ω , just erase all the wedges.

Theorem 30 (change of variables). Suppose $D, E \subseteq \mathbb{R}^n$ and $F : \overline{D} \rightarrow \overline{E}$ is a smooth map that restricts to a diffeomorphism from D to E . For every continuous function $f : \overline{E} \rightarrow \mathbb{R}$, $\int_E f dV = \int_D (f \circ F) |\det dF| dV$.

Proposition 36. Let $F : M \rightarrow N$ be a smooth map. If (x^i) and (y^j) are smooth coordinates on open subsets $U \subseteq M$ and $V \subseteq N$, and u is a continuous real-valued function on V , then the following holds on $U \cap F^{-1}(V)$:

$$F^*(u dy^1 \wedge \dots \wedge dy^n) = (u \circ F)(\det dF) dx^1 \wedge \dots \wedge dx^n$$

Remark. Theorem 29 is a fact from calculus where dF represents the Jacobian of F . Proposition deals with the pullback. In Proposition 24, we saw the pullback of real-valued functions (0-forms) like $F^*u = u \circ F$ and the pullback of 1-forms like $F^*(du) = d(u \circ F)$. Now we combine them together with $d(y^i \circ F) = dF^i$ and get $F^*(u dy^1 \wedge \dots \wedge dy^n) = (u \circ F) dF^1 \wedge \dots \wedge dF^n$. If we evaluate this on $(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$, we get $(u \circ F)(\det dF)$. This is the same as $(u \circ F)(\det dF) dx^1 \wedge \dots \wedge dx^n(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$.

Proposition 37. Let $F : \overline{D} \rightarrow \overline{D}'$ be a diffeomorphism of domains in $\mathbb{R}^n$. For some n -form $\omega \in \Omega^n(\mathbb{R}^n)$,

$$\int_{D'} \omega = \pm \int_D F^* \omega$$

"+" means F is orientation preserving while "-" means F is orientation reversing.

Proof. Let (y^i) be coordinates on D' and (x^i) be coordinates on D . Assume $\omega = f dy^1 \wedge \dots \wedge dy^n$. Then

$$\begin{aligned} \int_{D'} \omega &\stackrel{12.2}{=} \int_{D'} f dV \stackrel{\text{thm 29}}{=} \int_D (f \circ F) |\det dF| dV = \pm \int_D (f \circ F)(\det dF) dV \\ &= \pm \int_D (f \circ F) |\det dF| dV = \pm \int_D (f \circ F)(\det dF) dx^1 \wedge \dots \wedge dx^n \stackrel{\text{prop 37}}{=} \pm \int_D F^* \omega \end{aligned}$$

$\square$

Vocabulary 12.3 (integrating over M). Suppose M is a compact oriented n -manifold with smooth oriented charts $\{U_i, \varphi_i\}$. Choose a partition of unity $\{\psi_i\}$ subordinate to this cover. Then

$$\int_M \omega := \sum_i \int_{\varphi_i(U_i)} \psi_i(\varphi_i^{-1})^* \omega|_{U_i} = \sum_i \int_M \psi_i \omega$$

Remark. If M is compact, the sum is finite. We can also take ω compactly supported so that the sum is automatically locally finite. In fact, it is finite because ω is compactly supported.

Proposition 38. $\int_M \omega$ is well-defined.

proof sketch. We must show the integral is independent of the choice of cover and partition of unity. Suppose we have another partition of unity $\{\tilde{\psi}_i\}$ subordinate to some other cover $\{\tilde{U}_i, \tilde{\varphi}_i\}$,

$$\int_M \omega = \sum_i \int_M \psi_i \omega = \sum_i \sum_j \int_M \tilde{\psi}_j \psi_i \omega = \sum_j \sum_i \int_M \psi_i \tilde{\psi}_j \omega = \sum_j \int_M \tilde{\psi}_j \omega$$

Some lemma along the line tells us we have compatibility on $\tilde{U}_i \cap U_i$. So the equation above makes sense. $\square$

Vocabulary 12.4 (integrating over submanifolds). Let $i : S \hookrightarrow M$ denotes an oriented compact k -submanifold and $\omega \in \Omega(M)$. Set $\int_S \omega := \int_S i^* \omega$. There is a far more interesting special case. When $i : \partial M \hookrightarrow M$ is the boundary of an oriented n -manifold M , $\omega \in \Omega^{n-1}(M)$ defines

$$\int_{\partial M} \omega := \int_{\partial M} i^* \omega$$

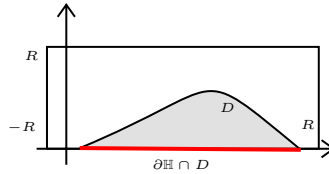
Proposition 39. Suppose M and N are non-empty oriented smooth n -manifolds with or without boundary, and ω, η are compactly supported n -forms on M .

1. Linearity: If $a, b \in \mathbb{R}$, then $\int_M a\omega + b\eta = a \int_M \omega + b \int_M \eta$.
2. Orientation Reversal: If $-M$ denotes M with the opposite orientation, then $\int_{-M} \omega = - \int_M \omega$.
3. Positivity: If ω is a positively oriented orientation form, then $\int_M \omega > 0$.
4. Diffeomorphism Invariance: If $F : N \rightarrow M$ is an orientation-preserving or reversing diffeomorphism, then $\int_M \omega = \pm \int_N F^* \omega$. "+" if F is orientation-preserving. "-" otherwise.

Proposition 40. For a domain $D \subseteq \mathbb{H} \subseteq \mathbb{R}^n$,

$$\int_D d\omega = \int_{\partial \mathbb{H} \cap D} \omega$$

proof idea. We start with a picture:



Then we mess with the signs a little bit and get

$$\int_D d\omega = \int_0^{+R} \int_{-R}^{+R} \cdots \int_{-R}^{+R} d\omega = \int_0^{+R} \int_{-R}^{+R} \cdots \int_{-R}^{+R} (-1)^{\text{some\#}} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n \stackrel{\text{FTC}}{=} \int_{\partial \mathbb{H} \cap D} \omega$$

$\square$

Theorem 31 (Stokes). If M is an oriented compact n -manifold, then for some $\omega \in \Omega^{n-1}(M)$ compactly supported

$$\int_M d\omega = \int_{\partial M} \omega$$

proof idea. Let $\{U_i, \varphi_i\}$ be smooth charts and $\{\psi_i\}$ be the partition of unity subordinate to this cover. We obtain

$$\begin{aligned} \int_{\partial M} \omega &\stackrel{12.3}{=} \sum_i \int_{\partial M} \psi_i \omega \stackrel{\text{prop } 41}{=} \sum_i \int_M d(\psi_i \omega) \stackrel{\text{prop } 32}{=} \sum_i \int_M ((d\psi_i) \wedge \omega + \psi_i d\omega) \\ &= \int_M d\left(\sum_i \psi_i \equiv 1\right) \wedge \omega + \int_M \left(\sum_i \psi_i \equiv 1\right) d\omega = \int_M d\omega \end{aligned}$$

□

Corollary. If M is an oriented compact manifold without boundary, then for some $\omega \in \Omega^{n-1}(M)$, $\int_M d\omega = 0$. In other words, the integral of every exact form over M is zero.

Proof. $\int_M d\omega = \int_{\partial M} \omega = \int_{\emptyset} \omega = 0$. □

Corollary. Let M be an oriented compact manifold with boundary. If $d\omega = 0$ for some $\omega \in \Omega^{n-1}(M)$, then $\int_{\partial M} \omega = 0$. In other words, the integral of a closed form over the boundary is zero.

Corollary. Suppose M is a smooth manifold with or without boundary, $S \subseteq M$ is an oriented compact smooth k -dimensional submanifold (without boundary), and ω is a closed k -form on M . If $\int_S \omega \neq 0$, then

1. ω is not exact on M ($\omega \neq d\eta$ for any $\eta \in \Omega^{k-1}(M)$).
2. S is not the boundary of an oriented compact smooth submanifold with boundary in M . In other words, $S \neq \partial T$ where $T \hookrightarrow M$ is a $(k+1)$ -dimensional submanifold of M .

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Theorem 32 (Green's Theorem). Suppose D is a compact regular domain⁶ in $\mathbb{R}^2$, and P, Q are smooth real-valued functions on D . Then

$$\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_{\partial D} P dx + Q dy$$

Proof. Apply Stokes' theorem to the 1-form $\omega = P dx + Q dy$. $d\omega$ is the integrand on the left hand side. □

Vocabulary 12.5 (Riemannian volume form). On an oriented Riemannian manifold (M, g) , we have a preferred top degree form (i.e. maximal nonzero form). The Riemannian volume form $d \text{vol}_g \in \Omega^n(M)$ has the local expression

$$d \text{vol}_g \stackrel{\text{loc}}{=} \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^n \qquad d \text{vol}_g \stackrel{\text{gl}}{=} \sum_i \psi_i d \text{vol}_g|_U$$

We can globalize this expression using ① a choice of orientation to fix the sign on the right hand side and ② the partition of unity. This globalized expression is shown above.

Vocabulary 12.6 (divergence). On an oriented Riemannian n -manifold (M, g) , define an isomorphism $\star : C^\infty(M) \xrightarrow{\sim} \Omega^n(M)$ given by $\star(f) = f d \text{vol}_g$. Recall that we defined $\beta : \mathfrak{X}(M) \rightarrow \Omega^{n-1}(M)$ by $\beta(X) = \iota_X d \text{vol}_g \in \Omega^{n-1}(M)$ previously. Finally, define

$$\text{div}(X) = \star^{-1}(d\beta(X)) \in C^\infty(M)$$

Or equivalently, $\mathcal{L}_X d \text{vol}_g = \iota_X(d(d \text{vol}_g)) + d(\iota_X d \text{vol}_g)$. Since $d \text{vol}_g$ is a top degree form, $d(d \text{vol}_g) \in \Omega^{n+1}(M)$ is zero on n -dimensional space. So $\mathcal{L}_X d \text{vol}_g = \text{div}(X) d \text{vol}_g$.

Lemma. Let S be an oriented codimension 1 submanifold of an oriented manifold M with $i : S \hookrightarrow M$. Then

$$i^* \beta(X) = i^*(\iota_X d \text{vol}_g) = \langle X, N \rangle d \text{vol}_{g,S} \in \Omega^{n-1}(S)$$

where N is a unit outward-pointing normal vector field along $S \subseteq M$ and (g, S) is the induced metric on S .

Theorem 33 (divergence theorem). Let M be a compact oriented manifold, then for any $X \in \mathfrak{X}(M)$

$$\int_M \text{div}(X) d \text{vol}_g = \int_{\partial M} \langle X, N \rangle d \text{vol}_{g,\partial M}$$

where N is a unit outward-pointing normal vector field along $\partial M \subseteq M$.

⁶ D is a closed, embedded, codimension-0 submanifold of M with a boundary.

Remark. Suppose M is a compact n -manifold without boundary. Let $\omega \in \Omega^n(M)$, $d\omega = 0$. Then $\int_M \omega = \int_M (\omega + d\eta)$ for some $\eta \in \Omega^{n-1}(M)$. This is because $\int_M d\eta = \int_{\partial M=\emptyset} \eta = 0$ by Stokes. This implies for some other form ω' the integral only depends on the equivalence class (ω, ω') in the same class if differing by an exact form)

$$[\omega] = [\omega'] \iff \omega \sim \omega' \iff \omega - \omega' = d\eta \quad (\spadesuit)$$

Vocabulary 12.7 (closed p -form). The vector space of closed p -forms is the subspace $\mathcal{Z}^p(M) \subseteq \Omega^p(M)$ such that for any $\omega \in \mathcal{Z}^p(M)$ we have $d\omega = 0$. Or equivalently, $\mathcal{Z}^p(M) = \ker(d)$.

Vocabulary 12.8 (exact p -form). The vector space of closed p -forms is the subspace $\mathcal{B}^p(M) \subseteq \Omega^p(M)$ such that for any $\omega \in \mathcal{B}^p(M)$ there exists some $\eta \in \Omega^{p-1}(M)$ that satisfies $d\eta = \omega$. Or equivalently, $\mathcal{B}^p(M) = \text{im}(d)$.

Vocabulary 12.9 (de Rham cohomology group). The p^{th} de Rham cohomology group is a real vector space

$$H_{\text{dR}}^p(M) := \mathcal{Z}^p(M)/\mathcal{B}^p(M) = \ker(d)/\text{im}(d) \quad H^\bullet := \bigoplus_{p \in \mathbb{Z}} H^p(M) = \bigoplus_{p \in \mathbb{Z}} H_{\text{dR}}^p(M)$$

where M is a smooth manifold with or without boundary. Note that $d^2 = 0$ implies $\mathcal{B}^p(M) \subseteq \mathcal{Z}^p(M)$. We denote $H_{\text{dR}}^n(-)$ as $H^n(-)$ for convenience. $H^p(M)$ is nonzero only when $0 \leq p \leq \dim(M)$.

Vocabulary 12.10 (de Rham cohomology class). This is defined as the equivalence class of forms. Suppose M is a smooth n -manifold. Let $\omega, \omega' \in \Omega^n(M)$ be closed. They are in the same class if they differ by an exact form $(\spadesuit)$.

Vocabulary 12.11 (cup product). $H^\bullet(M)$ has the structure of a graded commutative ring with ring structure. We can define the cup product between two forms as

$$\smile: H^p(M) \times H^q(M) \rightarrow H^{p+q}(M) \quad \smile: ([\omega], [\eta]) \mapsto [\omega \wedge \eta]$$

We denote $[\omega] \smile [\eta] := [\omega \wedge \eta]$. This is the product between de Rham cohomology classes.

Lemma. This cup product thing makes sense.

Proof. We need to check the map is well-defined. Suppose $\omega, (\omega + d\eta) \in [\omega]$ and $\omega', (\omega' + d\eta') \in [\omega']$. Then $[\omega + d\eta] \smile [\omega' + d\eta'] = [(\omega + d\eta) \wedge (\omega' + d\eta')] = [\omega \wedge \omega'] + [\omega \wedge d\eta' + d\eta \wedge \omega' + d\eta \wedge d\eta']$. We want to show if we can write $[\omega \wedge d\eta' + d\eta \wedge \omega' + d\eta \wedge d\eta']$ as some $d\alpha$. Indeed,

$$\begin{aligned} d\left((-1)^{|\omega|}\omega \wedge \eta' + \eta \wedge \omega' + \eta \wedge d\eta'\right) &= (-1)^{|\omega|}d\omega \wedge \eta' + \omega \wedge d\eta' + d\eta \wedge \omega' + (-1)^{|\eta|}\eta \wedge d\omega' \\ &\quad + d\eta \wedge d\eta' + (-1)^{|\eta|}\eta \wedge d(d\omega') \end{aligned}$$

$[\omega], [\omega'] \in H^p(M)$ implies $d\omega = 0$ and $d\omega' = 0$ because cohomology classes are defined on closed p -forms. So the above calculation simplifies to $\omega \wedge d\eta' + d\eta \wedge \omega' + d\eta \wedge d\eta'$. Lastly, we need to show that it is graded commutative: $[\omega] \smile [\omega'] = [\omega \wedge \omega'] = [(-1)^{|\omega||\omega'|}\omega' \wedge \omega] = (-1)^{|\omega||\omega'|}[\omega'] \smile [\omega]$. $\square$

Theorem 34. $H^\bullet$ is a contravariant functor from smooth manifolds to graded commutative rings satisfying

1. $H^\bullet(M \sqcup N) = H^\bullet(M) \oplus H^\bullet(N)$.
2. For $F_0, F_1 : M \rightarrow N$ smoothly homotopic, $F_0^* = F_1^* : H^\bullet(N) \rightarrow H^\bullet(M)$.

Proof. Suppose we have a smooth map $F : M \rightarrow N$. We need to show $F : H^p(N) \rightarrow H^p(M)$ given by $[\omega] \mapsto [F^*\omega]$ is well-defined. Recall that $F^*d\alpha = dF^*\alpha$. It follows that F^* sends closed forms to closed forms and exact forms to exact forms. This implies F^* is well-defined on cohomology because $[\omega] = [\eta]$ if ω and η differ by an exact form. To show it is an algebra map, we need to check its compatibility with products:

$$F^*[\omega \wedge \eta] = [F^*(\omega \wedge \eta)] \stackrel{\text{prop 31}}{=} [(F^*\omega) \wedge (F^*\eta)] = [F^*\omega] \smile [F^*\eta] = F^*[\omega] \smile F^*[\eta]$$

We conclude that F^* is an algebra map. It is a functor because $(F \circ G)^* = G^* \circ F^*$ and $(\text{id})^* = \text{id}$. (1.) easily follows from $\Omega^p(M \sqcup N) = \Omega^p(M) \oplus \Omega^p(N)$. For (2.), suppose $F_0, F_1 : M \rightarrow N$ are homotopic via $F : M \times I \rightarrow N$. We need to show that for $\omega \in \Omega^k(N)$ closed, $[F_0^*\omega] = [F_1^*\omega]$ if and only if $F_1^*\omega - F_0^*\omega = d\eta$. We must use the fact that $F^*\omega \in \Omega^k(M \times I)$. Consider the construction

$$h : \Omega^k(M \times I) \rightarrow \Omega^{k-1}(M) \quad \text{satisfying} \quad h(d\alpha) + d(h\alpha) = i_1^*\alpha - i_0^*\alpha \quad (\otimes)$$

For $\alpha = F^*\omega$ representing a cohomology class, $h(dF^*\omega) + d(h(F^*\omega)) = d(h(F^*\omega))$ because α is assumed to be a closed form. Note that $d(h(F^*\omega))$ is the $d\eta$ we wish to find earlier. We first want to think about simpler cases. If M is a surface with coordinates (x, y) and $\alpha = a(x, y, t)dx + b(x, y, t)dy + x(x, y, t)dt$ is a 1-form, $(h\alpha)(p) = \int_0^1 c(x, y, t)dt = \int_0^1 \alpha\left(\frac{\partial}{\partial t}\right)dt$. If α is a k -form and $v_1, \dots, v_{k-1} \in T_p M$, then $(h\alpha)(v_1, \dots, v_{k-1}) = \int_0^1 \alpha_{(p,t)}\left(\frac{\partial}{\partial t}, v_1, \dots, v_{k-1}\right)dt$. So define

$$h(\alpha) := \int_0^1 i_t^* \left(\iota_{\frac{\partial}{\partial t}} \alpha \right) dt$$

We can derive $\circledast$ using Cartan's magic formula and the fundamental theorem of calculus. The intuition is that integration is compensating the missing dimension by adding the t -axis. $\square$

Corollary. Homotopy equivalent smooth manifolds M, N have the same de Rham cohomology: $H^\bullet(M) \simeq H^\bullet(N)$.

Corollary. If M is contractible (homotopy type of a point), then

$$H^\bullet(M) = \begin{cases} \mathbb{R} & \text{if } \bullet = 0 \\ H^\bullet(\{\text{pt}\}) = 0 & \text{else} \end{cases}$$

Week 13

Additional applications of de Rham cohomology including Poincaré lemma, compactly supported cohomology, calculations, etc. Introduced degree theory. Turned out singular cohomology is isomorphic to de Rham cohomology. Theorems about symplectic manifolds and Hamiltonian vector fields.

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Lemma (Poincaré lemma). Let M be a star-like region, then $H^\bullet(M) = \begin{cases} \mathbb{R} & \text{if } \bullet = 0 \\ 0 & \text{else} \end{cases}$.

Corollary. Locally on M , every closed form is exact above degree 0.

Proof. Locally, every manifold is contractible. So locally, $\omega|_U = \begin{cases} \text{constant function} & \text{if } |\omega| = 0 \\ d\eta & \text{if } |\omega| > 0 \end{cases}$ $\square$

Proposition 41. $H^0(M)$ is a vector space such that $\dim(H^0(M)) = [\# \text{ of connected components}]$.

Proof. $H^0(M)$ is the set of locally constant functions. So it is the vector space $\mathbb{R}(\# \text{ of connected components})$. $\square$

Example. Now we compute some real life examples such as a torus $S^1 \times S^1$

1. $H^0(S^1 \times S^1) = \mathbb{R}$.
2. $H^0(S^2 \sqcup \{\text{pt}\} \sqcup (S^1 \times S^1)) = \mathbb{R}$.

Vocabulary 13.1 (compactly supported form and cohomology group). Let M be a smooth manifold with or without boundary and let $\Omega_c^p(M)$ denote the vector space of compactly supported smooth p -forms on M . The p^{th} compactly supported de Rham cohomology group of M is $H_c^p(M) = \ker(d)/\text{im}(d)$ where d maps compactly supported forms.

Proposition 42. Let $\dim(M) = n$. $H_c^n(M) = \begin{cases} \mathbb{R} & \text{if } M \text{ is connected orientable} \\ 0 & \text{if } M \text{ is connected nonorientable} \end{cases}$

proof sketch. Assume M has no boundary. If M is orientable, consider a map $I : \Omega_c^n(M) \rightarrow \mathbb{R}$ given by $I(\omega) = \int_M \omega$. Due to Stokes, the integral of the exterior derivative of a compactly supported $(n-1)$ -form is zero. This shows that I is linear. Let (U_i, x^i) be some oriented chart and let f be a smooth bump function compactly supported in U_i . Then the n -form $\omega = f dx^1 \wedge \dots \wedge dx^n$ is nonzero in U_i . It follows that $I(\omega) > 0$. We can scale ω to make it surject onto all of $\mathbb{R}$. Then we claim that $\ker(I)$ consists of exact n -forms. This proves the orientable case. If M is nonorientable, the result follows from the fact that a closed form is always exact. $\square$

Remark. Given a two-piece cover $\{U, V\}$ of M ,

$$\begin{array}{ccc} & M & \\ \swarrow & & \searrow \\ U & \longleftrightarrow U \cap V & \longleftrightarrow V \end{array} \implies \begin{array}{ccccc} & \Omega^\bullet(U \cap V) & & & \\ \swarrow & & \searrow & & \\ \Omega^\bullet(U) & \longleftarrow \Omega^\bullet(M) & \longrightarrow & \Omega^\bullet(V) & \end{array}$$

translates to the sequence $0 \rightarrow \Omega^\bullet(M) \xrightarrow{\alpha} \Omega^\bullet(U) \oplus \Omega^\bullet(V) \xrightarrow{\beta} \Omega^\bullet(U \cap V) \rightarrow 0$. α is given by $\omega \mapsto (\omega_U, \omega_V)$ and β is given by $\beta : (\eta_U, \eta_V) \mapsto (\eta_U - \eta_V)$.

Vocabulary 13.2 (exact sequence). A sequence $\dots \rightarrow V^{k-1} \xrightarrow{F_{k-1}} V^k \xrightarrow{F_k} V^{k+1} \xrightarrow{F_{k+1}} V^{k+2} \rightarrow \dots$ is exact when $\text{im}(F_k) = \ker(F_{k+1})$ for all k .

Theorem 35 (Mayer-Vietoris). There exists a map $\delta : H^k(U \cap V) \rightarrow H^{k+1}(M)$ such that the sequence is exact:

$$\begin{array}{ccccccc} \dots & \xrightarrow{\delta} & H^k(M) & \xrightarrow{[\alpha]} & H^k(U) \oplus H^k(V) & \xrightarrow{[\beta]} & H^k(U \cap V) \xrightarrow{\delta} H^{k+1}(M) \longrightarrow \dots \\ & & & & [\omega] \longmapsto ([\omega_U], [\omega_V]) \longmapsto [\omega_U - \omega_V] \longmapsto \delta[\omega_U - \omega_V] & & \end{array}$$

Remark. The proof of this theorem uses homological algebra. Explicitly, $\delta[\eta] = \begin{cases} -d(\psi_V \eta) & \text{on } U \\ d(\psi_U \eta) & \text{on } V \end{cases}$ where $\{\psi_U, \psi_V\}$ is the partition of unity subordinate to $\{U, V\}$.

Lemma. If M is a connected smooth manifold with a finite fundamental group, then $H^1(M) = 0$.

Example. Now we compute the de Rham cohomology of S^n . Recall that we can use a two-piece chart $\{U, V\}$ to cover S^n where U covers the northern hemisphere and V covers the southern hemisphere. So for each patch $U \simeq \mathbb{R}^n \simeq V$. $U \cap V \simeq S^{n-1}$ describes the "equator". We use " $\simeq$ " for homotopy equivalence. By theorem 34,

$$\begin{array}{ccccccc} \dots \xrightarrow{\delta} H^k(S^n) & \longrightarrow & H^k(\mathbb{R}^n) \oplus H^k(\mathbb{R}^n) & \longrightarrow & H^k(S^{n-1}) \xrightarrow{\gamma} H^{k+1}(S^n) & \longrightarrow & H^{k+1}(\mathbb{R}^n) \oplus H^{k+1}(\mathbb{R}^n) \longrightarrow \dots \\ & & \downarrow \simeq & & & & \downarrow \simeq \\ & & H^k(\{\text{pt}\}) \oplus H^k(\{\text{pt}\}) & & & & H^{k+1}(\{\text{pt}\}) \oplus H^{k+1}(\{\text{pt}\}) \end{array}$$

By proposition 42, $H^0(S^1) = \mathbb{R}$. Recall that S^1 is oriented, connected, and compact. Due to general topology, any closed subset of a compact topological space is compact. It follows that the support of any differential form on S^1 is compact. Now it is safe to apply proposition 43 and conclude $H^1(S^1) = \mathbb{R} = H^n(S^n)$. By the lemma above, $H^k(S^n) = 0$ for any $0 < k < n$. This shows that γ is an isomorphism. We conclude that

$$H^k(S^n) \simeq \begin{cases} \mathbb{R} & \text{if } k = 0 \text{ or } k = n \\ 0 & \text{if } 0 < k < n \end{cases}$$

Vocabulary 13.3 (de Rham complex). The de Rham complex of a smooth n -manifold M is given by

$$0 \rightarrow \Omega^0(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^p(M) \xrightarrow{d} \Omega^{p+1}(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^n(M) \rightarrow 0$$

whose cohomology groups are the de Rham groups of M .

Vocabulary 13.4 (degree). Let M, N be compact connected oriented n -manifolds. For some smooth function $f : M \rightarrow N$, the degree of f is

$$\deg(f) = \sum_{x \in f^{-1}(q)} \text{sign}(\det(df_x))$$

where $q \in N$ is a regular value.

Example. Interpret $(\mathbb{R}, +)$ and $(\mathbb{Z}, +)$ as groups. We identify $S^1 = e^{2\pi i\vartheta}$ with $\mathbb{R}/\mathbb{Z}$ using the map $\varphi(\vartheta) = e^{2\pi i\vartheta}$. Now consider a map $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ given by $f(x) = kx$, $k \in \mathbb{Z}$. Since f is a local diffeomorphism whenever $k \neq 0$, every point on S^1 is a regular value. For some $y \in f^{-1}(q)$, $y = ((\text{fraction part of } q) + j)k^{-1} \bmod 1$. There are k options for j . So the size of $f^{-1}(q)$ is exactly $|k|$. With the consideration of signs, $\deg(f) = |k|\text{sign}(k) = k$. Signs determine the orientation. Coincidentally, we represent $z = e^{i\vartheta}$ with the usual angle $\vartheta \in [0, 2\pi)$. Then $f(z) = z^k$. Now write $d\vartheta = \frac{1}{i}d\log(z)$. This number can be obtained through integration:

$$\int_{S^1} f^*d\vartheta = \int_{S^1} f^*\left(\frac{1}{i}d\log(z)\right) = \int_{S^1} \frac{1}{i}d\log(f(z)) = \int_{S^1} \frac{1}{i}d\log(z^k) = k \int_{S^1} d\vartheta$$

Theorem 36. Let $f : M \rightarrow N$ be the function in Vocabulary 13.4.

$$\int_M f^*\omega = \deg(f) \int_N \omega$$

where $\omega \in \Omega^n(N)$.

Proof. The proof comes in two steps.

Step 1: We will show the existence of a *fixed* constant $a \in \mathbb{R}$ such that for all $\omega \in \Omega^n(N)$, $\int_M f^*\omega = a \int_N \omega$. Choose some $\vartheta \in \Omega^n(N)$ such that $\int_N \vartheta = 1$. Set $a := \int_M f^*\vartheta$. Since we assume the manifold is compact, $H_c^n(M) \simeq H^n(M)$. Proposition 43 tells us $H^n(M) \simeq \mathbb{R}$ is 1-dimensional. That is, $H^n(M)$ can be spanned by any top degree form (n -forms in this case). So for any other $\omega \in \Omega^n(N)$, we have $[\omega] = b[\vartheta]$. Then

$$\int_M f^*\omega = b \int_M f^*\vartheta = b \left(a \int_N \vartheta \right) = a \int_N b\vartheta = a \int_N \omega$$

Step 2: We choose a special ω that allows us to verify the theorem. Choose a regular value q of f . Then f is a local diffeomorphism near q . Since $f^{-1}(q)$ is a properly⁷ embedded 0-submanifold of M , each point in $f^{-1}(q)$ has a neighborhood that is homeomorphic to a point. Such a neighborhood only contains that one point. Suppose $f^{-1}(q)$ is infinite. Since M is compact, $f^{-1}(q)$ has some accumulation point p . But as we discussed earlier, neighborhoods around such a p cannot exist. More specifically, $f(p) = q$ due to continuity. This violates the injectivity conclusion of the inverse function theorem for a neighborhood of p . Therefore, $f^{-1}(q)$ is finite. Suppose $f^{-1}(q) = \{x_1, \dots, x_k\}$. By the inverse function theorem, there is a neighborhood Y_i for each x_i such that $f : Y_i \rightarrow W_i$ for some neighborhood W_i of q . With some necessary shrinking, we assume Y_i are pair-wise disjoint. Then $K := M \setminus (\bigcup_i Y_i)$ is closed in M and thus compact. It follows that $q \notin f(K)$ is closed in N . Let U be the connected component of $W_1 \cap \dots \cap W_k \cap (N \setminus f(K))$ containing q . And let $V_i = f^{-1}(U) \cap Y_i$. We now have the diagram

$$\begin{array}{ccc} M & \xrightarrow{f} & N \\ \uparrow & & \uparrow \\ V_1 \sqcup \dots \sqcup V_k & \xrightarrow{f|_{V_i}} & U_q \end{array}$$

where $f|_{V_i} \simeq U_q$ is a diffeomorphism. Next, we choose some $\omega \in \Omega^n(N)$ with support in U and $\int_U \omega = \int_N \omega = 1$:

$$\int_M f^*\omega = \sum_{i=1}^k \int_{V_i} (f|_{V_i})^* \omega \stackrel{\text{prop 38}}{=} \sum_{i=1}^k (-1)^{\text{sign}(f_i)} = \sum_{i=1}^k (-1)^{\text{sign}(\det(df_x))} = \deg(f)$$

The sign tracks orientation. $x \in V_i$ and $f(x) = q$. □

Remark. This is the generalized version of the $\mathbb{R}/\mathbb{Z}$ example above.

Proposition 43. Some properties of degree include

1. $\deg(f \circ g) = \deg(g) \cdot \deg(f)$.
2. If f is a diffeomorphism, then $\deg(f) = \pm 1$.
3. Homotopic maps have same degree.

⁷A continuous map is proper if preimages of compact sets are compact.

Theorem 37. Suppose N is a compact, connected, oriented, smooth n -manifold, and X is a compact, oriented, smooth $(n+1)$ -manifold with connected boundary. If $f : \partial X \rightarrow N$ is a continuous map that has a continuous extension to X that is denoted as F in the diagram, then $\deg(f) = 0$.

$$\begin{array}{ccc} \partial X & \xrightarrow{f} & M \\ \downarrow & \nearrow F & \\ X & & \end{array}$$

Proof. Let $\omega \in \Omega^n(N)$. Then $d\omega = 0$ because it is an $(n+1)$ -form on an n -manifold. Now integrate

$$\int_{\partial X} f^* \omega = \int_{\partial X} F^* \omega \stackrel{\text{thm 30}}{=} \int_X d(F^* \omega) = \int_X F^* d\omega = 0$$

By Theorem 35, $\deg(f) = 0$. □

Example. We can let $\partial X = S^n$ and $X = D^{n+1}$.

Remark. Whitney approximation allows us to replace smooth maps by continuous maps.

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Vocabulary 13.5 (n -simplex). $\Delta^n \subseteq \mathbb{R}^n$ is given by the convex hull of the origin and standard basis vectors. For instance, a single point gives Δ^0 , a basis vector gives Δ^1 , and two basis vectors gives Δ^2 .

Vocabulary 13.6 (singular chain). Singular p -chains on a space X are the free vector space on the set of continuous functions $\Delta^p \rightarrow X$. $C_p(X)$ denotes the p -chains on X .

Vocabulary 13.7 (i^{th} face map). For each $i = 1, \dots, p$, we define the i^{th} face map in p to be the singular $(p-1)$ -simplex $F_{i,p} : \Delta^{p-1} \rightarrow \Delta^p$ defined by $F_{i,p} = (e_0, \dots, \hat{e}_i, \dots, e_p)$. This is a restricted affine map.

Vocabulary 13.8 (simplex boundary). Let M be a topological manifold. The boundary of a singular p -simplex $\sigma : \Delta^p \rightarrow M$ is the singular $(p-1)$ -chain defined by $\partial\sigma = \sum_{i=0}^p (-1)^i \sigma \circ F_{i,p}$. This extends uniquely to a map $\partial : C_p(X) \rightarrow C_{p-1}(M)$.

Lemma. $\partial \circ \partial = 0$.

Vocabulary 13.9 (singular (co)homology group). For the same M as above, its singular homology group and singular cohomology group are defined as

$$H_p^{\text{sing}}(M) = \frac{\ker(\partial : C_p(M) \rightarrow C_{p-1}(M))}{\text{im}(\partial : C_{p+1}(M) \rightarrow C_p(M))} \quad H_{\text{sing}}^p(M) = \frac{\ker(d : C^p(M) \rightarrow C^{p+1}(M))}{\text{im}(d : C^{p-1}(M) \rightarrow C^p(M))}$$

respectively. Note that $C^p(M)$ is a function on p -chains. The boundary map ∂ gives a differential.

Theorem 38 (de Rham). Let M be a smooth manifold. The map $R : H^p(M) \rightarrow H_{\text{sing}}^p(M)$ is an isomorphism.

proof idea. Let $\sigma : \Delta^p \rightarrow M$ be some p -simplex. We define R to be $[\omega] \mapsto R([\omega]) = \int_{\Delta^p} \sigma^* \omega$. The main obstacle is that σ is only continuous. One approach is to show it suffices to take σ to be smooth. Another approach is to characterize functors $H^p(-)$ and $H_{\text{sing}}^p(-)$. □

Vocabulary 13.10 (nondegenerate form). Suppose M is a smooth manifold. $\omega \in \Omega^2(M)$ is nondegenerate if

$$\Gamma(TM) \longrightarrow \Gamma(T^*M) = \Omega^1(M)$$

$$X \longmapsto \iota_X \omega = \omega(X, -)$$

is an isomorphism.

Vocabulary 13.11 (symplectic manifold). A symplectic manifold (M, ω) is a smooth manifold with $\omega \in \Omega^2(M)$ closed and nondegenerate.

Example. Some symplectic stuff may include

1. Symplectic vector space $M = \mathbb{R}^{2n} = \mathbb{R}^n \oplus \mathbb{R}^n$:

$$\omega = \begin{bmatrix} 0 & \text{id}_{n \times n} \\ -\text{id}_{n \times n} & 0 \end{bmatrix} = \sum_{i=1}^n dp^i \wedge dq^i \in \Omega^2(M)$$

Here $p^1, \dots, p^n$ are coordinates on $\mathbb{R}^n$ and $q^1, \dots, q^n$ are coordinates on $\mathbb{R}^n$.

2. T^*M is symplectic for any M and $\omega = d\alpha$ where α is a "tautological 1-form" on T^*M . Let $\pi : T^*M \rightarrow M$ be the projection that brings $(x, \varphi) \mapsto x$. For some $\alpha \in \Omega^1(T^*M)$ and $V_{(x, \varphi)} \in \mathfrak{X}(T^*M)$, we define

$$\alpha_{(x, \varphi)}(V_{(x, \varphi)}) = \varphi(d\pi_{(x, \varphi)}(V_{(x, \varphi)})) \stackrel{(!)}{=} \sum_{i=1}^n \varphi_i dx^i(V_{(x, \varphi)})$$

to be the tautological 1-form on T^*M . From the definition given above, we know that $d\pi_{(x, \varphi)}(V) \in T_x M$ and $\varphi \in T_x^* M$. Therefore, this expression makes sense. It turned out that $d\alpha$ is nondegenerate (local calculation yields (!)). This implies $(T^*M, d\alpha)$ is symplectic.

Theorem 39 (Darboux). Locally, every $2n$ -dimensional symplectic manifold looks like a symplectic vector space (M, ω) . For all $p \in M$, there exist coordinates $(x^1, \dots, x^n, y^1, \dots, y^n)$ centered at p with $\omega = \sum_i dx^i \wedge dy^i$.

Remark. ① In particular, every symplectic manifold is even dimensional. ② Because every symplectic manifold is locally determined by its dimension, symplectic geometry is closer to topology than Riemannian geometry.

Vocabulary 13.12 (symplectomorphism). For symplectic manifolds (M_1, ω_1) and (M_2, ω_2) , a symplectomorphism is a smooth map $f : M_1 \rightarrow M_2$ such that $f^*\omega_2 = \omega_1 \in \Omega^2(M_1)$. This gives the category of symplectic manifolds.

Vocabulary 13.13 (symplectic vector field). For a symplectic (M, ω) , $X \in \mathfrak{X}(M)$ is symplectic if $\iota_X \omega$ is closed.

Vocabulary 13.14 (Hamiltonian vector field). X (from above) is Hamiltonian if $\iota_X \omega$ is exact. For any smooth f , the Hamiltonian vector field of f is $X_f = \hat{\omega}^{-1}(df)$ where $\hat{\omega} : TM \rightarrow T^*M$ is the bundle isomorphism determined by ω . Equivalently, $\iota_{X_f} \omega = df$. In Darboux coordinates:

$$X_f = \sum_i \left(a^i \frac{\partial}{\partial x^i} + b^i \frac{\partial}{\partial y^i} \right)$$

where a^i, b^i are coefficient functions.

Remark. Recall that we call $\iota_X \omega \in \Omega^1(M)$ exact if there exists some $f \in C^\infty(M)$ such that $\iota_X \omega = df$. Observe that for a function $f \in C^\infty(M)$, we get a Hamiltonian vector field using the fact that ω is nondegenerate:

$$\begin{array}{ccc} \Gamma(TM) & \xrightarrow[\omega(-, -)]{\sim} & \Gamma(T^*M) \\ X_f & \longleftarrow & df \end{array}$$

Hamiltonian vector fields are symplectic. If $H^1(M) = 0$, then all closed 1-forms are exact. It follows that all symplectic vector fields are Hamiltonian.

Proposition 44. The flow of a symplectic vector field is through symplectomorphisms $\vartheta_t : M \rightarrow M$. Time t flows along X and satisfies $\vartheta_t^* \omega = \omega \in \Omega^2(M)$

Proof idea. The statement is equivalent to $\mathcal{L}_X \omega = 0$. By Cartan's magic formula, $(d\iota_X - \iota_X d)\omega = d(\iota_X \omega)$. Using the fact that $d\omega = 0$ and the Vocabulary 13.13, $\mathcal{L}_X \omega = d(\iota_X \omega) = 0$. $\square$

Vocabulary 13.15 (Poisson bracket/algebra). Given $f, g \in C^\infty(M)$ on a symplectic manifold (M, ω) , we define their Poisson bracket to be $\{f, g\} = \omega(X_f, X_g) \in C^\infty(M)$. The bracket is a map $\{-, -\} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$. This is the structure of a Poisson algebra $(C^\infty(M), \{-, -\})$.

Vocabulary 13.16 (Hamiltonian (system)). A Hamiltonian system (M, ω, H) is a symplectic manifold and a Hamiltonian $H \in C^\infty(M)$.

Example. The n -body problem involves $M = \mathbb{R}^{3n} \oplus \mathbb{R}^{3n}$ with the standard symplectic structure. $m_1, \dots, m_n$ correspond to the masses of n objects. Each object has positive mass. Let $x_1, \dots, x_n$ be n points in $\mathbb{R}^3$. We have $p_i = m_i \dot{x}_i \in \mathbb{R}^3$. $\dot{x}_i$ represents the ordinary derivatives of component functions with respect to t . And finally,

$$\sum_{i=1}^n \left(\underbrace{\frac{1}{2m_i} p_i^2}_{\text{kinetic}} + \underbrace{V(x_1, \dots, x_n)}_{\text{potential}} \right)$$

Vocabulary 13.17 (conserved quantity). For a Hamiltonian system (M, ω, H) , a conserved quantity is a function $f \in C^\infty(M)$ constant on the integral curves of X_H .

Vocabulary 13.18 (infinitesimal symmetry). For a Hamiltonian system (M, ω, H) , an infinitesimal symmetry is a symplectic vector field X such that $X(H) = 0$.

Observation.

1. $f \in C^\infty(M)$ is conserved if and only if $\{f, H\} = 0$. The key is that for some $f, g \in C^\infty(M)$, $X_{\{f, g\}} = -[X_f, X_g]$.
2. For an infinitesimal symmetry X , the flow of X sends integral curves of X_H to integral curves of X_H .

Theorem 40 (noether). Let (M, ω, H) be a Hamiltonian system. If f is a conserved quantity, then X_f is an infinitesimal symmetry. f is a conserved quantity if and only if X_f is an infinitesimal symmetry and $H^1(M) = 0$.

Proof. $(\Rightarrow)$ f is conserved means $\{f, H\} = 0$. So $\{f, H\} = \{H, f\}$ in this case. But $\{H, f\} = \omega(X_H, X_f) = dH(X_f) = X_f(H) = 0$. This shows X_f is an infinitesimal symmetry. $(\Leftarrow)$ Given an infinitesimal symmetry X , we have $X_f = X$ for some f when $H^1(M) = 0$. This f is conserved. $\square$

Week 14

Fall break.

Week 15

...

Recall the degree of a smooth map and the $S^1 \rightarrow S^1$ example we did last week. Let $f : M \rightarrow N$ be a smooth map between manifolds where $n + k = \dim(M) \geq \dim(N) = n$. Both M, N are compact without boundary. We want to know how much we can learn from the k -dimensional submanifold $f^{-1}(p) \subseteq M$ associated to a regular value p .

Vocabulary 15.1 (framed (sub)manifold). A framed submanifold $Y \subseteq M$ is a submanifold together with a trivialization of the normal bundle...

Appendix

Topics here are written without formal introductions.

Vocabulary 16.1 (category). A category $\mathbf{C}$ consists of:

1. a class, denoted $\text{Ob}(\mathbf{C})$, of objects;
2. for each pair of objects X and Y , there exists a class of morphisms $f : X \rightarrow Y$, where X is called the domain and Y is called the codomain;
3. a composition operation, which gives, for each pair of morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, a morphism $g \circ f : X \rightarrow Z$,

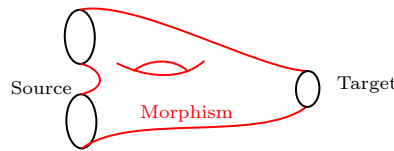
such that

1. given any $f : X \rightarrow Y$, $g : Y \rightarrow Z$, and $h : Z \rightarrow W$, we have the identity $(h \circ g) \circ f = h \circ (g \circ f)$;
2. for each object X , there exists an identity morphism $\text{id}_X : X \rightarrow X$ with the property that $f \circ \text{id}_X = f$ and $\text{id}_X \circ g = g$ for any $f : X \rightarrow Y$ and $g : Z \rightarrow X$.

The class of morphisms from X to Y is denoted by $\text{Hom}_{\mathcal{C}}(X, Y)$. A category $\mathcal{C}$ is a locally small category if $\text{Hom}_{\mathcal{C}}(A, B)$ is a set for all objects A and B in $\mathcal{C}$.

Example. Common examples of (object, morphism):

1. Every monoid is a category.
2. Given a category $\mathcal{C}$, the *dual category* $\mathcal{C}^{op}$ shares the same objects of $\mathcal{C}$ and the converse of morphisms of $\mathcal{C}$.
3. **Set**: (sets, maps of sets).
4. **Grp**: (groups, homomorphisms).
5. **Ring**: (rings, homomorphisms).
6. **Vect_k**: (vector spaces, linear maps).
7. **SmMfld**: (smooth manifolds, smooth maps)
8. **TopMfld**: (topological manifolds, continuous maps)
9. **Cob**: $((n-1)$ -dimensional compact manifolds without boundary, n -dimension compact manifolds with boundary). This is also called the cobordism category. Below is a visualization of cobordism between manifolds.



Vocabulary 16.2 (functors). A covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between two categories assigns

1. $F(X)$ an object of $\mathcal{D}$ for each object X of $\mathcal{C}$.
2. $F(f)$ a morphism in $\mathcal{D}$ from $F(X)$ to $F(Y)$ for each morphism $f : X \rightarrow Y$ in $\mathcal{C}$.

satisfying ① $F(\text{id}_X) = \text{id}_{F(X)}$ and ② $F(f \circ g) = F(f) \circ F(g)$.

Example. Some examples of functors

1. $\text{Grp} \rightarrow \text{Set}$ sends a group to its underlying set. This determines a function once we realize that a homomorphism of groups determines a map of sets. Such functors are called *forgetful functors*, similarly, one could define a list of forgetful functors $\text{Ring} \rightarrow \text{Grp} \rightarrow \text{Mon} \rightarrow \text{Set}$.
2. $\text{Set} \rightarrow \text{Grp}$ sends a set to the free group on the set.
3. Consider a morphism $u : C_* \rightarrow D_*$ that sends boundaries to boundaries and cycles to cycles. Each H_n is a functor from $\text{Ch}(\text{mod-R})$ to mod-R .
4. The *tangent functor* gives a functor $T : \text{Mfld} \rightarrow \text{Mfld}$ maps objects $M \mapsto TM$ and morphisms $(F : M \rightarrow N) \mapsto (dF : TM \rightarrow TN)$.

Formalisms in classical mechanics are based on differential geometry. In particular, we consider the configurational space of a mechanical system as a smooth manifold Q . In many cases $Q = \mathbb{R}^{3N}$ where N is the number of particles in the system. We call $3N$ the degree of freedom. There are two ways of formulation: 1. Lagrangian formalism 2. Hamiltonian formalism. The definition of Hamiltonian is given in the main text, hence we introduce Lagrangian mechanics here in the appendix.

Vocabulary 16.3 (Lagrangian). The Lagrangian of a system is a smooth map $L : TQ \times \mathbb{R} \rightarrow \mathbb{R}$

Let $q : \mathbb{R}(\text{aka, time}) \rightarrow Q$ be a smooth curve. Consider $\dot{q}(t_0)f := \frac{d}{dt}(f \circ \gamma(t))|_{t=t_0}$ for any $f \in C^\infty(Q)$ and $t_0 \in \mathbb{R}$. Hence any point in $TQ \times \mathbb{R}$ can be written as $(q(t), \dot{q}(t), t)$ for some $t \in \mathbb{R}$ and smooth curve q .

Vocabulary 16.4 (action functional). Given a Lagrangian L , the action functional is defined as

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t) dt$$

for some $t_0, t_1 \in \mathbb{R}$.

Fix two points $q(t_0) = q_0$ and $q(t_1) = q_1$, then the physical trajectory of a system extremizes (note that some textbooks use the word minimize, which might not be true all the time...) $S[q]$ (There's no way we can "prove" this statement. In classical mechanics it's assumed to be true). The extremization lead to the following theorem, or the Euler-Lagrange equation.

Theorem 41. $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = \frac{\partial L}{\partial q}$

You might wonder how Hamiltonian formalism is connected to Lagrangian mechanics. In fact, they are equivalent and can be derived from one to the other through Legendre transform.

Vocabulary 16.5 (Legendre transform). Given the Lagrangian $L : TQ \times \mathbb{R} \rightarrow \mathbb{R}$, $\mathbb{F}L : TQ \rightarrow T^*Q$ given by $(q(t), \dot{q}(t)) \mapsto (q, p := \frac{\partial L}{\partial \dot{q}})$

When the Legendre transform is a diffeomorphism, then $H : T^*Q \rightarrow \mathbb{R}$ given by $(q, p) \mapsto \sum_i q^i p^i - L(q, \dot{q}(q, p))$ is a well-defined smooth function. This gives us the Hamiltonian. When the Legendre transform is not a diffeomorphism, then a more complicated theory is needed...

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